NASA Mission Concept Academy

**Preliminary Design Review**

**Team 27**
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## Table of Acronyms

| | |
|---|---|
| VMC | Venus Monitoring Camera |
| SOIR | Solar Occultation InfraRed Spectrometer |
| PM | Project Manager |
| DPMR | Deputy Project Manager of Resources |
| AOTF | Acousto-optical turntable filter |
| CRE | Readout electronic |
| DPU | Digital Processing Unit |
| EDI | Entry Descent and Insertion |
| MCCET | Mission Concept Cost Estimate Tool |
| CER | Cost Estimating Relationship |
| D2D | Door-to-door |
| CCD | Charge-Coupled Device |
| VOC | Volatile Organic Compound |
| OSHA | Occupational Safety and Health Administration |
| RPN | Risk Priority Number |
| SRB | Standing Review Boards |
| Sev | Severity |
| Occ | Occurrence |
| KDP | Key Decision Point |
| EPS | Electrical Power System |
| PCDU | Power Conditioning and Distribution Unit |
| PMAD | Power Management and Distribution |
| TCS | Thermal Control System |
| OBS | Onboard Computer |

## 1.1. Mission Statement

The Venera Nova mission aims to investigate how Venus's surface and interior influence and interact with its atmosphere by studying the planet's superrotation, lack of magnetic field, and potential for life in the planet's upper atmosphere. Venus poses numerous questions for scientists due to its unique atmospheric and geological processes that differ significantly from other planets within the Solar System. Venus's atmosphere rotates significantly faster than the planet itself, with winds circling the planet at speeds of up to 360 kilometers per hour (Kashimura et al, 2019). This atmospheric phenomenon is known as superrotation, for which the causes and atmospheric effects on Venus are largely unknown. Furthermore, Venus notably lacks a magnetic field, exposing the planet to more high-energy solar particles from the Sun (Rehm, 2021). Due to Venus' high wind speeds and extreme temperatures, few spacecraft have measured the precise effects and composition of Venus' atmosphere as it is affected by increased solar radiation. Additionally, scientists have frequently detected phosphine, a gas that is associated with biological processes, in Venus's atmosphere; however, the precise location and concentration of phosphine on the planet is still unknown (Howell, 2023). To further our understanding of these phenomena, the Venera Nova mission will deploy an Aerobot System from Venus's orbit down to the surface. As it descends from the upper atmosphere, the Aerobot system will consist of multiple scientific instruments that will measure the planet's wind speeds, fluctuations in brightness, Kelvin waves, and atmospheric spectra over a 48 hour period. At the conclusion of this mission, data concerning Venus's geologic and atmospheric properties will be collected and relayed to Earth for further analysis. As this valuable data is collected and processed, this will enable the team to make crucial conclusions about the effects of superrotation, consequences of a lack of magnetic field, and the potential for life of Venus - furthering understanding of the unique composition and environment of Venus.

### 1.2 Science Traceability Matrix

The science objectives that will be accomplished by the Venera Nova mission include determining the influence of Venus' superrotation on its surface and atmosphere, investigating how a lack of magnetic field affects atmospheric composition and particle escape, and analyzing the probability of life existing within Venus' upper atmosphere through gas detection. These three objectives align with NASA's Decadal Question 6.4 of "How do planetary surfaces and interiors influence and interact with their host atmospheres?" The Venus Monitoring Camera (VMC) will do this by tracking cloud motion, brightness variations of the clouds, and Kelvin waves in the form of capturing images of Venus's clouds in Ultraviolet (365 nm), Visible (513 nm), and Near-Infrared (965 nm) wavelengths. The other instruments involved in the mission are the Ultraviolet Spectrometer and Solar Occultation InfraRed Spectrometer (SOIR) which are both part of a suite called SPICAV. The UV Spectrometer will collect ionized species such as O+, H+, CO2+ in the UV range (118-320 nm) over a 1 km range every hour. The SOIR will detect phosphine gas between the 190 and 220 nm range. The detection of phosphine could indicate the presence of life in Venus' upper atmosphere. All science operations shall be completed within 48 hours of the Aerobot's descent above Ovda Regio.

| Science Goals | Science Objectives | Science Measurement Requirements | | Instrument Performance Requirements | | Predicted Instrument Performance | Instrument | Mission Requirements |
|---|---|---|---|---|---|---|---|---|
| | | Physical Parameters | Observables | | | | | |
| Determine how Venus' planetary surface and interior influence and interact with its atmosphere. “Decadal Question 6.4: How do planetary surfaces and interiors influence and interact with their host atmospheres ?” | Determine the influence of Venus' superrotation on its surface and atmosphere. | Track cloud motion, brightness variations, and Kelvin waves every 10 seconds. | Capture images of Venus's clouds every 10 seconds in UV, visible, and near-infrared wavelengths. Collect wind speed data by tracking cloud motion every 10 seconds. Record periodic fluctuations in clouds. | Wavelength Range | 365 - 1000 nm | 365 - 1000 nm | Venus Monitoring Camera (VMC) | Mission shall measure cloud dynamics including wind speed and brightness variations through high-quality atmospheric data which will be obtained through the Aerobot's descent. |
| | | | | Integration Time | 0.5 ms - 30 s | 10 s | | |
| | | | | Sensitivity | 0.3 to 1.0 µm | 0.3 to 1.0 µm | | |
| | | | | Spatial Resolution | 0.2 - 45 km | ~ 1.5 km | | Mission shall measure atmospheric spectra, including ion composition and absorption features, as the |
| | Determine how a lack of magnetic field affects atmospheric | Measure the composition, ionization, and the escape rates | Collect ionized species such as O+, H+, CO2+ in the UV range (118-320 nm) | Wavelength Range | 118 - 320 nm | 118 - 320 nm | SPICAV (Ultraviolet Spectrometer) | |
| | | | | Integration | 80 - | 80 - 450 ms at | | |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | composition. | of particles and solar photons in the upper atmosphere. | over a 1 km range every hour. | Time | 450 ms at 1 km | 1 km | | Aerobot descends. |
| | | | | Sensitivity | 200 - 300 nm | 200 - 300 nm | | All science operations shall be completed within 48 hours after deployment of the EDI system. |
| | | | | Spatial Resolution | 100 km | 100 km | | |
| | Determine the probability of life existing within Venus' upper atmosphere. | Confirm presence of phosphine gas in the upper atmosphere. | Detect phosphine peaks between 190 and 220 nm. | Wavelength Range | 2.2 to 4.3 μm | 2.2 to 4.3 μm | SPICAV (Solar Occultation InfraRed Spectrometer - SOIR) | |
| | | | | Integration Time | 5 s | 5 s | | Aerobot will descend from 70 - 50 km above the Ovda Regio region. |
| | | | | Sensitivity | 2257–4430 cm−1 | 2257–4430 cm−1 | | |
| | | | | Spatial Resolution | 0.025-cm−1 | 0.025-cm−1 | | |

### 1.3 Summary of Mission Location

The region of interest (ROI) selected for the Venera Nova mission is a highland region named Aphrodite Terra. Aphrodite Terra is composed of two plateaus—Thetis Regio and Ovda Regio (Squyres, 2012). For this mission, Venera Nova's aerobot will be traveling to Ovda Regio as it sits closer to the equator. Ovda Regio is an ideal site for this mission as locations closest to the equator appear to be where superrotation is strongest. Superrotation is a phenomenon that is exhibited by Venus, where its atmosphere rotates faster than the planet itself. Venus' atmosphere orbits the planet in four days, while it takes 243 days for the surface of the planet itself to rotate completely. (Machemer, 2020). Superrotation is thought to be a result of thermal tides which are atmospheric waves that are driven by heat from the Sun, causing variations in pressure that then generate strong enough winds to accelerate the atmosphere (JPL, 2012). Mission Venera Nova will specifically be exploring the influence of Venus' superrotation and the effects it has on the surface and atmosphere by tracking cloud motion, brightness variations, and measuring Kelvin waves.

Operations will begin in the lower haze layer of the atmosphere, estimated around 70 km in altitude, and descend down to 50 km above the surface. During this time, the aerobot will be collecting measurements of atmospheric spectra in the UV range to determine how an induced magnetic field affects the atmospheric composition of Venus. Lastly, the aerobot will be measuring for phosphine gas peaks, which acts as a biosignature that could potentially indicate the probability of life existing within Venus' atmosphere.

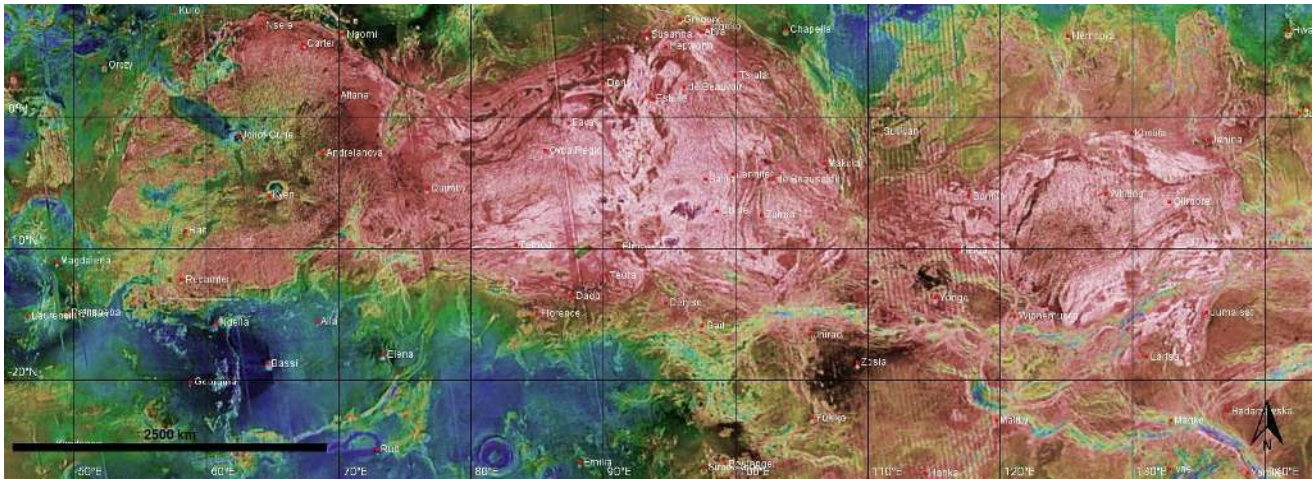



*Figure 1 displays a JMARS map of Ovda Regio (-2.8°N, 85.6°E) at 16 pixels-per-degree.*

### 1.4 Mission Requirements

Numerous requirements for this spacecraft are defined following customer constraints and overall mission task objectives as demonstrated below:

| Req # | Requirement | Rationale | Parent Req | Child Req | Verification Method | Req met? |
|---|---|---|---|---|---|---|
| 0.1 | System shall investigate the impact of Venus' lacking magnetic field on geological landforms and atmosphere. | Fundamental objective of mission. | Fundamental Objective | SCI-002, SCI-004, PAY-005, CDH-8 | Demonstration | Met |
| 0.2 | System shall investigate the impact of Venus' superrotation on atmospheric systems. | Fundamental objective of mission. | Fundamental Objective | SCI-002, SCI-003, PAY-002, PAY-003, PAY-004, CDH-8 | Demonstration | Met |
| 0.3 | System shall investigate the probability of life existing within Venus' upper atmosphere. | Fundamental objective of mission. | Fundamental Objective | PAY-006, CDH-08 | Demonstration | Met |
| SYS-1 | Mass Constraint: The system shall not exceed a mass of 50 kg. | Works with launch vehicle capacity. | Mission Constraints | MECH-001, PAY-001, CDH-001, TH-1, PWR-001, PAY-001 | Inspection | Met |
| SYS-2 | Dimension Constraint: System design shall fall within dimensional constraints before deployment (1m x 1m x 1m). | System must be compatible with the descent module. | Mission Constraints | PAY-007 | Inspection | Met |

| | | | | | | |
|---|---|---|---|---|---|---|
| SYS-3 | Prohibited Materials: System shall not use a Radioisotope Thermoelectric Generator (RTG) or any radioactive material over 5g. | System shall comply with regulations and laws. | Safety Regulations | | Inspection | Met |
| SYS-4 | Cost Constraint: The system shall not exceed a cost of $200 million. | Complies with NASA Discovery Program constraints. | Budget Limitations | PAY-008 | Inspection | Met |
| SYS-5 | System Integration Date Constraint: System shall be ready for integration at Goddard Space Flight Facility by October 1st, 2028. | System shall be ready for integration with other components on schedule. | Project Schedule | | Schedule Basis of Estimate | Met |
| SYS-6 | Launch Readiness Date (LRD) Constraint: System shall be ready for launch at Cape Canaveral by March 1st, 2029. | System shall comply with launch vehicle schedule and planetary alignment of Venus. | Project Schedule | | Schedule Basis of Estimate | Met |
| SYS-7 | The system shall survive for at least 48 hours in the Venusian atmosphere. | Scientific instrumentation needs to survive the entire duration of the Aerobot's descent. | Mission Constraints | PAY-009 | Demonstration | Met |
| MECH-3 | The mechanical system shall survive for at least 48 hours in the Venusian atmosphere | This mission constraint ensures that the mechanical system can wholly support the system and payload during the mission duration | SYS-007 | MECH 3.1, MECH 3.2, MECH 3.3, MECH 3.4, MECH 3.5, MECH 3.6 | Test | Met |

| | | | | | | |
|---|---|---|---|---|---|---|
| PWR-1 | There shall be an adequate amount of power to sustain all subsystems for the entirety of the mission | Electronics include a variety of sensors and controls that shall be functional within planetary entry until total system failure at Venus' surface | SYS-001 | PWR-002 | Testing | Met |
| CDH-8 | The control and data handling subsystem shall contribute to fulfilling the science objectives of the mission | Fundamental objective of mission | 0.1, 0.2, 0.3 | CDH-1 | Analysis | Met |
| TH-2 | The TCS shall maintain all thermal control components within the system's dimensional constraints. | Ensures all passive and active thermal components fit within the system's dimensional constraints. | SYS-002 | | Inspection | Met |
| PAY-9 | All science operations shall be completed within 48 hours after deployment of EDI system. | After 48 hours the SS will uncontrollably fall and crash into Venus' surface, discontinuing science operations. | SYS-007 | | Demonstration | Met |

### 1.5 Concept of Operations

The aerobot will be deployed at 70 km above Ovda Regio. During the first 20 minutes of the mission, the team will check various systems to ensure they are operational. During the first stage after deployment, 5 minutes will be spent calibrating the communications system, along with instruments, telemetry, and electrical calibrations after this for 5 minutes each in the order stated above. Once each system is operational, the deployment stage will end and the operations stage begins. During this stage, the instruments will begin taking measurements simultaneously. The SPICAV UV Spectrometer will collect ionized species such as O+, H+, and CO2+ over a 1 km range and transmit up to 18.18 MB of data back to the orbiter every hour. Every 30 seconds, phosphine peaks between 190 and 220 nm will be recorded and transmitted back to the orbiter by SOIR. VMC will capture images of Venus' clouds in visible, near-infrared, and UV wavelengths every 10 seconds and transmit this data back to the orbiter. It will also track cloud motion and collect wind speed data during this timeframe. Electric propulsion will begin immediately after the aerobot is deployed and will be used to assist the aerobot in descending in a direct path with minimal drift. It will be controlled by unmanned software that will engage the thrusters when needed. Every two hours after the initial checks, two of the four discharging batteries will be switched out with two batteries that were previously charging. This is done to ensure that there will always be four batteries providing power to the system at any given time and that none of the batteries are ever completely depleted. At the 24-hour mark, a mid-way check will be performed to confirm each system is still fully functioning. This will take approximately 10 minutes, with the instruments continuing to take measurements during this time period. During the last several minutes of the mission's 48th hour, the operations stage will end and the aerobot will begin the end of operations stage. The aerobot will send its final communications back to Earth. This will occur when the aerobot is around 50 km from Venus's surface. Exactly when this final communication occurs depends on when the balloon material degradation causes the aerobot to crash-land on the surface, which will render the instruments inoperative.

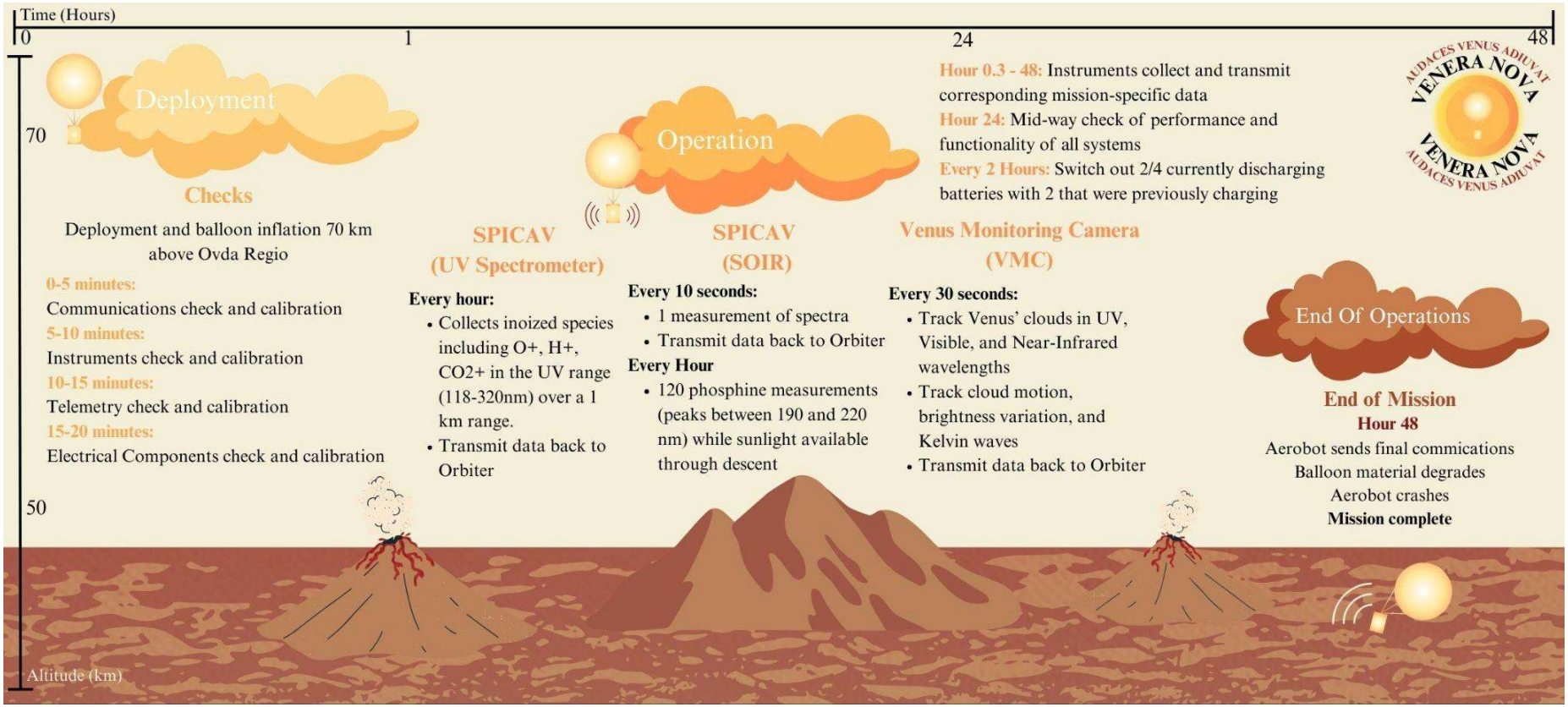



*Figure 2: Graphical representation of the Venera Nova's mission. Time is represented horizontally and the Aerobot's altitude relative to Venus' surface is represented vertically. Scientific operations throughout the mission are detailed throughout the descent process.*

### 1.6 Vehicle Design Summary

The Venera Noba Aerobot is a robust, autonomous probe designed to withstand the extreme environment of Venus. Its primary aim is to descend into Venus' atmosphere above Aphrodite Terra in order to investigate the effects of atmospheric superrotation, the impact of Venus' weak magnetic field, and the potential for life in the upper atmosphere. The vehicle's architecture is centered around a durable chassis, complex thermal protection, and subsystem integrations to ensure that the scientific instrument payload has optimal performance during the Aerobot descent and data collection period.

The key subsystems include the mechanical system - which provides the structural support for the vehicle, the power subsystem - which provides power storage and distribution systems, the command and data handling system - which ensure reliable command and data processing and transmission, the thermal management system - which provides components to maintain operational temperatures for components, and the payload - which consists of the main scientific instruments.

The team's vehicle design aims to be compact and streamlined in order to fit under the dimension and mass constraints while fully supporting and protecting the instruments required for valuable scientific data collections. The chassis and framework are designed for structural integrity, with materials selected in order for the framework to best withstand the high pressures, temperatures, and radiation of Venus. The remaining systems in turn are designed to integrate with this overarching mechanical structure, fully integrating and connecting with one another while fitting under the necessary constraints. The total mass of the system comes to 45, meeting the requirement of being under 45 kg. The overall dimension of the system (focusing on the mechanical structure) is 1 m^3, with all other subsystems lying within this frame.

The Aerobot's power subsystem is built around high-efficiency solar cells which help generate the electricity required for all onboard systems and subsystems. This includes powering the scientific instruments, on-board computer, thermal management, and data handling systems. The dedicated Power Management and Distribution (PMAD) system helps manage and distribute the power that is generated, storing this energy in the case of low solar input or for recovery mechanisms. Overall, this subsystem is designed to be able to handle Venus' temperature and pressure extremes, aiming for continuous operation of all the critical components. The total power provided by the system is 1000 W and the total power used by each of the subsystems is ___, ensuring that the system as a whole consistently has enough power to function and collect data during the Aerobot's descent.

The following depicts the overall design for the vehicle focusing on the main Aerobot system housing the science instruments and necessary subsystems to ensure survival and efficient science collection of the system

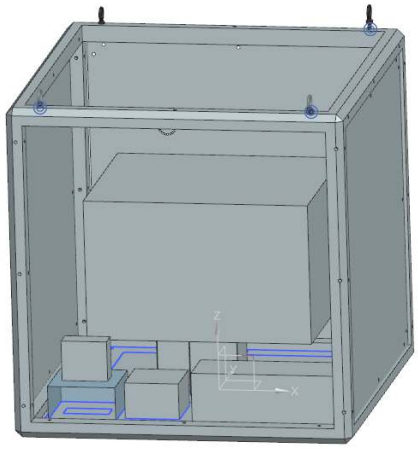


Figure 3: Depiction of the Vehicle Design

### 1.7 Science Instrumentation Summary

The science instruments included on the aerobot are the Venus Monitoring Camera (VMC) and the SPICAV suite, specifically the Ultraviolet Spectrometer and Solar Occultation InfraRed Spectrometer (SOIR). VMC was the instrument selected to help determine the influence of Venus' superrotation on its surface and atmosphere. It will be responsible for capturing images of clouds every 10 seconds in 3 different wavelengths: Ultraviolet, Visible, and Near-Infrared. Distinct features such as swirls or bright spots in the clouds can be identified, then tracked across several images at different altitudes to determine how far they have moved horizontally over 48 hours. This allows for wind speed calculations specifically in the cloud tops where superrotation is most prominent. VMC will also be tracking fluctuations in brightness and Kelvin waves which both provide insight on how energy is transferred. The SPICAV Ultraviolet Spectrometer was selected to help determine how a lack of magnetic field affects atmospheric composition. The spectrometer will measure the composition, ionization, and escape rates of atmospheric particles in the upper atmosphere. More specifically, it will detect concentrations of ions like O+, H+, CO2+ to gain a deeper understanding of Venus' atmosphere and processes causing atmospheric loss. By tracking the concentration of ionized species, scientists can calculate escape rates by observing how UV light from the Sun is absorbed or scattered at different altitudes. Data collected will help further our knowledge and come to a conclusion on how solar energy impacts Venus' atmosphere modern day and in the future. The SPICAV SOIR was selected to determine the probability of life existing within Venus' upper atmosphere. The instrument will target the 190-220 nm range where phosphine - a potential indicator of life, has known absorption features. SOIR will be detecting phosphine peaks and monitor concentration fluctuations between 70-50 km.

## 1.8. Programmatic Summary

### 1.8.1. Team Introduction



**Project Manager:** Rachel Brown studies Physics at the University of California, Irvine and specializes in astrophysics. She has extensive experience in teaming, Python programming, and science communication.



**Chief Scientist:** Valerie Miranda studies Physical Science with an emphasis in Geology at the College of Southern Nevada. She has experience with fossil prepping and excavating late Jurassic Morrison Formation fossils, as well as Pleistocene megafauna.



**Lead System Engineer:** Manahil Syeda studies Electrical Engineering & Computer Science and Data Science at University of California, Berkeley in Berkeley, CA and is highly specialized in machine learning integrations. She has experience working with deep learning platforms, integrated circuits, and DevOps technologies.



**Deputy Project Manager of Resources:** Tyree Nguyen studies Mechanical Engineering at California State Polytechnic University, Pomona and is experienced in mechanical systems/devices that involve physics, mathematics, and sciences. He has experience in design, Java programming, team work, and problem solving.




**Primary Scientist I, Outreach Officer:** Alaina Claudio studies Space Studies at American Military University in Charles Town, West Virginia and specializes in Astronomy. She has experience in teaming, problem solving and scientific writing.



**Thermal Engineer, Scientist III:** Elias Pedroza studies Mechanical Engineering with a minor in Electrical Engineering at Santa Clara University in Santa Clara, California. He has experience in thermal and structural simulations, lab work, and system modeling in both team and personal settings. He plans to further his knowledge in mechatronic, electromechanical, and robotic systems through advanced coursework, hands-on projects, and graduate study.



**Computer Hardware Engineer, Electrical Engineer:** Gregory Jabido is an Electrical and Computer Engineering student at Washington State University, Pullman, with a focus on robotics and mechatronics, autonomous systems, and guidance, navigation, and control (GNC) engineering. He has hands-on experience with the Robot Operating System (ROS), Verilog, and programming controllers for autonomous vehicles.



**Mission Assurance Specialist, Electrical Engineer:** Isaac Munoz studies Mechanical Engineering at California State Polytechnic University, Pomona and is highly specialized in mechanical design, thermodynamical systems, and static/dynamic systems. He has experience with mathematical modeling and mechanical design.



**Electrical Engineer:** Andrew Hoang studies Aerospace Engineering at the University of California, Los Angeles in Los Angeles, CA and is highly specialized in aerodynamic structures. He has experience in Finite Element Analysis and Computational Fluid Dynamics Modeling and manufacturing processes.



**Mechanical Engineer:** Patrick Nguyen studies Aerospace Engineering at the University of California, Irvine. He is specialized in Aerodynamics design with having experience in Low/high speed wind tunnels as well as simulation experience in Ansys. He is also proficient in CAD design and 3D fabrication of aircraft parts.



**Program Analyst, CDH Engineer:** Victor Chen studies Computer Science at Chabot College in Hayward, CA and specializes in software engineering. He has experience in web development programming tools and research in mathematics/aerospace.



**Primary Mechanical Engineer, Scientist II:** Isabela Benavidez studies Astronautical Engineering at the University of Southern California in Los Angeles, California, and is highly specialized in composites. She has experience with lab work and programming in MATLAB and Python.

### 1.8.2. Team Management Overview

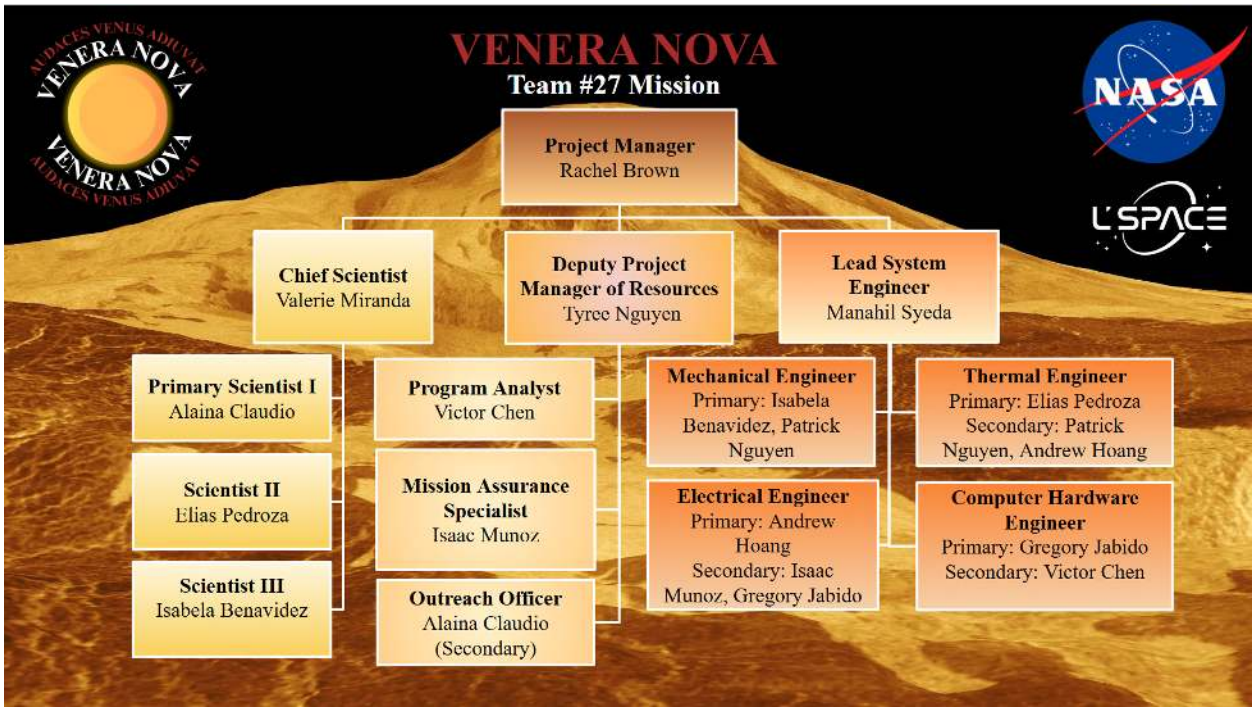



*Figure 4: Graphical representation of the team's organizational structure including team leaders and primary and secondary roles.*

**Handling Workload:**
Similar to previous deliverables, the team schedules weekly meetings to determine and complete tasks that must be fulfilled. Specific work is split between relevant teams, and members that finish early are encouraged to assist others. Members are not expected to attend each meeting due to constraints, but are expected to complete their tasks within their own time and before the next team meeting. There have been no changes that were made to workload handling.

**Organization:**
As shown in Figure 4 above, the organization of team #27 is from top to bottom, top having the PM (Project Manager) followed by the leads CS (Chief Scientist), DPMR (Deputy Project Manager of Resources), and LSE (Lead System Engineer) followed by members of subteams for science, programmatics, and engineering. The project manager sets up meetings and announcements for specific tasks the team needs to complete or oversee. The leads report to the project manager if there are problems/concerns. Regarding the delegation of tasks, the team leaders delegate tasks to members based on experience, availability, and volunteering from members. There have been no changes in team organization from the start until now in the mission life cycle.

**Decision-Making:**
No changes have been made as of now.

- Science-related or value-based decisions use trade studies.
- Time-sensitive decisions, use autocratic (the leader makes the final decision with minimal to no input)
- If there is a lot of time to decide, use consultative ways to discuss and come up with a final decision.
- If there is no time constraint, use a democratic method to discuss with the whole team to come up with a decision.
- If the team is not equipped to handle the project, team leaders/mentors with knowledge will be told to ensure things proceed smoothly.
- The team will handle differences in opinions/ideas by consulting with each other respectfully and calmly. If additional help is needed to resolve problems, the leader/team leads will work together to find the best option based on the current issue.

**Handling differences in ideas/opinions:**
No changes have been made as of now.

- In terms of differences or opinions, all team members must be willing to share their ideas and eventually come to a compromise.
- Differences in ideas and opinions can be settled through messages or in a voice chat
- Anyone on the team is encouraged to talk about their relevant opinions
- Any resentment between teammates should be settled immediately
- No one is to be cut off or interrupted on the basis that another member doesn’t want to hear their opinion or idea.

**Conflict resolutions:**
No changes have been made as of now.

- Allow everyone to talk and actively listen to come up with a resolution.
- Be understanding and compromise to reach an agreement.
- Asking a leader, MCA mentor, or the Student Success Advisor for help

**Future Changes:**
Regarding changes for the future, it is hard to pinpoint exactly what changes need to be made, but the team will proceed and adapt as necessary. The team has collectively had multiple meetings to plan around and have open communication when met with problems that limit the team's ability to meet soft deadlines set by the project manager and team leads. The team will work together, making sure the mission is organized and progresses smoothly to succeed not just alone, but together.

**Challenges:**

- Scheduling

Regarding scheduling, it has been a bit of a problem as not everyone's schedules mesh with one another, causing problems with scheduling subteams or general meetings. However, these have been addressed by recording notes and setting aside another time for whoever was not able to make certain scheduled events.

- Team members dropping
- Time management

Both these challenges have been addressed through meetings and open communication encouraged by the project manager and leads. Regarding the team members that had dropped, they had not selected their roles yet so for the most part team leads and members stepped up to take on additional workload to ensure the team would be able to succeed even with fewer people on the team.

### 1.8.3. Schedule Overview

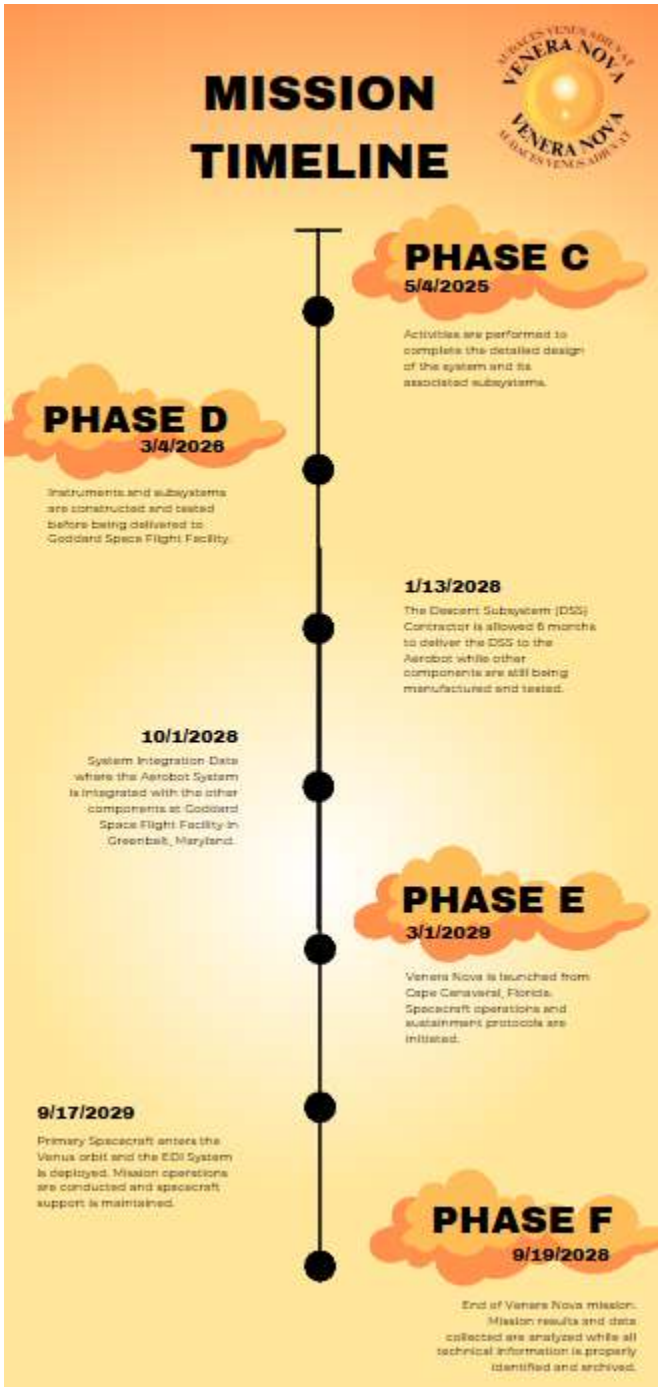



Figure 5: Depicts schedule overview for the Venera Nova Mission

The timeline depicts Phases C, D, E, and F and provides a brief overview of the events that occur and milestones along the way. These are further expanded on in the Mission Schedule and Gantt Chart.

### 1.8.4. Cost Overview

The team's total mission budget of $199,393,691 falls within stakeholder constraints of $200 million. These costs are calculated for Phases C-F and margins included vary for different costs, generally around the range of 10%.

| FINAL COST CALCULATIONS | | | | | |
|---|---|---|---|---|---|
| **Total F&A** | $ 2,199,274 | $ 2,263,585 | $ 2,327,895 | $ 2,392,206 | **$ 9,182,960** |
| | | | | | |
| **Total Projected Cost** | $ 42,438,139 | $ 43,867,825 | $ 45,891,116 | $ 47,196,611 | **$ 179,393,691** |
| **Total Cost Margin** | $ 5,588,368 | $ 5,588,368 | $ 5,588,368 | $ 5,588,368 | **$ 22,353,474** |
| | 13.2% | 12.7% | 12.2% | 11.8% | |
| | | | | | |
| **Total Project Cost** | $ 42,438,139 | $ 43,867,825 | $ 45,891,116 | $ 47,196,611 | **$ 179,393,691** |

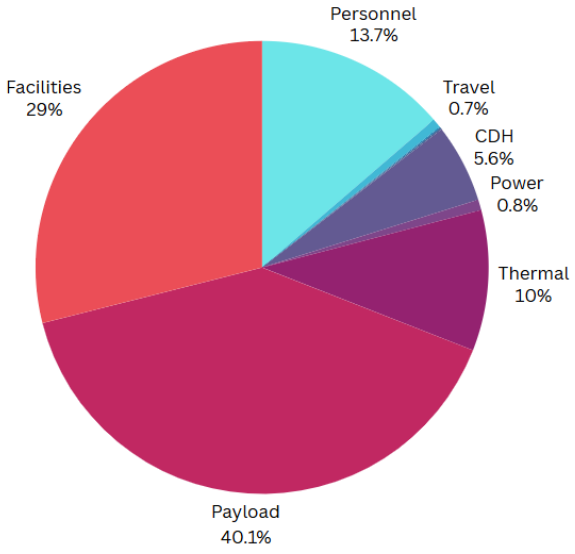



**Figure 6: Depicts pie chart illustrating the spread of costs across different mission aspects**

## 2. Overall Vehicle and System Design

### 2.1 Spacecraft Overview

The success of the Venera Nova mission relies heavily on the design and functionality of the spacecraft's subsystems, including the mechanical, power, command and data handling, thermal, and science instrumentation subsystems. Each subsystem contributes a unique, critical role to the survival and efficient operation of the spacecraft, ensuring the system as a whole can not only withstand the harsh environment of Venus but also collect and transmit valuable scientific, mission specific data and findings through its instruments, ultimately accomplishing this mission's scientific objectives.

The payload subsystem consists of the crucial and unique science instrumentation required to fulfill mission goals, each being integrated with internal systems designed to collect specific data about Venus' atmosphere and environment in accordance with the mission's science objectives. These instruments are the primary reason behind the mission and are carefully selected and integrated to ensure they not only fulfil the required mission goals but also are able to collect and transmit valuable scientific data while withstanding the harsh Venusian environment. This subsystem is then supported and integrated into the Aerobot system by means of the other engineered subsystems. The mechanical subsystem is crucial in providing structural integrity and mobility of the Aerobot system, being carefully designed such that it is robust enough to withstand the high pressures and temperatures of Venus. This system will ensure the payload subsystem is well protected and able to operate at full capacity. The power system is then responsible for generating, storing, and distributing energy throughout the spacecraft. This system ensures not only that the instruments are efficiently and sufficiently powered to correctly complete their science tasks, but also that the spacecraft as a whole can operate for extended periods and can power the essential systems required to do so. The command and data handling (CDH) subsystem acts as the brain of the spacecraft, organizing and executing communication and commands between the spacecraft and Earth. This subsystem is critical in processing, managing, and transmitting the science data from the instruments, controlling spacecraft operations, and executing commands from Earth - all in an autonomous, self-sufficient manner. In relation to the payload system, the CDH system will ensure the spacecraft can adapt in face of unexpected situations. Finally, the thermal subsystem is vital in maintaining a stable temperature environment within the spacecraft, involving cooling and absorption systems to protect the sensitive instruments and spacecraft components, and ensure integrity of the system as a whole.

**Top-Level Subsystem Requirements:**

The following depicts the top-level subsystem requirements required to be fulfilled to ensure the subsystems and their integration have features and aspects that all correctly fall under customer and spacecraft requirements. This includes ensuring each subsystem meets not only the mass, dimension, and expense requirements as provided by the mission itself, but also fulfil the temperature (heat dissipation), power usage, and science objective requirements as laid out by each subsystem team.

| ID | Requirement | Rationale | Parent Req | Child Req | Verification Method | Req Met? |
|---|---|---|---|---|---|---|
| MECH-1 | The Aerobot is at a fixed measurement of 1.0 x 1.0 x 1.0 meters<br>1.0 x 1.0 x 1.0 meters | This specification allows enough space for all the required payload to fit while staying within the requirement for aerobot size. | SYS-002 | | Inspection | Met |
| MECH-2 | The weight of the structure must not exceed 25 kg. | This ensures the weight requirement is not exceeded and that the other systems are accounted for. | SYS-001 | | Inspection | Met |
| MECH-3 | The mechanical system shall survive for at least 48 hours in the Venusian atmosphere | This mission constraint ensures that the mechanical system can wholly support the system and payload during the mission duration | SYS-007 | MECH 3.1, MECH 3.2, MECH 3.3, MECH 3.4, MECH 3.5, MECH 3.6 | Test | Met |
| MECH-3.1 | The structure must withstand a minimum | This requirement ensures the mechanical systems will not suffer extreme heat damage | MECH-3 | MECH 3.1.1, MECH | Test | Met |

| | | | | | | |
|---|---|---|---|---|---|---|
| | temperature of 344.7 K. | during mission operations up to 50 km above the surface. | | 3.1.2, MECH 3.1.3 | | |
| PWR-1 | There shall be an adequate amount of power to sustain all subsystems for the entirety of the mission | Electronics include a variety of sensors and controls that shall be functional within planetary entry until total system failure at Venus' surface | SYS-001 | PWR-002 | Testing | Met |
| PWR-2 | Power shall be available to all necessary electronics systems through power supply networks | Electronics requiring power shall be connected to a power supply, in order for electronics to perform set functions | PWR-001 | PWR-003, PWR-004 | Testing | Met |
| PWR-3 | The EPS shall be able to generate and distribute at least 51.4 W at a time to sustain all instrumentation for the entirety of the mission | Electronics include a variety of sensors and controls that shall be functional within planetary entry until total system failure at Venus' surface | PWR-001, PWR-002 | PWR-006 | Testing | Met |
| CDH-1 | The Aerobot shall implement an Onboard Computer (OBC) | The OBC handles the main operations of the Aerobot, interlinks the various subsystems, and handles data/data collection | CDH-8 | CDH-4, CDH-1.1, CDH-1.2, CDH-1.3 | Inspection | Met |
| CDH-2 | A Data Storage System shall be implemented within the Aerobot payload | Data collected from instruments and regarding Aerobot status must be stored reliably before being | CDH-1.3 | CDH-2.1 | Inspection | Met |

| | | | | | | |
|---|---|---|---|---|---|---|
| | | transmitted | | | | |
| CDH-3 | The Aerobot system shall check on the statuses of other subsystems through a Data Interface | Ensures overall Aerobot functionality and determines activation of the Redundancy and Recovery System | CDH-1 | | Demonstration | Met |
| CDH-4 | Aerobot system shall implement a Communication System | Allows for transmission of telemetry data and telecommands between Aerobot, orbiter, and ground systems on Earth | CDH-1 | CDH-4.1, CDH-4.2, CDH-4.3, CDH-4.4 | Inspection | Met |
| TH-1 | The total mass of all thermal control components shall not exceed 18 kg. | The TCS must be designed with light materials to ensure compliance with the mass constraint. | SYS-001 | | Inspection | Met |
| TH-2 | The TCS shall maintain all thermal control components within the system's dimensional constraints. | Ensures all passive and active thermal components fit within the system's dimensional constraints. | SYS-002 | | Inspection | Met |
| TH-3 | The costs of the TCS shall not exceed $200 million. | Funding must be distributed according to the customer's budget. | SYS-004 | | Inspection | Met |
| TH-4 | The TCS shall withstand environmental temperatures ranging from 225.8 K to 344.7 K (at altitudes of 70 to 50 km) during the | Scientific instrumentation and electronics must be at operating temperatures for the entirety of the SS's descent. | SYS-007 | TH-4.1 TH-4.2 | Test & Analysis | Met |

| | | | | | | |
|---|---|---|---|---|---|---|
| | mission duration. | | | | | |
| PAY-1 | Science instruments shall weigh no more than 30 kg. | Payload cannot be allocated any more mass or the other subsystems may not function appropriately. | SYS-001 | | Inspection | Met |
| PAY-8 | The payload shall not exceed a cost of $70,000,000 | Cost Constraints | SYS-004 | | Analysis | Met |
| PAY-9 | All science operations shall be completed within 48 hours after deployment of EDI system. | After 48 hours the SS will uncontrollably fall and crash into Venus' surface, discontinuing science operations. | SYS-007 | | Demonstration | Met |

**Spacecraft Subsystem Overview**

The Venera Nova mission through the Aerobot System aims to descend into Venus' atmosphere to fulfill specific goals exploring how planetary surfaces and interiors interact with host atmospheres, targeting the Aphrodite Terra as its region of interest. Tasked with the main objectives of determining the influence of Venus's superrotation on its surface and atmosphere, exploring the effects of a lack of magnetic field on atmospheric composition, and determining the probability of life within Venus' upper atmosphere, the primary spacecraft system is equipped with three main scientific instruments - The Venus Monitoring Camera (VMC), the SPICAV suite that includes the Ultraviolet Spectrometer and the Solar Occultation Infrared Spectrometer (SOIR). The engineered subsystems each in turn will support these instruments in fulfilling the relevant requirements, providing the necessary structure, protection, and power to ensure each can operate at its full capacity.

Mechanical System:

The **mechanical subsystem** ensures structural integrity for the Aerobot system, supporting the integrated components, withstanded stresses from operation, launch, and descent in the environment of Venus, and providing the necessary attachment points and foundations for the subsequent subsystems. The main components of this subsystem include: the chassis and framework for the system serving as the foundation of the Aerobot; population system for the Aerobot to maneuver while descending in the atmosphere; thermal protection for the payload, allowing the Aerobot system and its payload to be preserved within the harsh environment of Venus, and the power assembly for the equipment to operate once the Aerobot is launched. The mechanical subsystem first and foremost chooses the payload's shape and design to ensure smooth descent and minimal undesired movement during the descent. Then, this subsystem takes caution in choosing the optimal material for the system to best withstand the harsh pressure, temperature, and radiation facets. Overall, this subsystem ensures structural integrity for the system as a whole.

Power Subsystem:

The **power subsystem** is critical in ensuring each system receives sufficient power to correctly and efficiently operate through a carefully chosen energy power system (EPS). This subsystem provides the necessary power generation subassembly (PGS) required to generate the total power requirement for all components existing in the Aerobot System, including the science instruments and mechanisms, at any given time. The power generated from these solar cells will then be distributed by the chosen power management and distribution (PMAD) system, effectively and accurately distributing power to the relevant components as needed without overwhelming the system. The Power Storage Subassembly further stores excess power, saving this for later use. These power systems must be able to handle high temperature variations, withstand under high pressure conditions, and be able to correctly function in the unique Venus environment.

Commands and Data Handling:

Command reception, data processing, and transmission capabilities are provided by the **commands and data handling subsystem (CDH)**. The CDH subsystem is responsible for managing, coordinating, and executing onboard systems and payloads, facilitating reliable communication of system status and crucial science data collection back to ground stations on Earth, and managing data flow between components. This subsystem provides the necessary antennas, memory bus, computer housing, data storage, interfaces, cables, and relevant electronics needed in order to receive and execute commands from Earth, process data from the science instruments, and then transmit this data back to Earth. The optimal onboard computers and memory are chosen in order to ensure efficient data management and communication between the Aerobot System and Earth. The CDH subsystem will further ensure that redundancy and recovery systems are in place in face of hazards, both allowing existing systems to be reinitialized or realigned in case of failures to ensure proper function as well as a redundant system to take the place of an existing one to ensure that all communication and control needs are met even in the face of risks.

Thermal Management:

The **thermal management** subsystem ensures that optimal temperatures and conditions are maintained for all components such that each can operate at its full capacity without malfunctioning or deteriorating. This subsystem provides the thermal control system (TCS) that will both be able to withstand the environmental temperatures of Venus itself, but also be able to minimize net heat gain/loss for the other system components through choosing ideal materials for construction and necessary protections through thermal coatings, cooling systems, and passive radiators. The Multi-Layer Insulation (MLI) component and white thermal coating will allow active cooling for the Aerobot system. The Loop Heat Pipe (LHP) allows for passive transfer of the heat generated by internal components, and a flexible heater helps counteract heat loss in the case that the system begins losing heat.

An automated thermal control (ATC) system will also be designed and integrated by this subsystem to maintain overall thermal stability within the components of the Aerobot system to ensure it does not degrade or fail to correctly function as it traverses Venus' atmosphere and environment.

Payload Subsystem:

The **payload subsystem** lies at the core of the Venera Nova mission, providing the instruments required to fulfil the intended science goals. As the Aerobot descends through Venus' atmosphere, this suite of instruments will have the capabilities of measuring wind speeds and Kelvin waves, tracking cloud movement, identifying magnetic activity, and discerning chemical composition of this atmosphere. These key instruments will fulfil the science objectives of analyzing the influence of superrotation, explore potential signs of life existing in Venus' upper atmosphere, and identify the relationship between the lack of a magnetic field and atmospheric composition.

The Venus Monitoring Camera (VMC) instrument contributes to the supperrotation analysis objective of the mission, providing imaging and spectral data collection

capabilities allowing the instrument to capture image data from the relevant optic, store this data, then transmit for processing. This image collection will allow scientists to better track cloud movements through the Aerobot's descent in the atmosphere, allowing analysis on changes in wind speeds. This way, the VMC instrument provides information on how Venus' surface and atmosphere is affected by superrotation.

The UV Spectrometer, a component of the SPICAV suite, fulfills the objective of analyzing the atmospheric composition of Venus to determine how it is influenced by a lack of magnetic field. This instrument gathers atmospheric spectroscopy data in various wavelength ranges to better illustrate the influences of cosmic ray flux and solar radiation caused by the lacking magnetic field.

Finally, the astrobiological aspect of our mission involving determining the probability of life existing within Venus' upper atmosphere will be supported by the Solar Occultation Infrared Spectrometer (SOIR). This instrument provides the unique ability to determine the presence and location of phosphine, a potential biosignature and indicator of biological processes, in the atmosphere of Venus. Through solar occultation capabilities, this instrument will collect and send data on these chemical concentrations in the upper and briefly the lower atmosphere of Venus, ultimately providing a preliminary hint on signs of life on Venus.

All specified instruments integrate as part of the payload system, in turn supported by the other subsystems required for the Aerobot System to function and fulfil the goals of the Venera Nova mission

**Preliminary Spacecraft Design**

The following is a CAD depiction of the integration of the subsystem assemblies for the full spacecraft design. This assembly includes the chassis on the outer framework. The science payload exists within the framework, taking a significant portion of this structure to provide maximum protection for the instruments. At the center of the base plate, the propulsion system exists pointing downwards to allow stability for the Aerobot system as it descends and collects data. The front of the structure depicts the thermal subsystems. The remaining subsystems, including CDH and Power are also integrated within this framework, protected by the chassis against the harsh Venusian environment.

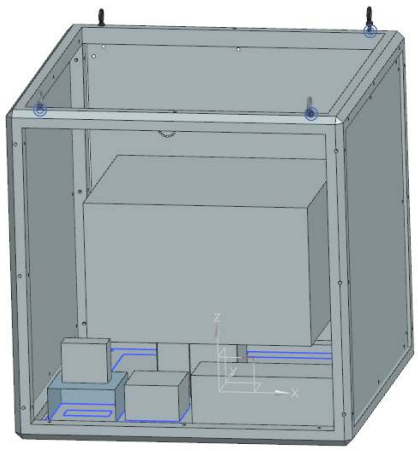


Figure 7: Preliminary Spacecraft Design

**System Mass, Dimensions, and Max Power Draw**

In accordance with the top-level subsystem requirements, the following figure depicts the distribution of mass, dimension, and maximum power draw across each subsystem. This ensures that the subsystems of the Venera Nova mission have relevant value totals that all correctly align under the mission constraints. The spacecraft's total mass comes to 45 kg, with dimensions 1 m^3, and a maximum power draw of ___ W, fitting under the mission requirements.

| Subsystem | Mass (kg) | Dimension | Maximum Power Draw (W) |
|---|---|---|---|
| Mechanical | 8.469 | 1 (Overarching) | 700 |
| Power | 5.81 | 0.039962 | Max power output: 1000W |
| CDH | 1.161 | 0.001077 | 16.56 |
| Thermal | 13.86 | 0.257 | 150 |
| Payload | 15.42 | 0.01 | 79.4 |
| Totals | 44.72 | 1 x 1 x 1 | 945.96 |

All subsystems will be within the overarching Mechanical Subsystem, thus lying within the 1m x 1m x 1m structure that is the chassis within the subsystem.

### 2.1.1 Mechanical Subsystem Overview

The Mechanical Subsystem Overview provides a detailed breakdown of the Aerobot design and mechanical subsystems that are implemented to enable the Aerobot to perform tests on Venus. The system is designed to deal with physical environmental hazards from the atmosphere during descent, as well as on the surface of Venus. Factors such as high temperatures, radiation, and extreme pressures are accounted for when designing the Aerobot. Each component of the subsystem was measured and placed inside the Aerobot accordingly. The main components are the chassis and framework, which embody the Aerobot, the propulsion for maneuvering, thermal protection for the payload, and power subassembly for the equipment to operate once launched. The mechanical subsystem will feature a metal framework held together by a chassis. The framework and chassis are in a cubic configuration,.

Each subassembly was researched to ensure it could withstand the conditions of Venus. Technology Readiness Levels (TRL) were used to help understand where the equipment stands in terms of being mission-ready. The table provides a visual breakdown of the dimensions, mass, TRL, and max power of each subassembly. With this information, it will be ensured that the Aerobot can launch autonomously into harsh environments.

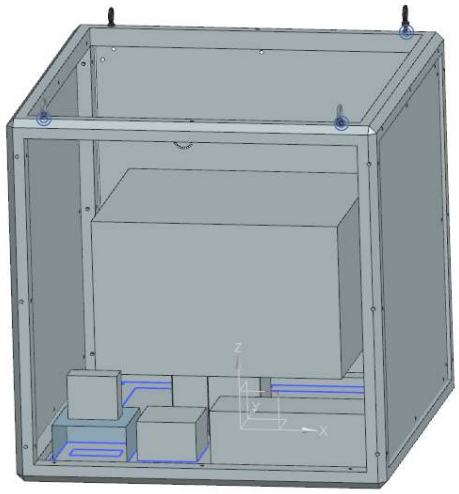


**Figure 8: Aerbot Assembly**
**(top frame and side frame taken out for display purposes)**

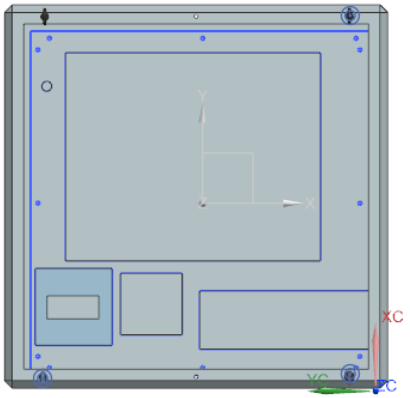



**Figure 9: Side View**

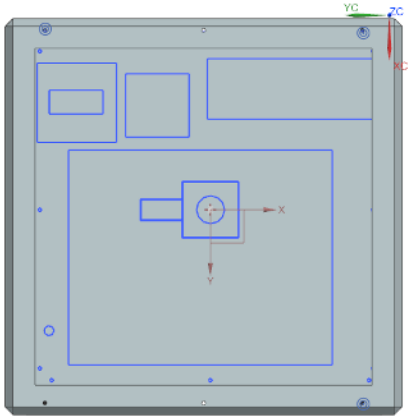



**Figure 10: Bottom View**

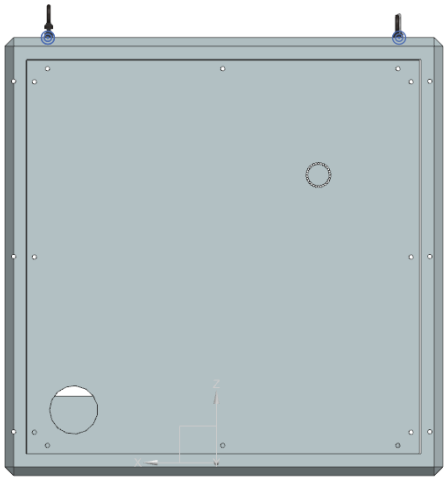


**Figure 11: Back View**

The chassis and airframe of the Venus Aerobot are built to handle the planet's extreme heat, pressure, and corrosive atmosphere. They're made from titanium alloy, which is both strong and lightweight, and can withstand high temperatures without corroding. The structure holds and protects everything inside the Aerobot, like the electronics, payload, and propulsion system. It's designed in a modular way to make assembly and maintenance easier, and it's covered with thermal insulation and reflective coating to reduce heat exposure. Shock mounts are added to protect key components during launch and descent. Right now, this part of the system is at Technology Readiness Level 5, meaning it's been tested in lab conditions similar to Venus. More testing is planned to make sure it's ready for actual use.

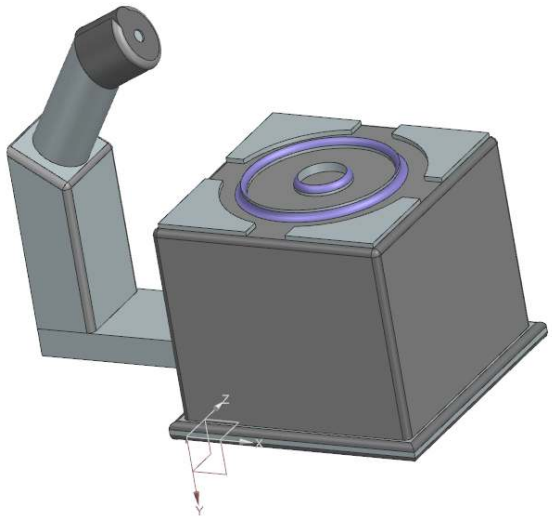


**Figure 12: Electric Propulsion System**

The BHT-600 Hall Effect Thruster is the main propulsion system for our Aerobot. It's a small and lightweight thruster that has been tested for over 7,000 hours, making it a solid choice for long missions like our trip to Venus. It runs on xenon, krypton, or iodine and uses about 600 watts of power during normal operation. At just 2.6 kg, it helps keep the Aerobot light. The team plans to mount it with a gimbal so they can control its direction during flight. Since it's already been used in space, it's at a high Technology Readiness Level and ready for flight.

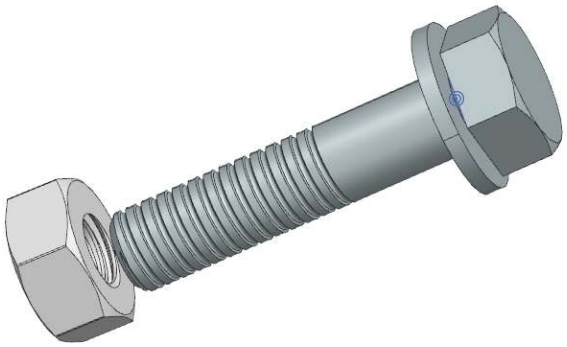


**Figure 13: Hex Bolt, Hex Nut, Washer**

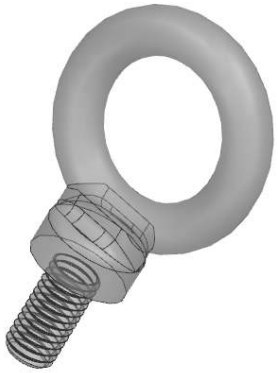


**Figure 14: Closed Eye Bolt with Locking Nut)**

We’re using titanium bolts, nuts, and washers in the Aerobot’s structure because they’re strong, lightweight, and built to withstand the extreme conditions on Venus. Titanium resists corrosion and handles high temperatures, which is critical for operating in Venus’s dense and acidic atmosphere. These fasteners help keep the Aerobot lightweight without compromising structural strength. The eye bolt is also made from titanium and includes a locking nut to keep it secure during vibration from flight and descent. Since titanium fasteners have been used in many aerospace applications before, the team considers this hardware to be at Technology Readiness Level 9 (TRL 9)—fully flight-proven and ready for use.

### 2.1.1.1 Mechanical Subsystem Requirements

The following depicts the top-level subsystem requirements specific to the mechanical subsystems ensuring that each component meets both the customer and the mission requirements. This includes mass and dimension constraints, survival time period for the structure - especially as the mechanical system is key to the entirety of the Aerobot system - and specific environmental conditions the structure must be able to withstand.

| ID | Requirement | Rationale | Parent Req | Child Req | Verification | Req Met? |
|---|---|---|---|---|---|---|
| MECH-1 | The Aerobot is at a fixed measurement of 1.0 x 1.0 x 1.0 meters<br>1.0 x 1.0 x 1.0 meters | This specification allows enough space for all the required payload to fit while staying within the requirement for aerobot size. | SYS-002 | | Inspection | Met |
| MECH-2 | The weight of the structure must not exceed 25 kg. | This ensures the weight requirement is not exceeded and that the other systems are accounted for. | SYS-001 | | Inspection | Met |
| MECH-3 | The mechanical system shall survive for at least 48 hours in the Venusian atmosphere | This mission constraint ensures that the mechanical system can wholly support the system and payload during the mission duration | SYS-007 | MECH 3.1, MECH 3.2, MECH 3.3, MECH 3.4, MECH 3.5, MECH 3.6 | Test | Met |
| MECH-3 .1 | The structure must withstand a minimum temperature of 344.7 K. | This requirement ensures the mechanical systems will not suffer extreme heat damage during mission operations up to 50 km above the surface. | MECH-3 | MECH 3.1.1, MECH 3.1.2, MECH 3.1.3 | Test | Met |
| MECH-3 .1.1 | Temperature regulation system integrated within the | Keeping a constant temperature within the aerobot is a must and at higher altitudes, | MECH-3.1 | | Test | Met |

| | | | | | | |
|---|---|---|---|---|---|---|
| | Aerobot. | the temperature changes. | | | | |
| MECH-3.1.2 | There shall be thermal expansion joints within the aerobot. | This will help reduce the stress of the Aerobot under immense heat so that it doesn't break | MECH-3.1 | | Test | Met |
| MECH-3.1.3 | Cooling system for the payload inside the Aerobot | A strong cooling system prevents external temperatures from destroying equipment and ensures internal systems do not generate too much heat. | MECH-3.1 | | Test | Met |
| MECH-3.2 | The structure must withstand 2601.3 watts per square meter of radiation. | This ensures the framework and chassis will have adequate radiation protection so it will not suffer damage from the solar radiation during mission operations. | MECH-3 | | Analysis | Met |
| MECH-3.3 | The structure must withstand a minimum pressure of 1.2 bars. | This specification ensures the structure will not deform under the pressure at altitudes higher than 50 km. | MECH-3 | MECH-3.3.1 | Test | Met |
| MECH-3.3.1 | The mechanical frame will be built with Grade 5 titanium. | Using a strong material like titanium allows for better strength against pressures | MECH-3.3 | | Inspection | Met |
| MECH-3.4 | The structure must withstand vibration from launch and entry | This should be accounted for in the chassis and frame of the aerobot as disruption can cause fasteners to come loose. | MECH-3 | | Test | Met |
| MECH-3.5 | Redundant structural support system | Ensures the Aerobot will maintain structural integrity. | MECH-3 | | Test | Met |
| MECH-3.6 | All materials on the subsystem should be non-coated | This can cause erosion, leading to degradation depending on the environment conditions | MECH-3 | | Inspection | Met |

| | | | | | | |
|---|---|---|---|---|---|---|
| MECH 4 | Shape of the Aerobot must accommodate to Aerodynamic properties | This is to ensure the descent is smooth and direct so the aerobot is as aerodynamic as possible | MECH-1 | MECH-4.1 | Analysis | Met |
| MECH-4 .1 | The aerobot must descend with minimal rotational movement during descent. | This requirement ensures the instrumentation is not influenced by drag and turbulence. | MECH-4 | | Test | Met |
| MECH 5 | The Aerobot must be designed with modular components | This allows for easy repairs and upgrades when needed. | MECH-1 | | Test | Met |

#### 2.1.1.2 Mechanical Sub-Assembly Overview

The Mechanical Subsystem Overview provides a detailed breakdown of the Aerobot design and mechanical subsystems that are implemented to enable the Aerobot to perform tests on Venus. The system is designed to deal with physical environmental hazards from the atmosphere during descent as well as on the surface of Venus. Factors such as high temperatures, radiation, and extreme pressures are accounted for when designing the Aerobot. Each component of the subsystem was measured and placed inside the Aerobot accordingly. The main components are the chassis and framework, which embody the Aerobot, the propulsion for maneuvering, thermal protection for the payload, and power subassembly for the equipment to operate once launched. The mechanical subsystem will feature a metal framework held together by a chassis. The framework will feature square-shaped sides on all surfaces.

Every part of the mechanical subsystem of the Aerobot will be constructed out of the titanium alloy Ti-6Al-4V. Titanium's properties allow it to withstand the physical environmental hazards the aerobot would encounter in Venus's atmosphere at an altitude of 70-50 km. This alloy was chosen because of its desirable properties and common use in the aerospace industry. During the mission, the aerobot will experience temperatures ranging up to 344.7° K. Ti-6Al-4V has a melting point of around 1877.15° K and a low specific heat capacity of 0.5263 J/g °C, which ensures it will maintain structural integrity and prevent excess heat from entering the interior of the aerobot. Additionally, Ti-6Al-4V has a compressive yield strength of 9700 bars, which is more than enough to withstand the 1.2 bars of pressure the chassis and framework will experience during mission operations. Titanium also had the lowest cost of materials that offered the same or better performance. Each frame will be attached to the chassis using twelve titanium hex nuts (1/4"-20) and titanium flat washers (1/4"). The aerobot will attach to the balloon using four closed eye bolts with a locking nut positioned on each corner of the top frame. The mechanical system also features an electric thruster that will ensure the aerobot has a steady descent through the atmosphere. The specific model is the BHT-600 Hall Effect Thruster, which has a lifetime of over 7000 hours and a thrust strong enough to maneuver the aerobot (Buskek).

The bottom frame features compartments that will contain the payload and other subsystems. The CAD image demonstrates where each subsystem will reside within the interior of the aerobot. The largest box represents SPICAV, and the two cubes directly underneath it will contain the propulsion system. The small cylinder is a EAP-12015B thermal battery which is part of the power subassembly. The box with an open face will contain the thermal subassembly, while the small cube resting on top of that is PMAD, another power component. The CDH subassembly is in the box next to these two

cubes. Lastly, the large, long box near the front represents VMC. The framework panel that VMC faces will have a hole for the camera to view the atmosphere and small honeycomb-like grating that will allow the spectrometers to take measurements.

#### 2.1.1.3 Mechanical Subsystem Recovery and Redundancy Plans

The mechanical subsystem is responsible for ensuring the payload and other subsystems are protected from the extreme conditions of Venus' atmosphere. If the mechanical subsystem fails, then the structural integrity of the entire aerobot system will be compromised. These factors were taken into account as shown by the system's design. Ti-6Al-4V was used for the framework, chassis, nuts, washers, and bolts, as it has structural properties that allow it to easily withstand the temperatures and pressure of Venus' atmosphere. Redundancy was incorporated into the sub-assembly through the design of the fasteners securing the nuts onto the framework. Each frame is secured to the chassis in 12 areas, ensuring that if one of the screws becomes loose during mission operations, the aerobot will stay intact. Adding a redundant system to the mechanical design would make the interior of the aerobot much smaller, leading to the high likelihood of there not being enough space for the payload and other associated systems.

Recovery was a factor that was extensively researched and considered within the mechanical design. Despite this, it was found that it is not possible to implement recovery operations into the mechanical subsystem. If the mechanical system becomes compromised during mission operations, it will not be possible to recover and repair the system. Manufacturing and implementing a second framework and chassis would take space from the interior of the aerobot, which is already constrained. It is currently not possible for the aerobot to repair itself if the framework or chassis is damaged, or if multiple screws become unfastened during descent.

#### 2.1.1.4 Mechanical Subsystem Manufacturing and Procurement Plans

The following table highlights the key components for the mechanical subsystem, focusing on the needed supplier and back-up suppliers as well as estimated lead times for each component.

| Component | Supplier | Back-Up | Estimated Lead Times |
|---|---|---|---|
| Electric Propulsion System - BHT-600 Hall Effect Thruster | Busek Space Propulsion and Systems | L3harris | 6-12 months |
| Closed Eye Bolt with Locking Nut ⅜-16 | McMaster-Carr (Off-the-shelf) | E-Rigging | 2 weeks |

| | | | |
|---|---|---|---|
| Titanium Flat Washer (1/4") - Ultra-Corrosion-Resistant Alloy 20 Stainless Steel | McMaster-Carr (Off-the-shelf) | MetTec Titanium | 2 weeks |
| Titanium Hex Nut (1/4"-20) | McMaster-Carr (Off-the-shelf) | MetTec Titanium | 2 weeks |
| Grade 5 Titanium (Ti-6Al-4V) | OnlineMetals | Tiger Titanium | 1 week |

**Component**: Electric Propulsion System
This electric propulsion system is small enough to accommodate the 50kg Aerobot. Since it has electric propulsion, no fuel tank is necessary, which is beneficial for saving space. Busek Space makes a compact, small propulsion system that is powerful enough to manuever the aerobot. The BHT-600 Hall Effect Thruster is the model that will be used. While the company does not specify their pricing, the cost of the thruster was estimated to be $150,000 based on thrusters with a similar design that were used in the aerospace industry. An estimated 6-12 months of lead time are needed for manufacturing.

**Component**: Closed Eye Bolt with Locking Nut ⅜-16
This component will be purchased off the shelf from McMaster-Carr, which is a major supplier of hardware with reasonable pricing and fast shipping. The bolt is priced at $13.84 for each unit, making the total cost $55.36 for four bolts. 2 weeks of lead time are required for delivery. This was determined by the shipping time estimate for the aerobot's assembly location, which was found on their website. The backup supplier is E-Rigging, which provides a variety of metal hardware at a similar price as McMaster-Carr. Their shipping estimate is around the same as well, so the lead time will not change for this component.

**Component**: Titanium Flat Washer (1/4")-Ultra-Corrosion-Resistant Alloy 20 Stainless Steel.
The flat washer will also be purchased from McMaster-Carr. The washer is $1.94 per unit and 60 will be purchased, leading to a total cost of $116.4. This component has the same estimated lead time as other components purchased from McMaster-Carr, with 2 weeks required for delivery after the order has been placed. The backup manufacturer is MetTec Titanium, which provides titanium fasteners and bolts for aircraft for a reasonable price. The lead time will stay the same, if not decrease, since the company has a shipping time of 1 week.

**Component**: Titanium Hex Nut (1/4"-20)

McMaster-Carr has a multitude of different parts that fit the criteria needed for the Aerobot. The fast shipping and pricing make them an ideal manufacturer for the hardware required for our mission. Specific details such as corrosion prevention allows for better performance on the Aerobot. The hex nut is priced at $2.73 per unit, with a total cost of $163.8 for 60 units. 2 weeks of lead time are required for delivery, which was determined using the same method as the other components purchased from this manufacturer. This component will use MetTec Titanium as a backup supplier, with an unchanged lead time.

**Component**: Grade 5 Titanium (Ti-6Al-4V)
The titanium alloy will be purchased from OnlineMetals. This manufacturer provides a material that is both durable and light which allows for limiting weight for the Aerobot. Their reviews describe them as being reliable and providing good quality materials for a reasonable price. The supplier has a variety of different titanium sheets that can be chosen from, but the specific Grade 5 Titanium (Ti-6Al-4V) meets the needs for the mission. The framework will be constructed out of 6 titanium sheets. The cost of one sheet with the required thickness and similar dimensions to that of the Aerobot is $1,839.63. Multiplying this price by 6 for each side and rounding up for a slight difference in cost due to dimensioning results in a total cost of $11,100.00. For the chassis, the price of one square meter of Ti-6Al-4V is $4,070.74. This cost was multiplied by the area of the chassis, which was 3.52 $m^2$, making the total cost $14,341. This manufacturer claims to have 1-2 day delivery, but a week will be the estimate used so as to take into account manufacturing time. The backup supplier will be Tiger Titanium, which sells the required titanium alloy in a variety of sizes for a similar price. This manufacturer also claims to have next day shipping, so the 1 week lead time will be unchanged.

### 2.1.1.5 Mechanical Subsystem Verification Plans

The following table indicates the specific verification methods alongside justification for each listed requirement for the mechanical subsystem.

| Req # | Requirement Summary | Verification Method | Rationale for Method | Preliminary Verification Plan |
|---|---|---|---|---|
| MECH-1 | The Aerobot shall not exceed 1.0 x 1.0 x 1.0 meters. | Inspection | This allows for the Aerobot to stay within customer constraints while providing enough space for the payload and additional subsystems | Once the chassis and framework have been assembled, the assembly will be measured to ensure it fits the required measurements. |
| MECH-2 | The weight of the structure must not exceed 25 kg. | Inspection | This ensures the total weight requirement is not exceeded and that there is enough weight for the other systems. | Each component of the mechanical sub-assembly will be individually weighed and added to ensure the weight requirement isn't exceeded before assembly. Once the mechanical subsystem has been fully assembled, it will be weighed again to confirm the weight requirement. |
| MECH-3 | The mechanical system shall stay structurally intact in the atmosphere for at least 48 hours. | Test | Ensures that the system can wholly support the payload and other subsystems during the mission duration. | Will be placed in a thermal vacuum chamber to simulate the Venusian environment. |

| | | | | |
|---|---|---|---|---|
| MECH-3.1 | The subassembly shall withstand a minimum temperature of 344.7 K. | Test | Ensures mechanical systems will not suffer heat damage during mission operations. | Will be placed in a thermal vacuum chamber to expose mechanical components to the required temperatures. |
| MECH-3.1.1 | Temperature regulation system integrated within the Aerobot mechanical system. | Test | A constant temperature must be maintained within the aerobot. | Will be placed in a thermal vacuum chamber to test thermal hardware in Venus-like conditions. |
| MECH-3.1.2 | There shall be thermal expansion joints. | Test | This will help reduce the stress the Aerobot will experience under immense heat. | Will be placed in a thermal vacuum chamber to expose mechanical components to the required temperatures. |
| MECH-3.1.3 | Cooling system for the payload inside the Aerobot. | Test | A strong cooling system prevents external and internal temperatures from destroying equipment. | Will be placed in a thermal vacuum chamber with Venus-like temperatures to test cooling system. Thermal cycling will be performed to check durability. |
| MECH-3.2 | The Aerobot must withstand 2601.3 watts | Analysis | Ensures the framework and chassis will have adequate radiation protection so it will not suffer damage from the solar radiation. | Radiation analysis and modeling will be performed using simulation tools like Siemens NX and Ansys. |

| | | | | |
|---|---|---|---|---|
| | per square meter of radiation. | | | |
| MECH-3.3 | The structure must withstand a minimum pressure of 1.2 bars. | Test | This specification ensures the structure will not deform under pressure. | Will be placed in a thermal vacuum chamber to expose mechanical components to the required pressures. |
| MECH-3.3.1 | The mechanical frame will be built with Grade 5 titanium materials. | Inspection | Titanium is corrosive resistant and maintains structural integrity against pressures. | Metals received from manufacturers will be thoroughly inspected in a sterile lab environment to ensure the correct titanium alloy was used. |
| MECH-3.4 | The structure shall withstand vibration from launch and entry. | Test | This should be accounted for in the chassis and frame of the aerobot as disruption can cause fasteners to come loose. | The mechanical subsystem will be placed on a vibration table and inside an acoustic chamber. |
| MECH-3.5 | Redundant structural support | Test | Ensures the Aerobot will maintain structural integrity. | The mechanical subsystem will be placed on a vibration table and inside an acoustic chamber to |

| | | | | |
|---|---|---|---|---|
| | system. | | | ensure structural integrity. |
| MECH-3.6 | All materials on the subsystem should not be coated. | Inspection | Coating can cause erosion, leading to degradation depending on the environment conditions. | Metals received from manufacturers will be thoroughly inspected in a sterile lab environment to ensure the titanium is clean. |
| MECH-4 | Shape of the Aerobot must accommodate aerodynamic properties | Analysis | Ensures the descent is smooth and direct. | CAD models of the mechanical subsystem will be utilized to simulate mission operations to ensure the aerobot shape will comply with atmospheric conditions. |
| MECH-4.1 | The aerobot must descend with minimal rotational movement during descent. | Test | This ensures the instrumentation is not altered by turbulence. | CAD models of the mechanical subsystem will be utilized to simulate the decent in Venus' atmosphere. |
| MECH-5 | The Aerobot must be designed with modular components. | Test | This allows for components to be altered fairly easily. | Routine checks will take place during assembly to ensure constructed components are modular. A thorough, final check will be performed once the mechanical system is fully assembled. |

### 2.1.2 Power Subsystem Overview

The power requirement that the EPS needs to support is the total of all the scientific instrumentation and subsystem mechanism power draw. The UV Spectrometer: 17.6 W, Solar Occultation Infrared Spectrometer (SOIR): 51.4 W, and the Venus Monitoring Camera (VMC): 10.4 W give a total power requirement of 79.4 W, however, only one instrument will be used at a time, making the operational power requirement the highest of the 3 instruments: the SOIR at 51.4 W. The other systems operate continuously, with their power usage at a constant rate. CDH requires 187.5 W, propulsion systems require 75 W and the TCS requires 162.7 W, totaling to 476.6 W * 2 (Factor of Safety) to finally reach 953.2 W = 1000 W. Each subassembly: Power Generation, Power Storage, and Power Management and Distribution, is designed to meet that driving power requirement.

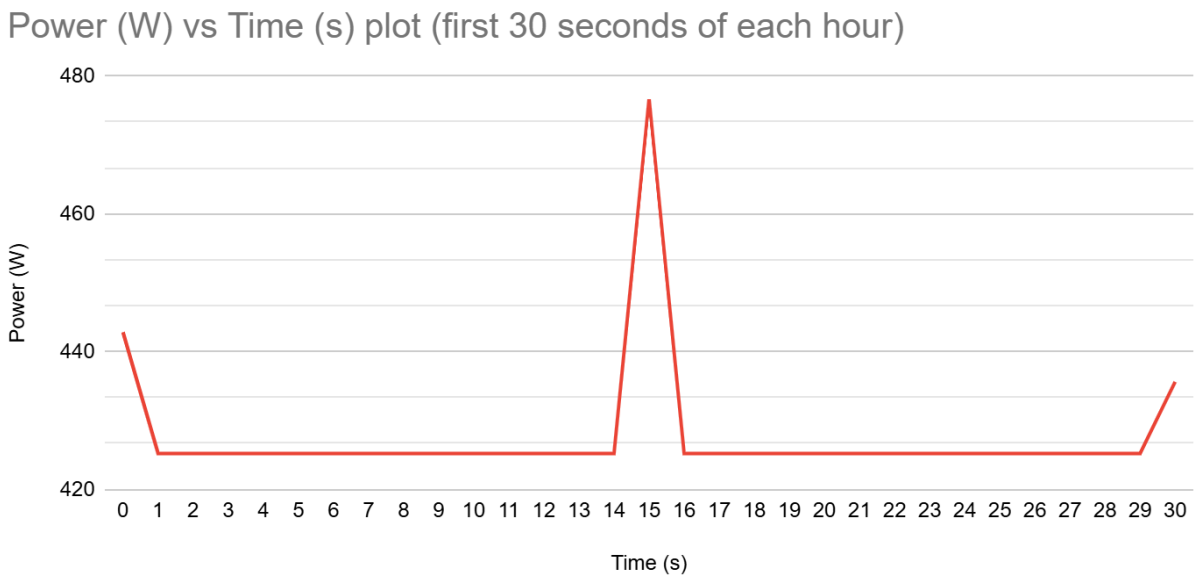



Figure 15: Graph of Power vs Time

To lower the maximum amount of power needed by the payload subsystem, the 3 scientific instrument measurement times are spread out, plotted above. The graph above plots only the first 30 seconds of each hour, so the power usage at 0 seconds will be at the minimum 425.2 W for the other 119 30-second intervals.

Another method to conserve power is the EPS will utilize rotational battery cycling, actively using only 4 out the 6 battery modules at a time and switching out the older 2 with the spare 2 every 2 hours. The thermal batteries contain intelligent gas gauge to monitor and predict battery status, health/aging and capacity. This telemetry is relayed

to the CDH system to monitor the battery health and cycle the batteries at the most efficient intervals.

### 2.1.2.1 Power Subsystem Requirements

Power subsystem requirements were determined based on system and customer constraints as well as the necessary physical parameters in order to withstand the harsh Venusian environment. Each requirement describes the system's ability to generate, regulate, and store power as required for the mission.

| ID | Requirement | Rationale | Parent Req | Child Req | Verification Method | Req Met? |
|---|---|---|---|---|---|---|
| PWR-001 | There shall be an adequate amount of power to sustain all subsystems for the entirety of the mission | Electronics include a variety of sensors and controls that shall be functional within planetary entry until total system failure at Venus' surface | SYS-001 | PWR-002 | Testing | Met |
| PWR-002 | Power shall be available to all necessary electronics systems through power supply networks | Electronics requiring power shall be connected to a power supply, in order for electronics to perform set functions | PWR-001 | PWR-003, PWR-004 | Testing | Met |
| PWR-003 | The EPS shall be able to generate and distribute at least 51.4 W at a time to sustain all instrumentation for the entirety of the mission | Electronics include a variety of sensors and controls that shall be functional within planetary entry until total system failure at Venus' surface | PWR-001, PWR-002 | PWR-006 | Testing | Met |
| PWR-004 | The EPS shall have multiple power generation sources (solar panels) | Redundancy is needed to prepare for failure | PWR-002, PWR-003 | PWR-006 | Inspection | Met |
| PWR-005 | The battery shall be able to contain the power generated during hours of sunlight, to save for power use during hours without sunlight | The battery needs to hold enough power so that all subsystems are supported throughout the mission | PWR-001, PWR-002 | PWR-006 | Testing | Met |

| | | | | | | |
|---|---|---|---|---|---|---|
| PWR-006 | The PCDU shall regulate the voltage before inputting from solar arrays and outputting to instrumentation and mechanisms | Voltage regulation is necessary to protect instruments from faults, and regulate load fluctuations | PWR-003, PWR-004, PWR-05, PWR-007, PWR-008 | PWR-010 | Analysis | Met |
| PWR-007 | The EPS shall be able to generate and distribute at least 187.5 W at a time to sustain all CDH operations for the entirety of the mission | CDH requires 187.5 W to operate, and is the responsibility of EPS to provide it with adequate power | PWR-001, PWR-002 | PWR-006 | Testing | Met |
| PWR-008 | The EPS shall be able to generate and distribute at least 75 W at a time to sustain all propulsion operations for the entirety of the mission | Propulsion requires 75 W to operate, and is the responsibility of EPS to provide it with adequate power | PWR-001, PWR-002 | PWR-006 | Testing | Met |
| PWR-009 | The EPS shall be able to generate and distribute at least 162.7 W at a time to sustain all TCS operations for the entirety of the mission | TCS requires 162.7 W to operate, and is the responsibility of EPS to provide it with adequate power | PWR-001, PWR-002 | PWR-006 | Testing | Met |
| PWR-010 | The EPS shall have sensors that monitor voltage, current, etc. | Communicate the health status of EPS (battery status, charging rate) to mission control | PWR-006 | | Demonstration | Met |

### 2.1.2.2 Power Sub-Assembly Overview

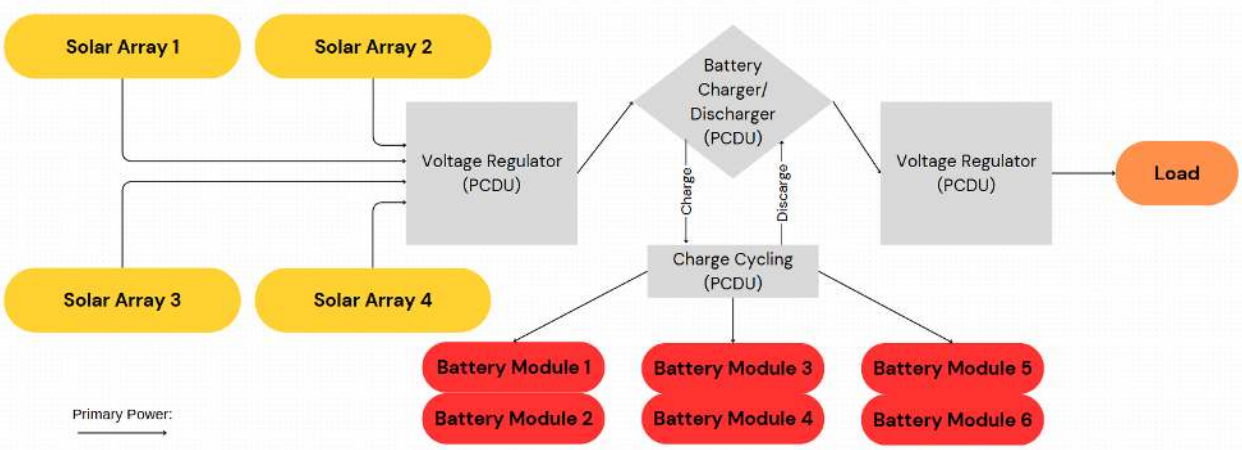



Figure #16: Diagram of the EPS

The power requirement that the EPS needs to support is the total of all the scientific instrumentation and subsystem mechanism power draw which is 1000 W. Each subassembly: Power Generation, Power Storage, and Power Management and Distribution, is designed to meet that driving power requirement. The TRL of the EPS is 4, with all components having a history of thorough testing and successful deployment in Low-Earth Orbiting applications.

The Power Generation Subassembly needs to generate the total power requirement for all scientific instruments and mechanisms (1000 W)  at a given time. It generates power using SpectroLab XTE-SF Solar Cells. Given that its W/cell is 1.17 W, there must be 856 solar cells to reach the 1000 W requirement. Since each cell is 27 cm^2, there will be a total surface area of 2.31 m² (23,112 cm²) in solar arrays. The cells will be arranged in 4 deployed arrays: 12 X 18 (with 2 empty spots in the corners), for a total of 856 solar cells 82.92 X 71.46 X 0.0225 cm. The cells will be wired and mounted on a 1 cm thick aluminum honeycomb structure with an epoxy adhesive that also has an electrical insulation strip embedded into it. On top of the cells, there is a 0.5 cm coverslide layer of Epoxy-Impregnated Fiberglass Insulators to protect the cells from ultraviolet and corpuscular radiation. The coverslide is treated with an anti reflective treatment to reduce the loss of sunlight in the wavelength range of 0.4 to 1.1 μm. (Spacecraft Solar Cell Arrays). The coverslide also contains a UV filter to reduce the photodegradation on the epoxy adhesive. Before deployment, the 4 solar arrays are folded onto the 4 walls of the aerobot cube, hinged at the top of its cube side. Using 4 small motors at the top of each 4 sides, the solar arrays make a 90 degree deployment so that they are in a horizontal position for more power generation. Power generated

from the solar cells will run through the power conditioning and distribution unit (PCDU) to be distributed and stored. While the solar cells have extensive testing and deployment in Low-Earth Orbit satellites, it has not been tested for the mission's specifications, so its TRL is a 4.

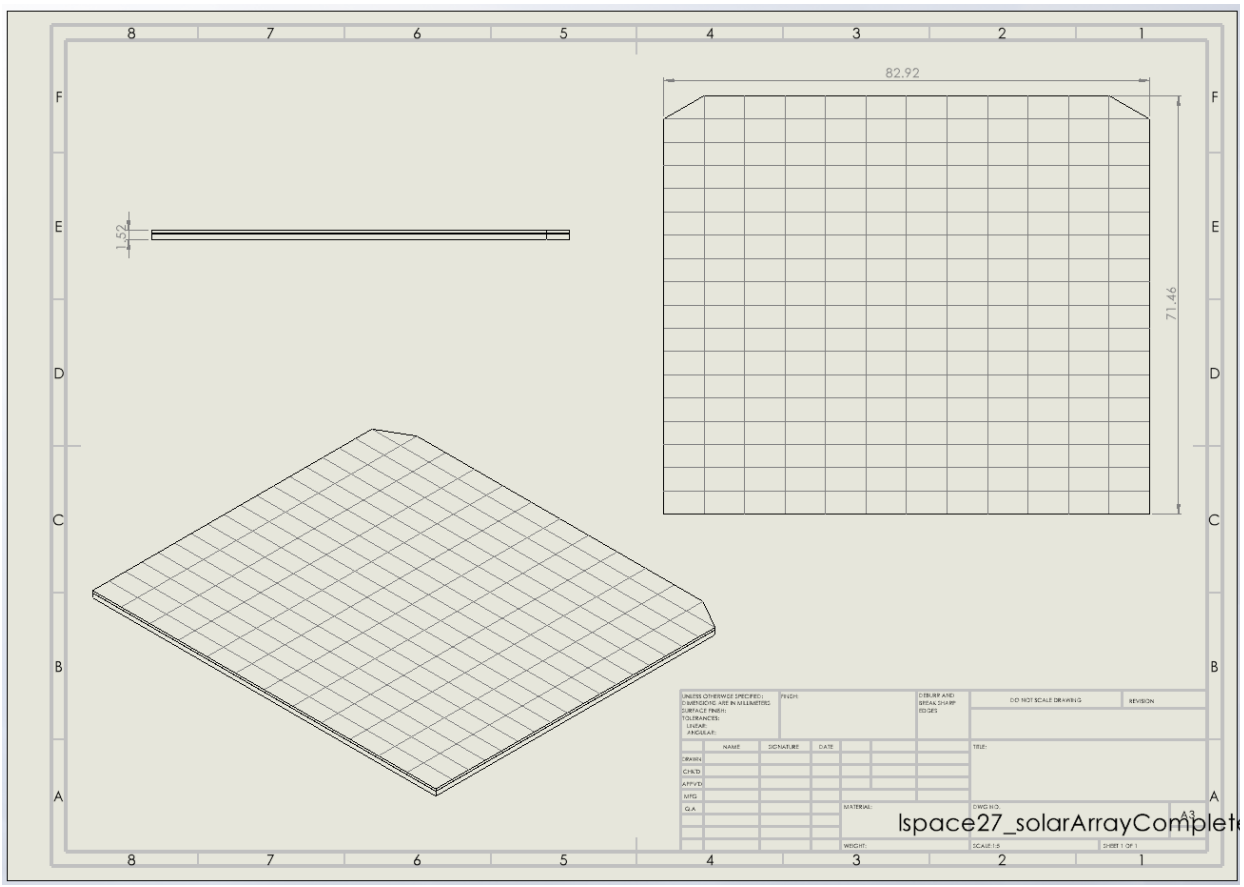



Figure 17: Design of Solar Cells

The Power Storage Subassembly stores excess power in to save for later use, helping with overconsumption and underconsumption, using 6 Pumpkin BM2 Thermal Batteries. Each battery has the maximum capacity of 168 W/unit * 6 units = 1008 W so it meets the 1000 W requirement for the instrumentation and mechanisms without shorting out. Inside the aerobot cube in the EPS compartment, the 6 batteries fit together in 2 columns, 3 rows high. To extend the battery life of the system, the EPS will utilize rotational battery cycling, actively using only 4 out the 6 battery modules at a time and switching out the older 2 with the 2 spare every 2 hours. The EPS will also implement a Depth of Discharge limit of 80% the total capacity and use supercapacitors for load balancing. The operational temperature of the battery goes up to 59°C, much lower than the temperature of Venus at 464°C, so it will need to be insulated and/or actively cooled.

Pumpkin Space's BM2 has been deployed on 6 missions, with the oldest one, "ASTRA LLC
SORTIE," in orbit since 2019 (Pumpkin Space On-orbit History). Because of its extensive history of testing and operation in cube satellite applications, its TRL is 7.

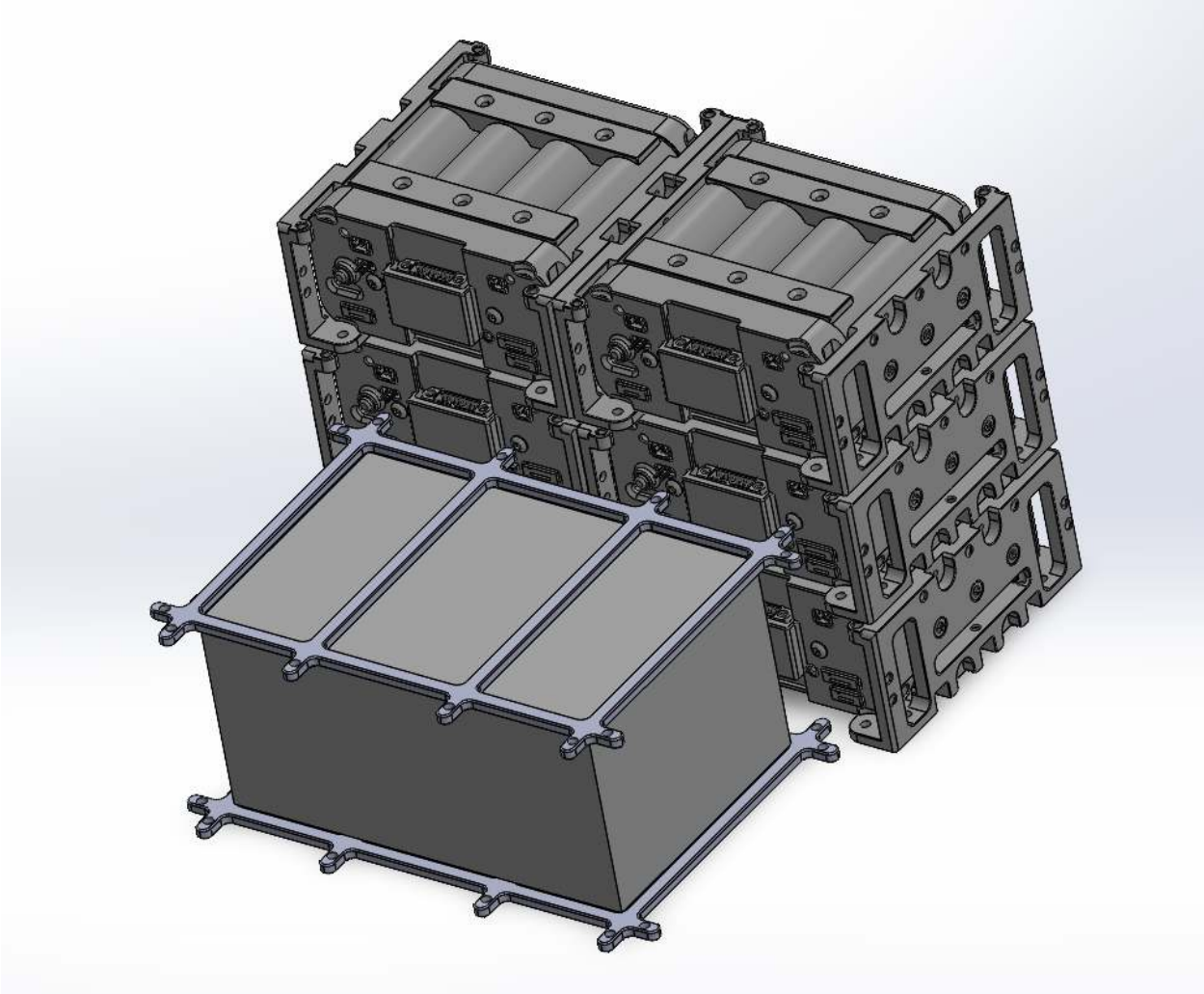


Figure 18: Design of Power Storage Subassembly

The Power Management and Distribution (PMAD) efficiently and accurately distributes power to every component using the Bradford Space SuperNova PCDU. The PCDU regulates the voltage using its Battery Charge Regulator, Array Power Regulators, and Regulated Power Distributors to prevent shorting out components, and regulates the input of power generated by the solar cells to avoid overvoltage. It has the capacity to input/output 1500 W, and will comfortably be able to distribute power to all instruments as well as aerobot mechanisms. It also has a wide variety of voltage outputs, so all the instruments and mechanisms are compatible with the PCDU. The PCDU contains a

Data and Control Interface to utilize battery separation switches for the rotational battery cycling procedure. It contains triple redundant CC, CV battery charge controller with controllable end-of-charge voltage for reduced battery fading, further protecting the power storage subassembly from overcharging and overstress. Its ability to regulate electrical power and protect load components from harmful disturbances from flowing downstream is vital to the EPS and mission operation. The PCDU is anchored next to the batteries, making their combined dimensions 20 X 20 X 14.29 cm, comfortably fitting within the aerobot chassis. Through its extensive testing and deployment in Low-Earth Orbit satellites, its TRL is a 5.

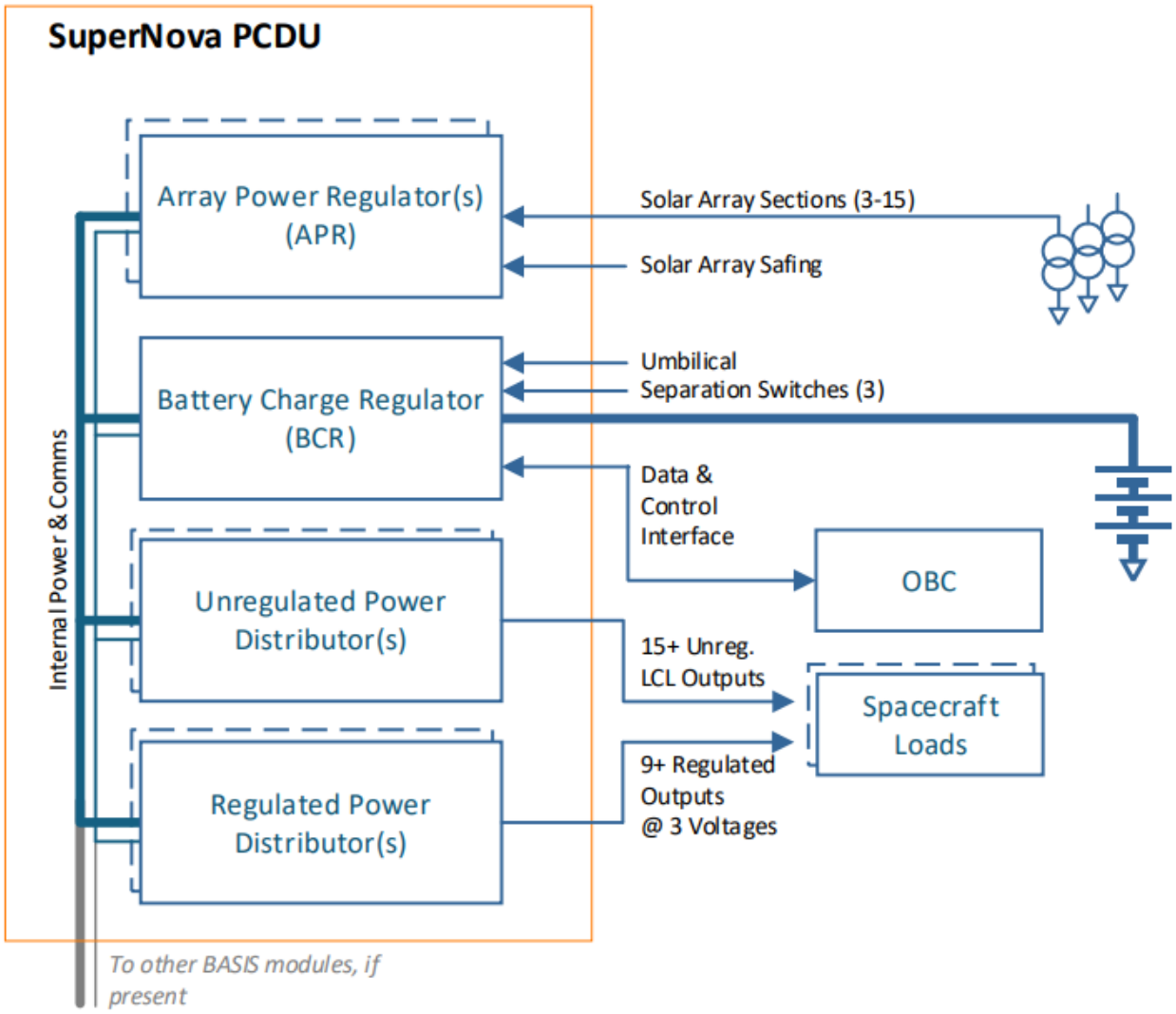



Figure 19: PCDU Breakdown

Dimension Calculations:
4 Solar Arrays each have dimensions 82.92 X 71.46 X 1.5225 cm for a total volume of 36096 cm^3
Batteries: With all 6 batteries stacked, the volume is calculated by (100mm * 100mm * 48.4 mm * 6 units) = 2904 cm3
The PCDU has dimensions (60 * 120 * 135 mm) = 972 cm³
Volume encompassing the PCDU and 6 batteries is calculated by 20 cm * 20 cm * 14.29 cm = 5716cm^3 = 0.005716 m^3
Therefore, the total volume is 39962.07 cm^3 = 0.039962 m^3.

### 2.1.2.3 Power Subsystem Recovery and Redundancy Plans

Recovery and Redundancy is of utmost importance in this mission. To counteract possible problems with the Aerobot System's power the team has implemented many forms of recovery and redundancy. This shall ensure that mission objectives are able to succeed and all necessary subsystems remain operational at a given time. Recovery and Redundancy is implemented to establish a threshold that can be referred to when the aerobot experiences a failure.

**Power Subsystem Redundancy:**
The Power Subsystem is to be composed of three smaller systems. Power generation, power distribution, and power storage.

For power generation the MCA team has implemented a redundancy plan. This shall be apparent in the 4 custom solar panels with each panel containing 214 solar cells. If the solar panels experience a failure that causes half of the 856 solar cells to be permanently unusable, the aerobot will still be able to maintain operation, since the solar arrays generate twice the power required from all the scientific instruments and mechanism without the factor of safety. The use of 4 solar panels also provides redundancy to the power generation subassembly, as it reduces cell loss from an individual panel failure.

With power storage, the Aerobot System shall contain 6 Thermal Batteries. 4 of 6 batteries shall be actively charged and discharged, while the other 2 shall be used as backup to maintain the Aerobot System, swapping out 2 used batteries every 2 hours. The use of 6 batteries provides redundancy to the power storage subassembly, and the bare minimum requirement for aerobot operation requires 3 of those batteries, allowing the power storage subassembly to tolerate a 50% loss.

For the Power Management and Distribution Subassembly, the PCDU utilizes 5 power modules, each managing 300 W. If 3 out of the 5 modules fail, the remaining 2 provide 600 W to the aerobot which is still more than the power required by all the scientific instruments and mechanism without the factor of safety.

**Power Subsystem Recovery:**
The Pumpkin BM2 battery has an intelligent gas gauge that is able to relay vital information on each battery. This information includes healthy/aging, capacity, battery temperature, and most importantly, battery status. This telemetry is relayed to the CDH system which prompts the PCDU to switch power routes to components that are not under stress. Battery heat sinks are also utilized to deal with overheating in the case of TCS failure.

### 2.1.2.4 Power Subsystem Manufacturing and Procurement Plans

**Direct Cost:** Electrical Power System
The direct cost is derived from the Mission Concept Cost Estimate Tool (MCCET), using the equation to estimate the cost of the EPS: 1516 * ElecMass^0.74 in thousands of dollars. The total mass of the EPS is (Power Generation: 1.5 kg + Power Storage: 3.31 kg + PMAD: 1 kg) = 5.81 kg. This makes the cost of the EPS: 5574.28975 * $1000 = $5,574,289.75.

| Component | Supplier | Back-Up | Estimated Lead Times |
|---|---|---|---|
| (COTS) Triple Junction Solar Cells integrated into a custom solar panel | SpectroLab | CAVU Aerospace UK | Supplier: 7-15 weeks<br>Backup: 8-16 weeks |
| (COTS) Thermal Battery | Pumpkin Space | Blue Canyon Technologies | Supplier: 6-12 weeks<br>Backup: 6-12 weeks |
| (COTS) Power Conditioning and Distribution Unit (PCDU) | Bradford Space | AAC Clyde Space | Supplier: 13 weeks (3 months)<br>Backup: 12-15 weeks |

**Component**: (COTS) Triple Junction Solar Cells integrated into a custom solar panel
SpectroLab was chosen as the primary supplier over CAVU Aerospace because of its cells' lower density. The total mass using SpectroLab would be (0.067 g/cm² * 23,015.5

cm² = 1,542.0385 g) = 1.5 kg, which is significantly lighter by (1,343.5865 g = 1.34 kg) than CAVU Aerospace's total mass of (0.125 g/cm² * 23,085 cm² = 2,885.625 g) = 2.9 kg, an extreme difference. However, SpectroLab does cost (856 cells * $250/cell = $214,000) - CAVU's (770 cells * $275/cell = $211,750) = $2250 more, but this difference in cost is negligible. The SpectroLab cells are slightly less efficient than CAVU, requiring 2.31 m² (23,112 cm²) - 2.30 m² (23,015.5 cm²) = 96.5 cm² (0.965 m²) more surface area, but this size difference is not as detrimental as the mass difference. SpectroLab is also the better manufacturer since it provides more personalized solar array arrangements than CAVU Aerospace. With SpectroLab, the estimated lead time is a total of 7-15 weeks (Cell Wiring: 2-4 weeks, Structural Assembly and Insulation: 1-3 weeks, Testing: 4-8 weeks). With backup supplier CAVU Aerospace UK, the estimated lead time is a total of 8-16 weeks (Cell Wiring: 2-4 weeks, Structural Assembly and Insulation: 1-3 weeks, Testing: 5-9 weeks).

**Component**: (COTS) Thermal Battery
Pumpkin Space was chosen as the primary supplier because of its thermal battery's lower mass, volume and cost compared to Blue Canyon Technologies. For Pumpkin Space, 6 battery modules are needed to meet the 1000W requirement: 168W/units * 6 units = 1008W. This makes the total mass: (551g/unit * 6 units) = 3.306kg, the total volume: (100mm * 100mm * 48.4 mm * 6 units) = 2904 $cm^3$, and the total cost: ($11,500/unit * 6 units) = $69,000. This is compared to Blue Canyon Technologies total mass: (1200g/unit * 5 units) = 6 kg, total volume: 46mm * 183mm * 89mm * 5 units = 3741.69 cm³, and total cost: ($70,488/unit * 5 units) = $352,440. This makes Pumpkin's total mass 2.694 kg lighter, total volume 837.69 $cm^3$ smaller, and total cost $283,440 cheaper. Pumpkin's superior specifications make it the better choice to meet the mission goals. The estimated Pumpkin Space is a total of 6-12 weeks (Wiring/Installing Battery Health Modules: 1-3 weeks, Insulation/Thermal Management Installation: 1-3 weeks, Testing: 4-6 weeks). The estimated lead time with Blue Canyon Technologies is a total of 6-12 weeks (Wiring/Installing Battery Health Modules: 1-3 weeks, Insulation/Thermal Management Installation: 1-3 weeks, Testing: 4-6 weeks).

**Component**: (COTS) Power Conditioning and Distribution Unit (PCDU)
Bradford Space was chosen as the primary supplier because of its PCDU's lower mass and volume compared to AAC Clyde Space. The Bradford Space SuperNova PCDU is 1kg and (60 * 120 * 135 mm) = 972 cm³ versus the Clyde's StarBuck-Mini being 3.3kg and (123 * 147 * 249 mm) = 4,497 cm³. Both suppliers offer a PCDU that manages the same amount of power (1,500 W) and has the same voltage range (22-34V), making mass and volume the deciding factor that puts Bradford Space ahead. The estimated lead time with Bradford Space is a total of 3 months or 13 weeks. (Supernova Power Subsystem.) The estimated lead time with AAC Clyde Space is a total of 7-13 weeks

(Wiring/Installing Current Health Monitoring Modules: 1-3 weeks, Insulation/Thermal Management: 1-3 weeks, Testing: 5-7 weeks).

### 2.1.2.5 Power Subsystem Verification Plans

Specific verification methods have been outlined for the power subsystem to ensure each component correctly meets the power needs of the remaining subsystems. These verification methods are outlined below.

| Req # | Requirement Summary | Verification Method | Rationale for Methods | Preliminary Verification Plan |
|---|---|---|---|---|
| PWR-001 | There shall be an adequate amount of power to sustain all subsystems for the entirety of the mission | Analysis and Testing | The analysis and testing ensures the EPS will perform adequately under the mission environment and duration. It also ensures the safe operation during a worse case loading. | Conduct a power budget analysis and simulated load testing over the different mission phases. |
| PWR-002 | Power shall be available to all necessary electronics systems through power supply networks | Inspection and Testing | The inspection confirms the correct physical connections. The testing ensures that the right amount of power is transferred. | Inspect the EPS wiring schematics for proper connecting. Conduct continuity tests and voltage measurements on each component. |
| PWR-003 | The EPS shall be able to generate and distribute at least 51.4 W at a time to sustain all instrumentation for the entirety of the mission | Testing and Analysis | Testing ensures that the scientific instruments are provided with a stable source of power of at least 51.4 W. The analysis records performance trends. | Conduct a simulated load for the highest instrumentation load. Analyze the telemetry to confirm stable power. |
| PWR-004 | The EPS shall have multiple power generation sources (solar panels) | Inspection and Demonstration | The inspection ensures the redundant power generation sources. The demonstration ensures that each solar panel is able to operate independent from the others. | Review the design documentation. Demonstrate the independence of each power generation source: each of the 4 solar panels |
| PWR-005 | The battery shall be able to contain the power generated during hours of sunlight, to | Testing and Analysis | Testing ensures the proper performance of the batteries and the amount of | Perform charging/discharging rotational cycling during sunlight |

| | | | | |
|---|---|---|---|---|
| | save for power use during hours without sunlight | | energy stored matches the amount needed during hours of no sunlight. | hours. Analyze the amount stored to be able to support dark mission periods. |
| PWR-006 | The PCDU shall regulate the voltage before inputting from solar arrays and outputting to instrumentation and mechanisms | Testing | Testing ensures that the PCDU controls the voltage and prevents the instrumentation and mechanisms from being shorted out. | Test the voltage input/output from the solar arrays to the load. |
| PWR-007 | The EPS shall be able to generate and distribute at least 187.5 W at a time to sustain all CDH operations for the entirety of the mission | Test | Testing ensures the proper power supply for the CDH subsystem and proper operation in the mission environment and duration. | Test a simulated CDH load and monitor its performance and power output. |
| PWR-008 | The EPS shall be able to generate and distribute at least 75 W at a time to sustain all propulsion operations for the entirety of the mission | Test | Testing ensures the proper power supply for the Propulsion subsystem and proper operation in the mission environment and duration. | Test a simulated Propulsion load and monitor its performance and power output. |
| PWR-009 | The EPS shall be able to generate and distribute at least 162.7 W at a time to sustain all TCS operations for the entirety of the mission | Test | Testing ensures the proper power supply for the TCS subsystem and proper operation in the mission environment and duration. | Test a simulated TCS load and monitor its performance and power output. |
| PWR-010 | The EPS shall have sensors that monitor voltage, current, etc. | Inspection and Testing | The inspection ensures that the electrical sensors are able to read the EPS status. Testing ensures that the readings are accurate. | Inspect that electrical sensors are placed and designed correctly. Test the sensor readings to match with calibrated external measurement instruments. |

### 2.1.3 CDH Subsystem Overview

The Command and Data Handling Subsystem manages the internal functions of the Aerobot system, primarily responsible for the communication between the spacecraft and orbiter/EDI, interconnection between the various subsystems and their technological components, and data handling regarding the mission objective. These tasks are divided amongst the subassemblies chosen, involving: an Onboard Computer (OBC), Data Storage, Data Interfaces, and Communications system. This also includes the S-BAND TT&C and High Gain  Patch Antenna, Micro SDR-C software defined radio, and a Crotex-M7, 384 kB RAM, 2 MB Flash Memory, and CAN/LVDS Data Interface.

The **Onboard Computer** is the main processing unit in the Aerobot system and performs the vital functions required for the mission objectives. The Aerobot system consists of two OBCs used to provide fault tolerance architecture in the case of hazards or unexpected difficulties, such as Venus' pressure or component malfunctions, that causes the backup OBC to become the primary OBC while the other is being repaired by mission personnel through telecommands or the OBC's semi-autonomous repair system. Each OBC has the same software and hardware configurations. Other software includes the Recovery and Redundancy system that allows for the Aerobot system to manage itself semi-autonomously while on Venus, executing programs and algorithms to navigate Venus' environment and activate the backup components in case the main one encounters a failure or error. The OBC is also responsible for the **Data Interface** that communicates between each subsystem and the various components. The OBC used has access to the CAN and LVDS Data Interfaces, providing flexibility at a high speed and low power to internal communications within the components, serving as a method for the OBC to connect with other subsystems.

The High Gain Patch antenna is designed for high throughput data downlink, which is useful for scientific telemetry. High gain antenna is particularly useful for high speed, long distance data transmission to earth or in this case, an orbiter. The TT&C antenna enables routine command and telemetry operations during all mission phases, specifically for health checks, system status and receiving commands. The Micro SDR-C functions as a versatile software defined radio for encoding, decoding and managing a variety of communication protocols in the S-band frequency. The Crotex M7 is responsible for mission autonomy, subsystem coordination and real time data logging.

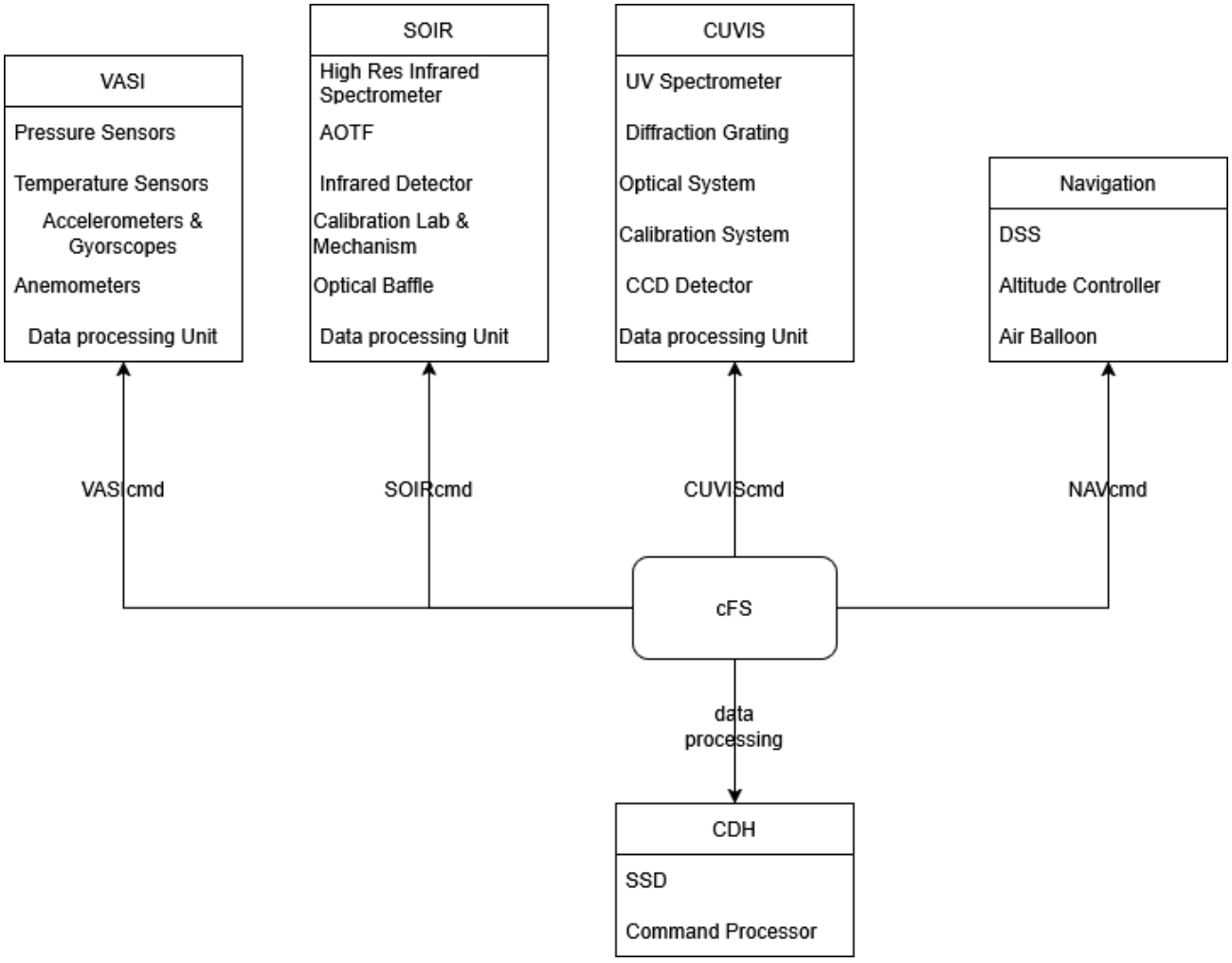



Figure 20: Software Connection Diagram

The above figure outlines the functional data flow within the CDH subsystem, mapping out how data and commands are transmitted, processed and routed between components and interfaces.

Inbound commands from ground control are received via the TT&C antenna, passed through the Micro SDR-C and forwarded to the OBC for execution.

Scientific instruments send data to the OBC for formatting and logging. Data is stored in onboard memory before being transmitted through the Micro SDR-C and High Gain Patch antenna.

The OBC polls other subsystems via the necessary buses, logging status/data and adjusts parameters as needed.

### 2.1.3.1 CDH Subsystem Requirements

Relevant CDH requirements were derived based on system and customer constraints as well as the need for the Aerobot system to be able to process and follow ground commands, collect and transmit data, and allow for the data collected fro the scientific instruments to be useful. The following requirements are listed below.

| Req # | Requirement | Rationale | Parent Req | Child Req | Verification method | Req met? |
|---|---|---|---|---|---|---|
| CDH-1 | The Aerobot shall implement an Onboard Computer (OBC) | The OBC handles the main operations of the Aerobot, interlinks the various subsystems, and handles data/data collection | CDH-8 | CDH-4, CDH-1.1, CDH-1.2, CDH-1.3 | Inspection | Met |
| CDH-1.1 | OBC shall have high performance specs | OBC requires sufficient specifications in order to manage Aerobot efficiently and remain within customer constraints | CDH-1 | | Demonstration | Met |
| CDH-1.2 | OBC shall implement a flight software that manages the Aerobot's flight and descent | Allows for adjustments in Aerobot's positioning and provides vertical stabilization during descent | CDH-1 | CDH-1.2.1 | Demonstration | Met |
| CDH-1.2.1 | OBC shall control the Descent Subsystem (DSS) | Attached to Aerobot and provides additional vertical descent options | CDH-1.2 | | Analysis | Met |

| | | | | | | |
|---|---|---|---|---|---|---|
| CDH-1.3 | OBC shall have a manner of handling information | Collects, organizes, and stores data within CDH subsystem before being downlinked to the EDI Orbiter | CDH-1 | CDH-1.3.1 | Inspection | Met |
| CDH-1.3.1 | The OBC shall take care of atmosphere, surface, and magnetic field composition data collection | Data regarding fundamental science objectives will be collected using instruments through OBC software | CDH-1.3 | | Analysis | Met |
| CDH-2 | A Data Storage System shall be implemented within the Aerobot payload | Data collected from instruments and regarding Aerobot status must be stored reliably before being transmitted | CDH-1.3 | CDH-2.1 | Inspection | Met |
| CDH-2.1 | Data Storage shall have enough memory and storage for offline/semi-autonomous data collection | Ensures all data is maintained when data transmission cannot unfeasible during the moment | CDH-2 | CDH-2.1.1 | Demonstration | Met |
| CDH-2.1.1 | Data Storage System shall have an additional method of storing data when | Instrument data and Aerobot subsystem statuses are stored within additional flash memory and RAM if primary | CDH-2.1 | | Test | Met |

| | | | | | | |
|---|---|---|---|---|---|---|
| | primary methods are unfeasible | data storage fails | | | | |
| CDH-3 | The Aerobot system shall check on the statuses of other subsystems through a Data Interface | Ensures overall Aerobot functionality and determines activation of the Redundancy and Recovery System | CDH-1 | | Demonstration | Met |
| CDH-4 | Aerobot system shall implement a Communication System | Allows for transmission of telemetry data and telecommands between Aerobot, orbiter, and ground systems on Earth | CDH-1 | CDH-4.1, CDH-4.2, CDH-4.3, CDH-4.4 | Inspection | Met |
| CDH-4.1 | Descent Subsystem (DSS) shall deliver no more than 5 Mbps of data to the CDH Subsystem during its operations | Constraint provided by the Interface Control Document (ICD) for the DSS | CDH-4 | | Test | Met |
| CDH-4.2 | Information delivered from the Aerobot System to EDI Orbiter shall be at 2-4 GHz frequency | Frequency constraint provided by the use of S-Bands | CDH-4 | | Test | Met |

| | | | | | | |
|---|---|---|---|---|---|---|
| CDH-4.3 | Status and data regarding Aerobot System shall be constantly transmitted to EDI orbiter | Provides means of checking Aerobot status | CDH-4 | | Test | Met |
| CDH-5 | The Aerobot shall implement a Recovery and Redundancy System | In the case of risks, hazards, and unexpected emergencies, the Aerobot will have a backup system to perform specific functions depending on the malfunctioning system | CDH-7 | | Inspection | Met |
| CDH-6 | Usage of CDH subassemblies/components shall not exceed 187.5 Watts within a given moment | Constraint provided by Power Subsystem | PWR-1 | | Test | Met |
| CDH-7 | The subsystem shall regulate itself and be hazard resistant | Allows Aerobot to survive Venus' harsh atmosphere and endure the entire mission duration | SYS-7 | CDH-5, CDH 7.1, CDH 7.2, CDH 7.3, CDH 7.4 | Test | Met |

| | | | | | | |
|---|---|---|---|---|---|---|
| CDH-7.1 | CDH Redundancy and Recovery System shall be autonomous and allow for human intervention in case of emergencies | Allows Aerobot to solve encountered problems on its own and function without constant human supervision | CDH-7 | | Test | Met |
| CDH-7.2 | The subsystem shall regulate its temperature | Methods provided by Thermal Subsystem | CDH-7 | | Test | Met |
| CDH-7.3 | The subsystem shall be vibration resistant | Methods provided by Mechanical Subsystem | CDH-7 | | Test | Met |
| CDH-7.4 | The subsystem shall be radiation resistant | CDH Subsystem plans to use radiation hardened components to combat Venus' radiation exposure, additional methods provided by Mechanical Subsystem | CDH-7 | | Test | Met |
| CDH-8 | The control and data handling subsystem shall contribute to fulfilling the science objectives of the mission | Fundamental objective of mission | 0.1, 0.2, 0.3 | CDH-1 | Analysis | Met |

### 2.1.3.2 CDH Sub-Assembly Overview

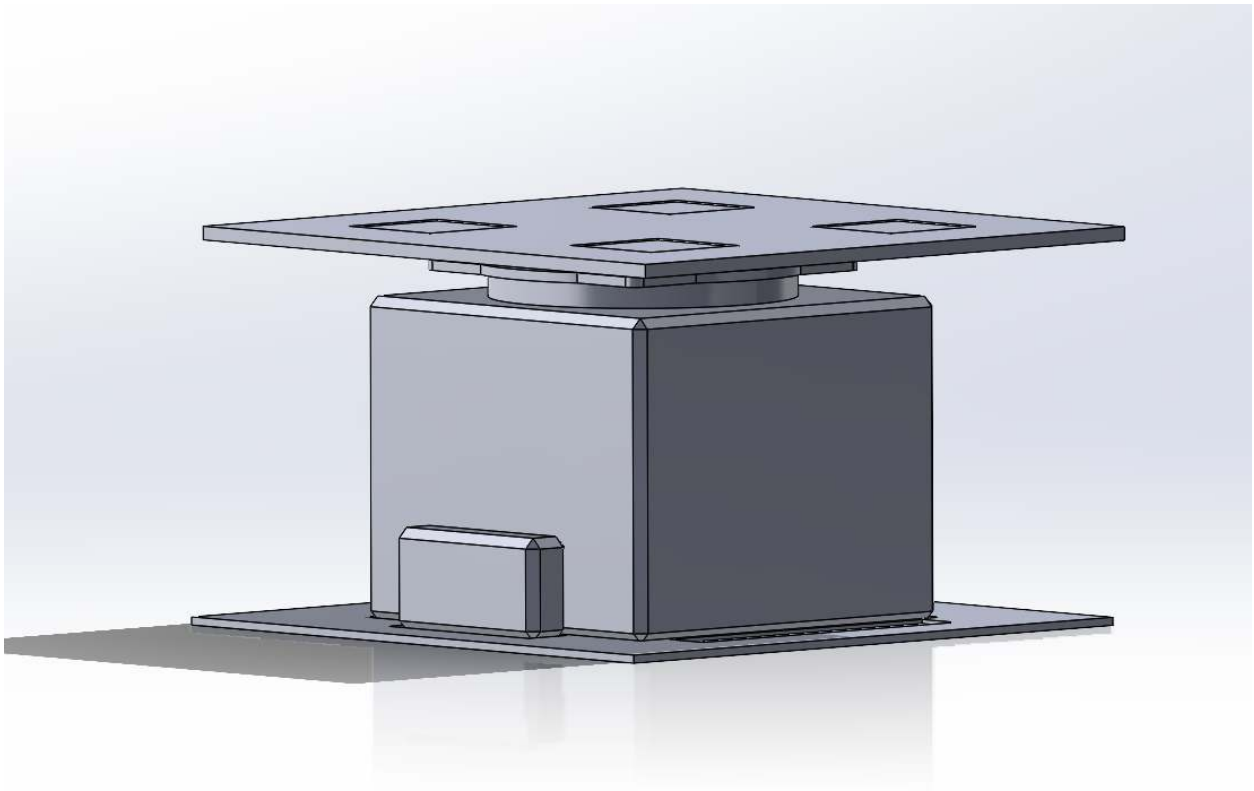


**Figure 21: OBC CAD Design**

The OBC modules assumes the team has the necessary IO and buses required for operations on an arbitrary PCB, mounted inside a protective module. The High Gain Patch antenna is mounted on the top of the assembly, with the TT&C antenna mounted directly under the High Gain Patch antenna. Under the TT&C antenna is the micro SDR-C. To the left of the Micro SDR-C is the computing module. All of these modules are mounted on a 2mm thick sheet of steel. The assembly is to be mounted inside the CDH compartment of the aerobot.

### 2.1.3.3 CDH Subsystem Recovery and Redundancy Plans

**Redundancy**

Data Storage Redundancy: A primary flash memory unit is supported by a secondary mirrored flash module. In the event of a primary memory failure, science and telemetry data are rerouted and buffered in the backup storage. This ensures no mission-critical

data is lost during the 48 hour descent. Additional flash memory is included to collect instrument data in the case the antenna signals have been significantly attenuated by the weather or other conditions. Additional RAM is included to provide short-term memory that can be overwritten, useful for transmitting data or collecting information about the Aerobot.

Power Cycling and Watchdog Timers: The M7 microprocessor is equipped with internal watchdog timers that monitor system health and trigger safe resets in case of processor hangs, instruction faults, or segmentation faults. If a software exception or radiation induced bit flip is detected, the watchdog resets the OBC to a known “safe” state.

Radiation-Tolerant Hardware: All CDH components are selected based on radiation tolerance and existing flight heritage.

Communication link backup: The CDH system includes two RF channels, a low gain S-BAND TT&C antenna for telemetry and command, and a high-gain patch antenna for science data transmission. If the high gain link fails or cannot establish lock with the orbiter, the TT&C channel will continue to transmit essential data and limited science information at reduced rates. The S-band Communication System chosen provides access to the Near Space Network (NSN) and to Tracking and Data Relay Satellites (TDRS), providing multiple methods of data communication that can serve as backups in addition to the EDI orbiter. The NSN also provides Direct-to-Earth (DTE) communication through their ground systems.

Subsystem health monitoring: The CDH system polls all major subsystems via necessary buses. If a fault is detected, the OBC logs the anomaly and executes recovery protocols.

**Recovery**

The CDH Subsystem implements a Recovery and Redundancy System for its main subassemblies, involving the OBC, Data Storage, and Communications in order to maintain constant communication and data handling throughout the mission. CDH Redundancy involves a duplicate component for main hardware systems, such as the OBC, flash memory, and antennas that take control over the primary components if the primary systems encounter an unexpected error or fail.

Autonomous fault handling: In the event of system errors, the OBC uses internal decision trees to isolate faulty modules, perform reboots, or transition to safe mode. Critical operations are prioritised, while non-essential processes are suspended. Similar to NASA’s Landsat 4’s Fail Check system, the Aerobot has an internal timer that sends commands to each major subsystem component every 3 seconds. If these components

fail to return a signal, the OBC will execute its Recovery software, an autonomous program that detects the type of problem and runs certain algorithms to fix them. For hardware and payload related issues, the Recovery software will autonomously troubleshoot these systems and if applicable, switch these components to their backups. Mission personnel on Earth are notified of the issue and can execute telecommands that take priority over the autonomous system. Other Aerobot related issues will be repaired autonomously through the Recovery software by utilizing the other subsystems, such as overheating and pressure failures.

Fallback Operation Modes: When high speed communication or storage fails, the system enters a degraded mode. This mode prioritizes essential telemetry, reduces science transmission rate and basic system health checks. This ensures the mission continues as long as power and structure allow.

Thermal and Vibration mitigation: Thermal sensors and vibration dampening mounts are included to prevent performance degradation during venusian environmental stress. In the even of overheating, the OBC can reduce processing frequency or disable non-critical tasks.

### 2.1.3.4 CDH Subsystem Manufacturing and Procurement Plans

**Direct Cost:** CDH System
The direct cost is derived from the Mission Concept Cost Estimate Tool (MCCET), using the equation to estimate the cost of the EPS: 236 * ElecMass^0.69 in thousands of dollars. The total mass of the EPS is (Power Generation: 1.5 kg + Power Storage: 3.31 kg + PMAD: 1 kg) = 5.81 kg. This makes the cost of the EPS: 794.6822895 * $1000 = $794,682.

| Component | Supplier | Back-Up | Estimated Lead Times |
|---|---|---|---|
| S-Band TT&C Antenna | Anywaves | SkyLabs S-band Patch Antenna | 10 - 12 weeks |
| S-Band High Gain Patch Antenna | IQ Spacecom | Haigh-Farr, Inc. P/N 21060 | 4 - 24 weeks |
| MicroSDR-C | Space Micro | CeisiumAstro SDR-1001 | 1 - 4 weeks |
| Crotex-M7, 384 kB RAM, 2 MB Flash Memory, CAN/LVDS Data Interface | ARM | STMicroelectronics ARM Cortex | 2 - 8 weeks |

### 2.1.3.5 CDH Subsystem Verification Plans

Specific verification methods have been outlined for the CDH subsystem to ensure each requirement is verified and for the overall system to be developed correctly based on the necessary constraints.

| Req # | Requirement Summary | Verification Method | Rationale for Methods | Preliminary Verification Plan |
|---|---|---|---|---|
| CDH-1 | The Aerobot shall implement an Onboard Computer (OBC) | Inspection and Test | The inspection confirms the OBC itself is maintained and up to date. The testing ensures programs and the necessary applications are usable on the OBC. | Inspect OBC connections, test OBC using programs and tools. |
| CDH-2 | A Data Storage System shall be implemented within the Aerobot payload | Inspection | The inspection ensures systems are up to date and will not lead to loss of data. | Inspect the system, collect and relay data. |
| CDH-3 | The Aerobot system shall check on the statuses of other subsystems through a Data Interface | Demonstration | The demonstration will show the system is operational to access other subsystems, ensuring they are operational. | Demonstrate the system works, accessing the data interface of all other subsystems. |
| CDH-4 | Aerobot system shall implement a Communication System | Inspection | The inspection will confirm proper installation, functionality, and compliance to codes and standards. | Inspect the system after installation, check for functionality, ensure system does not violate any codes or standards. |
| CDH-5 | The Aerobot shall implement | Inspection | The inspection will ensure the system functions correctly, and handling | Regular maintenance will be made, verify system to ensure if |

| | | | | |
|---|---|---|---|---|
| | a Recovery and Redundancy System | | failures while continuing operations. | backups are required and supplied to prevent failures when necessary. |
| CDH-6 | Usage of CDH subassemblies/components shall not exceed 187.5 Watts within a given moment | Test | The test confirms overload of power will not occur within the whole CDH system. | Test the voltage input/output from the systems. |
| CDH-7 | The subsystem shall regulate itself and be hazard resistant | Test | The test will ensure CDH systems are protected and highly resistant. | Test integrity of temperature regulators and subsystems. |
| CDH-8 | The control and data handling subsystem shall contribute to fulfilling the science objectives of the mission | Analysis | The analysis shall confirm that the CDH system is able to provide the necessary commands/communications. | Analyze data from CDH systems before operations to ensure no problems arise during operations. |

### 2.1.4 Thermal Control Subsystem Overview

The thermal management subsystem is responsible for maintaining the system's internal temperature at 300 K for the duration of the 48-hour mission, during which the aerobot will fly at altitudes ranging from 70 km to 50 km in Venus' atmosphere. The 20 km altitude range presents significantly varying temperatures, from 225.8 K at 70 km to 344.7 K at 50 km (Seiff et al., 57). Venus' extreme atmosphere, as well as the system's 45 kg and 167.2 W power limit, call for a balanced design using both passive and active thermal control components.

To tackle Venus' harsh conditions, the subsystem incorporates Multi-Layer Insulation (MLI) on five of the cubes' faces (top, bottom, and three sides) and Z93 white thermal coating on the remaining side face. An aluminum 6061 radiator will be coated in Z93 and mounted on the only non-MLI face to aid with heat rejection. Internally, the system features a Loop Heat Pipe (LHP) to passively transfer heat generated by internal components to the radiator when the system's temperature rises above the desired value (300 K), as well as a flexible heater to counteract heat loss when temperatures drop below the desired value.

The thermal control subsystem's passive and active thermal components allow the system to maintain an approximate net zero heat transfer load in the hot and cold extremes. The LHP and radiator dissipate up to 892 W of heat in the hot case, while the flexible heater generates the necessary 147 W of heat in the cold case. The heat flow maps in Figures 22 and 23 illustrate the energy balance for each scenario. The resulting heat transfer values were determined using the following equations and Venus-related assumptions.

**Hot Case (50 km altitude)**

<u>Solar Load</u>

Assumption(s):

- Only affects the top face.
    - Viewing factor, $F = 1$.
- Astronomical Units from the Sun to Venus, $AU_{Sun\ to\ Venus} = 0.723\ AU$, (Las Cumbres Observatory, 2025).

Equation(s) based on assumption(s):

$$q_{solar\ flux} = \frac{1367}{(AU)^2}$$

$$Q_{solar} = q_{solar\ flux} \cdot A \cdot \alpha_{MLI} \cdot F$$

<u>Heat Transfer due to Venus' Albedo</u>

Background:

- Albedo is the measure of radiation or incident light that reflects off a surface.
- Venus reflects approximately 75% of the sunlight that it receives; therefore, it has an albedo of ~0.75 (Encyclopaedia Britannica 2025).

Assumption(s):

- At 50 km, Venus' albedo will be at full effect due to its dense atmosphere, therefore$albedo_{50\ km} = 0.75$.
- Viewing factors will vary across the system's faces, $F_{top} = 0,\ F_{bottom} = 1,\ F_{side} = 0.3$.

Equation(s) based on assumption(s):

$$Q_{albedo} = q_{solar\ flux} \cdot albedo_{50\ km} \cdot A \cdot \alpha \cdot F$$

$$Q_{albedo} = q_{solar\ flux} \cdot albedo_{50\ km} \cdot A \cdot [3(\alpha_{MLI} \cdot F_{side}) + 1(\alpha_{Z93} \cdot F_{side}) + 1(\alpha_{MLI} \cdot F_{bottom})]$$

Heat Transfer due to Radiation

Assumption(s):

- The system will be surrounded by Venus' radiative atmosphere; therefore, the viewing factor $F = 1$.

Equation(s) based on assumption(s):

$$Q_{rad-venus} = \varepsilon\sigma FA(T_{sys}^4 - T_{Venus,\ hot}^4)$$

$$Q_{rad-venus} = \sigma A(T_{sys}^4 - T_{Venus,\ hot}^4)(5\varepsilon_{MLI} + 1\varepsilon_{Z93})$$

Heat Transfer due to Convection

Assumption(s):

- The system will be moving at the same speed as Venus' high winds; as a result, the system's relative velocity will be ~ 0.
    - Nusselt number, Nu, will equal 0.3, because the Reynolds' number is ~ 0.
    - The convective heat transfer coefficient will be defined as $h = Nu \cdot k$ because the distance from one end of the surface to the other is 1 m.
- Venus' atmosphere is 96% carbon dioxide (Encyclopaedia Britannica, 2025).
    - Use the thermal conductivity of $CO_2$; this value varies at different temperatures (Engineering Toolbox, 2018).

Equation(s) based on assumption(s):

$$Re = \frac{\rho v d}{\mu}$$

$$Pr = \frac{v}{\alpha} = \frac{\mu C_P}{k}$$

$$Nu = 0.3 + \frac{0.62\sqrt{Re}(Pr)^{1/3}}{[1+(0.4Pr)^{2/3}]^{1/4}}[1 + (\frac{Re}{282000})^{5/8}]^{4/5}$$

$$h = Nu\frac{k}{D}$$

$$Q_{conv} = hA(T_{sys} - T_{Venus,\,hot})$$

$$Q_{conv} = 6 \cdot Nu \cdot k_{CO2,\,344.7\,K} \cdot A \cdot (T_{sys} - T_{Venus,\,hot})$$

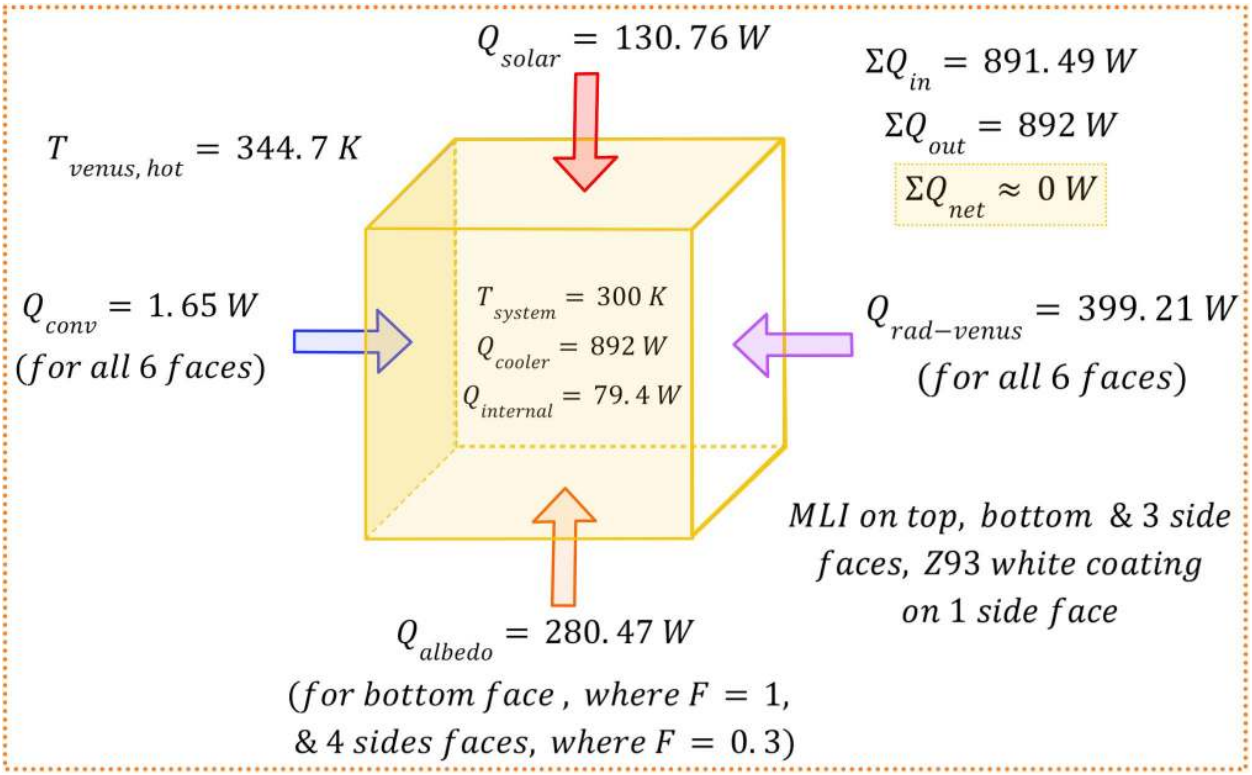



**Figure 22.** Heat flow map of the hot case (50 km above the surface).

**Cold Case (70 km altitude)**

In the cold case, the same assumptions and governing equations from the hot case are applied, with some differences due to the change in altitude and environmental conditions:

- Direct solar flux is neglected in the cold case because the system is assumed to be located on Venus' dark side. Where there is no heat contribution from direct sunlight, and the viewing factor equals zero.
- At 70 km in altitude, Venus' reflective cloud layer is less dense than in the hot case. As a result, the albedo is assumed to be 50% of the value used at 50 km. The governing albedo equation and the face-specific viewing factors remain

unchanged .

- The system remains surrounded by Venus' atmosphere. Therefore, a viewing factor of 1 is used for all faces. The same values are substituted into the Stefan-Boltzmann equation except for the temperature corresponding to Venus' atmosphere.
- The same assumptions about relative velocity and atmospheric composition are accurate. However, the thermal conductivity of CO2 is based on the temperature used for the cold case. The convection is summed over all six faces.

All assumptions, derivations, and equations used in the calculations of the heat flow maps are provided in Appendix [A.2].

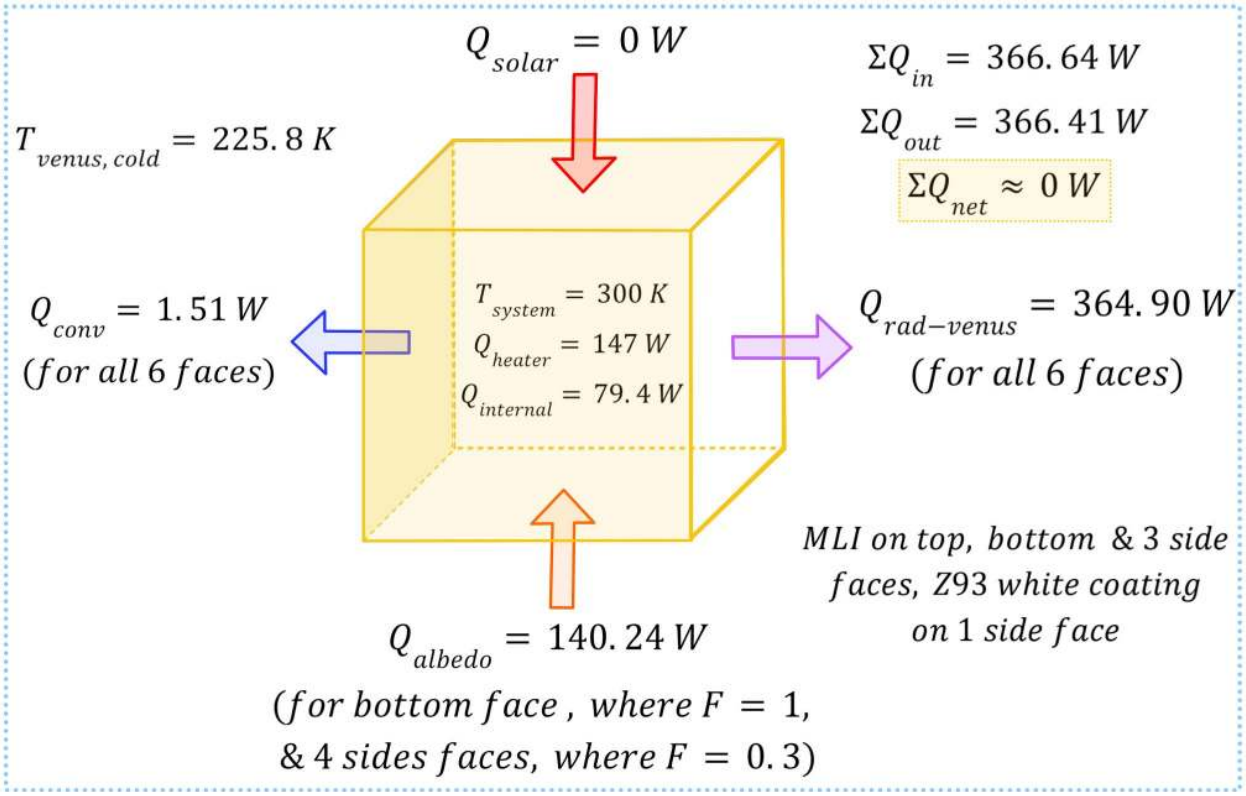



**Figure 23.** Heat flow map of the cold case (70 km above the surface).

#### 2.1.4.1 Thermal Control Subsystem Requirements

The Thermal Control Subsystem (TCS) requirements were derived to ensure that all thermal control components operate reliably under the mission's structural and environmental constraints. These requirements are organized using a hierarchical system. Regarding the TCS, the parent requirements (e.g., TH-1 through TH-4) are developed from broader system requirements (e.g., SYS-001). The child requirements further specify the needs of the thermal control components to satisfy the parent’s requirements. A mission can have as many iterations of requirements as needed to fulfill the overarching constraints. This structure ensures full traceability between system goals and subsystem design decisions.

The requirements address critical factors for the mission's overall success. Mission constraints present limitations on mass, volume, budget, and power consumption. Other requirements are necessary for the systems' survival in Venus' harsh atmosphere. These include leveraging active and passive thermal control components to maintain a healthy temperature for all instruments and electronics in the payload, as well as selecting durable materials and components with high degradation tolerances. Verification methods (e.g., inspection, analysis, and testing) are assigned to each requirement to evaluate the performance of components under mission-like conditions.

TCS Subsystem Requirements

Thermal requirements were outlined based on the customer and system requirements, providing parameters on the system's ability to protect the other subsystems while regulating the internal conditions of the system.

| Req # | Requirement | Rationale | Parent Req | Child Req | Verification Method | Req met? |
|---|---|---|---|---|---|---|
| TH-1 | The total mass of all thermal control components shall not exceed 18 kg. | The TCS must be designed with light materials to ensure compliance with the mass constraint. | SYS-001 | | Inspection | Met |
| TH-2 | The TCS shall maintain all thermal control components within the system's dimensional constraints. | Ensures all passive and active thermal components fit within the system's dimensional constraints. | SYS-002 | | Inspection | Met |
| TH-3 | The costs of the TCS shall not exceed $20 million. | Funding must be distributed according to the customer's budget. | SYS-004 | | Analysis | Met |
| TH-4 | The TCS shall withstand environmental temperatures ranging from 225.8 K to 344.7 K (at altitudes of 70 to 50 km) during the mission duration. | Scientific instrumentation and electronics must be at operating temperatures for the entirety of the SS's descent. | SYS-007 | TH-4.1<br>TH-4.2 | Test & Analysis | Met |
| TH-4.1 | The PTC components shall minimize net heat gain or loss (≈ 0 W) to reduce active thermal control power consumption in steady-state conditions. | Regulates temperatures using material properties instead of power consumption. | TH-4 | TH-4.1.1<br>TH-4.1.2<br>TH-4.1.3 | Test & Analysis | Met |

| | | | | | | |
|---|---|---|---|---|---|---|
| TH-4.1.1 | The PTC components should use MLI to reduce heat gain and loss. | MLI limits the amount of heat that leaves and enters the system, acting as a thermal insulator. | TH-4.1 | | Test & Analysis | Met |
| TH-4.1.2 | The PTC components should use thermal coatings to reduce heat gain and loss. | Selecting a thermal coating based on emissivity and absorptivity allows for further customization of TCS. | TH-4.1 | | Test & Analysis | Met |
| TH-4.1.3 | Passive radiators used for heat rejection should not exceed 1 m^2 to fit within dimensional constraints. | Ensures compatibility with the system's dimensional constraints. | TH-4.1 | | Inspection | Met |
| TH-4.2 | The ATC components shall regulate heat dissipation and retention only when the PTC cannot maintain operational temperature limits. | ATC systems rely on power-consuming components to regulate temperature, so using active components should be minimized to conserve energy. | TH-4 | TH-4.2.1 TH-4.2.2 | Analysis | Met |
| TH-4.2.1 | The ATC heaters and coolers shall activate when component temperatures exceed ±10 K of the desired 300 K steady-state temperature. | Defines when active heating or cooling is necessary, preventing unnecessary power use. | TH-4.2 | | Analysis | Met |
| TH-4.2.2 | ATC components shall generate or remove heat using a maximum of 165 W of power. | Ensures ATC components work efficiently within the set constraints of the power subsystem. | TH-4.2 | | Test & Analysis | Met |

### 2.1.4.2 Thermal Control Sub-Assembly Overview

The thermal control system consists of five components: Z93 white coating, multi-layer insulation (MLI), an Al 6061 radiator, a loop heat pipe (LHP), and a flexible polyimide thermofoil heater. These components work in unison to regulate the temperature of the cubic meter payload for the 48-hour duration of the mission. Over that period, the aerobot will experience Venus' cruel atmosphere, where temperatures range from 225.8 K to 344.7 K. The thermal control system will be required to maintain the system's internal electronics at a steady-state temperature of 300 K.

Z93 is a space-proven thermal control coating developed by AZ technology. The thermal coating's high emissivity ($\varepsilon = 0.92$) and low absorptivity ($\alpha = 0.16$) enable effective heat rejection and minimal heat gain (Kauder 2005, p. 42). In the current design, Z93 is applied to the side face where the radiator is located. This location enhances the radiator's passive heat rejection capabilities by increasing thermal emission while maintaining low exposure to solar and albedo loads. Z93 has been used on the surface of the International Space Station for over 10 years and has proven reliable in various other space missions (AZ Technology, 2025). Earning an estimated Technology Readiness Level (TRL) of 7. Also, note that the mass of the Z93 white coating is negligible due to its thin application layer.

Multi-layer insulation is crucial to the payload's passive thermal control strategy, as it minimizes radiative heat transfer between Venus' atmosphere and the system. MLI covers five of the cube's six outer faces, the top, bottom, and three sides. The insulation is composed of interwoven, alternating spacer and reflective layers, designed to produce surfaces with very low emissivity and absorptivity values. These characteristics help systems reflect incident thermal radiation while limiting the heat emitted by internal components. By reducing both heat gain in the hot case and heat loss in the cold case, the MLI allows scientific instruments and electronics to operate stably, while also reducing the work required for active thermal components. The MLI contributes 3.5 kg to the system and covers 5 $m^2$ of the payload (Birur, Siebes, and Swanson, 2004).

Venera Nova's MLI will be sourced from Northrop Grumman, a highly reputable company with extensive experience in a wide variety of spacecraft systems. Northrop has been involved in several high-profile missions, such as the James Webb Telescope. While MLI is space-flight proven, it has not been tested in Venus' dense and acidic atmosphere. Therefore, earning a Technology Readiness Level of 7.

To visualize the integration of these passive thermal components, Figure 24 displays the payload's exterior with visible applications of Z93 white coating, MLI, and the radiator. Note that this CAD model solely displays thermal control components; components from other subsystems are not shown.

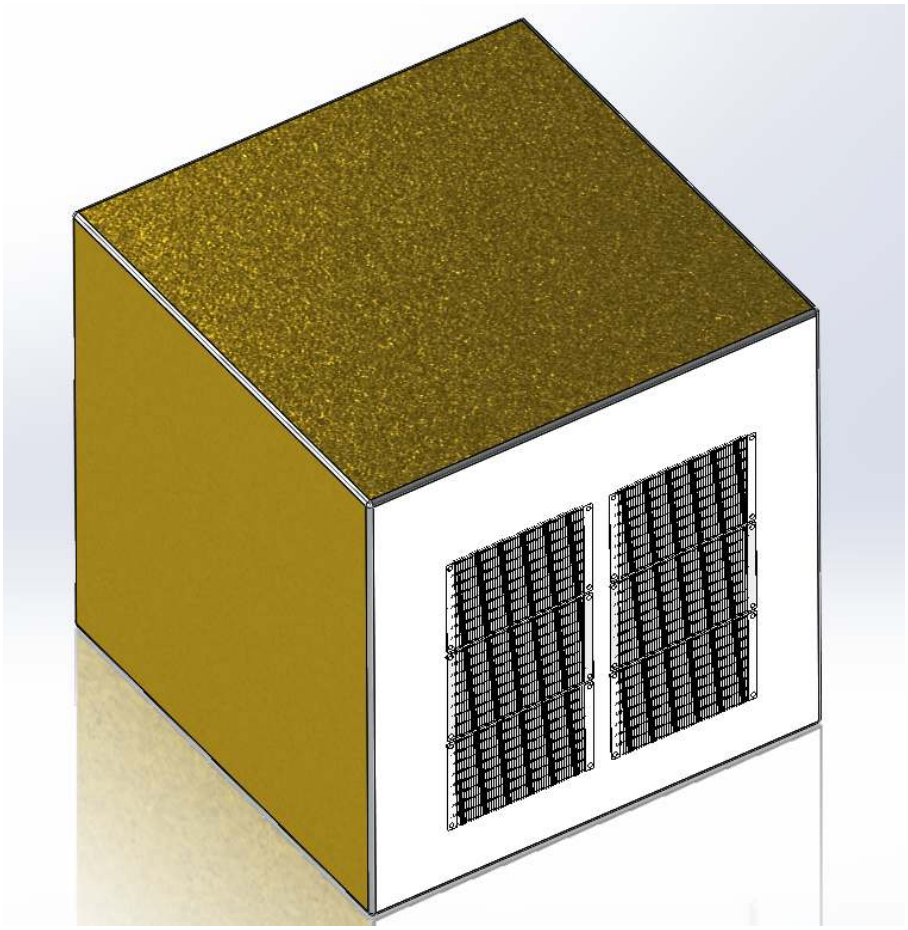


**Figure 24.** The CAD model of the cubic meter system with MLI on five faces and a Z93-coated radiator panel on one face.

The radiator is a passive thermal control component responsible for dissipating the heat transported by the loop heat pipe. The chosen design was constructed from aluminum 6061 and mounted on one side of the cubic payload. The radiator has an estimated mass of 6.36 kg and an exposed area of 0.4252 $m^2$. Its surface is coated in Z93 to improve its radiative performance further. The radiator CAD exhibited in Figure 25 includes two inlet and outlet pipes for integration with the LHP. These pipes enable the two-phase working fluid to circulate effectively between the payload's electronics and the radiator surface. Pairing these two components together allows the system to reject all 892 W of heat generated in the hot case. The radiator is a vital component of the thermal subsystem, as it continuously expels heat without using electrical power. However, while radiators are often used in spacecraft design, the lack of validation under Venus' harsh environmental conditions limits their Technology Readiness Level to TRL 6.

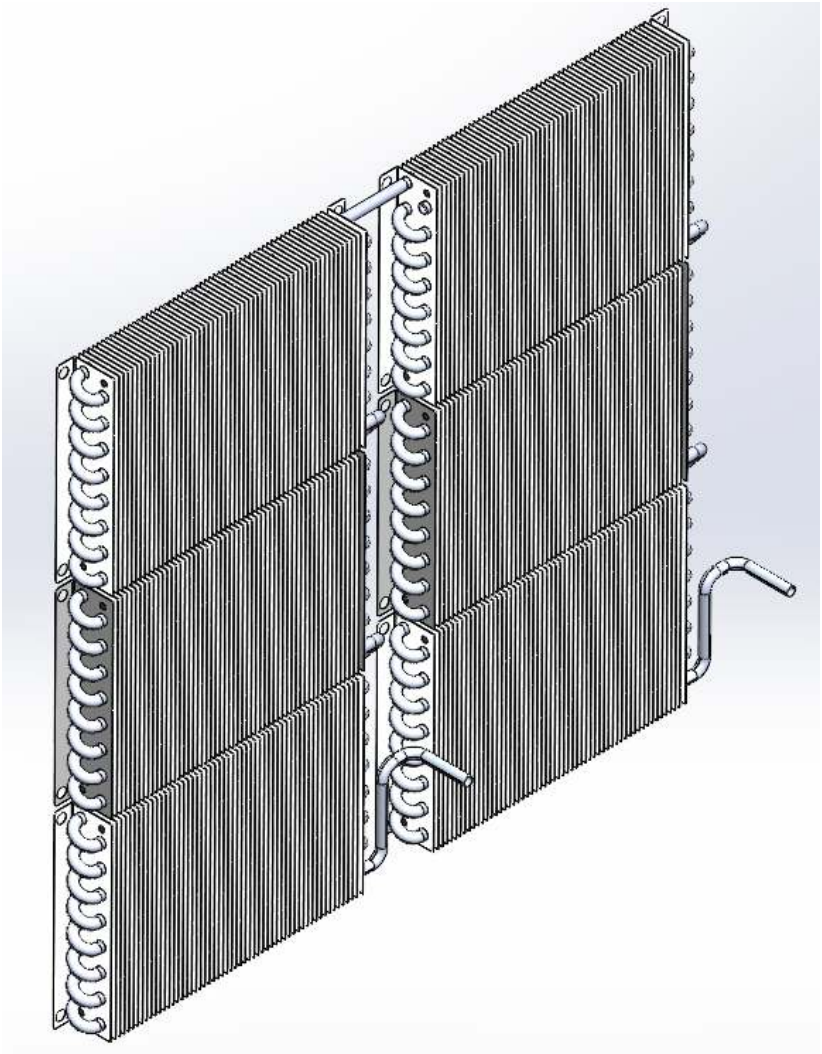


**Figure 25.** The CAD model of the Al 6061 radiator.

The loop heat pipe is the payload's primary cooling component. It is responsible for transporting excess heat from the scientific instruments and electronics to the radiator. Although technically a passive thermal component, LHPs operate just as efficiently as alternative active cooling components. It operates passively by leveraging capillary action within a wick structure to circulate as a working fluid (Ku, 1999). In other words, as internal components dissipate heat, the fluid in the LHP evaporates and travels toward the radiator where it then condenses once again and releases heat into the external environment. This phase-change cycle enables regulated heat transfer with zero power input.

Venera Nova's LHP will be sourced from Arquimea. Arquimea manufactures custom LHPs and has produced LHPs for prior space missions that have supported loads greater than 1000 W. The Venus-bound aerobot will require an LHP that, in partnership with the radiator, can reject 892 W of heat to ensure the payload's internal remains at 300 K. The LHP will occupy an estimated mass of 4 kg with a volume of 0.004 $m^3$ of the system's total physical constraints (Thayer et al., 2023). Its connection to the radiator is

facilitated by two inlet and outlet pipes, as shown in Figure 26. Though LHPs have been proven in spacecraft thermal control systems, they have not been tested in Venusian-like atmospheres. Therefore, earning a Technology Readiness Level of 6.

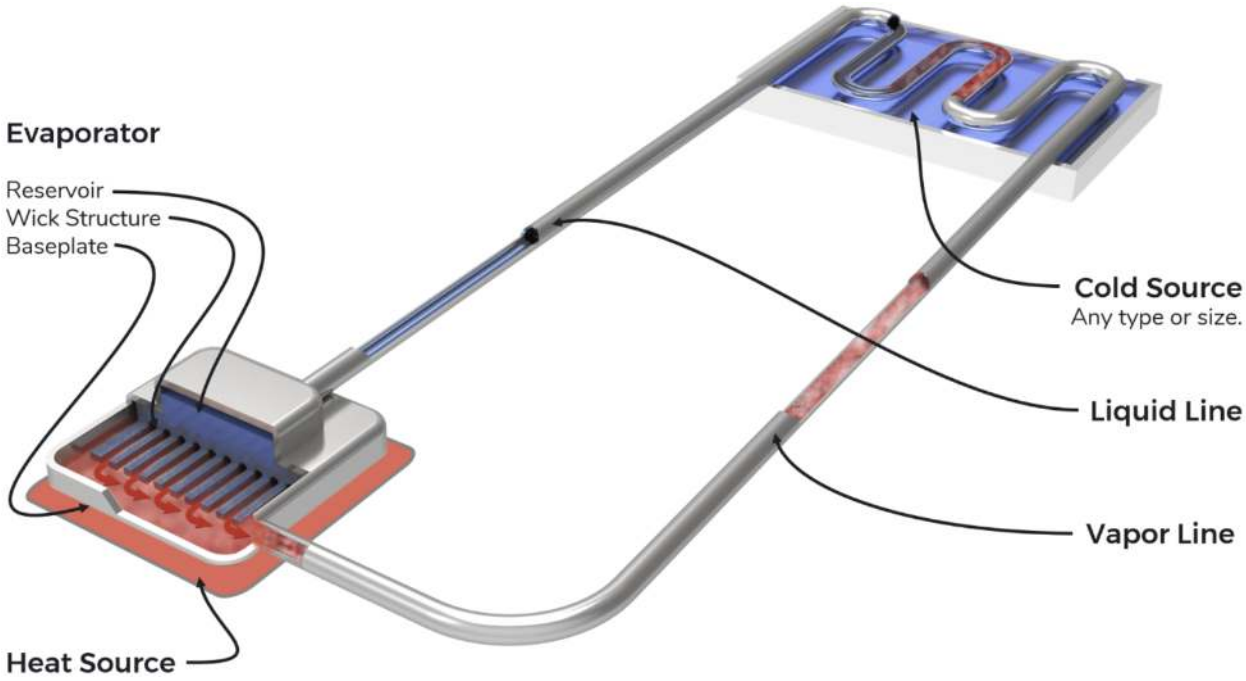



**Figure 26.** Phase-change cycle and components of a Loop Heat Pipe system (Calyos, 2025).

The All-Polyimide Thermofoil Heater (HAP6948-1) is a flexible heater manufactured by Minco that generates up to 170.9 W of heat and weighs only 3.18 g (0.00318 kg). To conserve power and operate within the thermal subsystem limits, the power subsystem will allocate a maximum of 162.7 W to the heater. This is more than sufficient, as the cold case requires a maximum of 147 W of supplemental heating to reach the system's target temperature of 300 K.

The flexible heater is mounted using a silicone pressure-sensitive adhesive, allowing it to securely stick to an internal structural surface (Minco, 2025). Two methods of integration were considered:

- to mount the heater on one of the internal titanium chassis walls
- to place the heater closer to or between the scientific instruments and electronics

The first integration method is the most optimal as it allows for heat to spread across the structure through conduction and gradually warm up the payload’s internals. Additionally, mounting the heater to the chassis wall avoids interference with other subsystem components or scientific instrumentation compartments, enabling a clean and more organized internal configuration. Due to its extensive use in aerospace systems, robust structural specifications, and operational temperatures ranging from 200 K to 450 K, conditions reflective of the Venusian atmosphere, the Thermofoil Heater has earned a Technology Readiness Level of 8.

A product drawing of the All-Polyimide Thermofoil Heater (HAP6948-1) is displayed in Figure 26, providing a visual reference of its dimensions, features, and layout as specified by Minco.

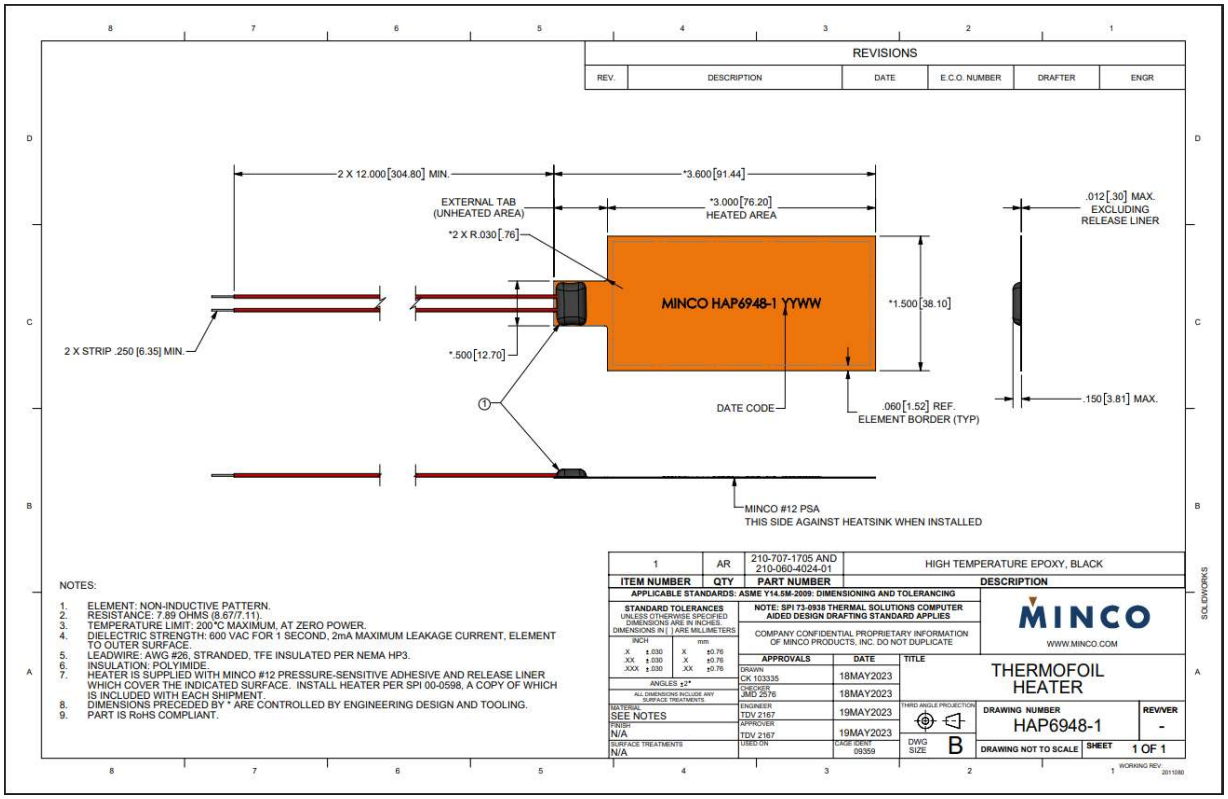



**Figure 27.** Engineering drawing of the All-Polyimide Thermofoil Heater (HAP6948-1) (Minco, 2023).

The thermal sub-assemblies form a functioning thermal system capable of maintaining the aerobot's payload at a steady-state temperature of 300 K in both hot and cold extremes. Passive components, such as Z93 white coating and MLI, reduce thermal load fluxes while the LHP and radiator passively remove heat from the hot case. In the cold case, the flexible heater activates when the temperature is near 10 K below the desired 300 K, drawing a minimal amount of power while heating the internal components. Although most of the thermal systems components are spaceflight-proven, the unique, harsh environment of Venus introduces uncertainties that the components have not been exposed to before. As a result, the Technology Readiness Level of the entire thermal subsystem is TRL 6, based on the lowest-rated component in the thermal configuration. The final component specifications are summarized in the following table, which lists each component's estimated mass, dimensions, power draw, and TRL.

**Thermal Subsystem Component Specifications**

| Thermal Subsystem Components | Mass (kg) | Dimensions (m^2) | Max Power (W) | TRL |
|---|---|---|---|---|

| | | | | |
|---|---|---|---|---|
| Z93 white coating | Negligible | 1 | 0 | 7 |
| MLI | 3.5 | 5 | 0 | 7 |
| Radiator | 6.36 | 0.4252 | 0 | 6 |
| Loop Heat Pipe | 4 | 0.16 | 0 | 6 |
| Polyimide Thermofoil Heater (HAP6948-1) | 0.00318 | 0.00256 | 150 | 8 |
| **Total** | **15.86** | **6.59** | **150** | **6** |

### 2.1.4.3 Thermal Control Subsystem Recovery and Redundancy Plans

Redundancy:

The thermal subsystem doesn't incorporate redundant hardware due to the use of highly reliable components and strict mass constraints. Passive components, such as Z93 white coating and multi-layer insulation, require no duplication as they contain no moving or failure-prone mechanisms. They have also proven to be reliable and durable in numerous space missions; their performance during Venera Nova's 48-hour mission on Venus will be consistent with prior space endeavors.

The thermal subsystem consists of three single-point failures: the loop heat pipe, the aluminum radiator, and the flexible heater. These are considered single-point failures, as duplicating them would exceed the allocated thermal mass constraint of 18 kg. This risk is mitigated by using highly credible components with previous space mission experience. For instance, the HAP6948-1 heater from Minco is space-rated and used routinely in the aerospace industry.

Limiting the thermal control system to one active thermal component emphasized simplicity and passive performance to minimize risk as much as possible. The passive radiator and LHP reject heat without any power input, making their operation fundamentally stable. Additionally, verification methods such as demonstrations, inspections, tests, and analyses completed during the developmental stages of the mission will validate steady-state performance under Venusian conditions, reducing the likelihood of unexpected thermal behaviors during the mission.

Recovery:

The thermal subsystem will rely on onboard autonomous software to maintain a net-zero heat load and respond to abnormal thermal behaviors. Temperature sensors placed across the electronics compartments will feed real-time data to the software, which will then compare observed temperatures to the safe operating range of 300 ± 10 K. If temperatures fall below the -10 K threshold, the software will activate the polyimide thermofoil heater, drawing up 147 W for the most extreme cold case. The thermal control system will operate passively for the hot case as the LHP relies on naturally occurring phase changes rather than power. Although the software will not be able to activate or improve the LHP's passive cooling, if the LHP is underperforming, the software will provoke the payload to enter an emergency cooling mode. In the emergency cooling mode, power and thermal loads will be rerouted to preserve the most essential instruments.

The combination of passive thermal controllers, highly reliable thermal components, and software-based monitoring systems ensures that the thermal subsystem is equipped to recover from thermal disturbances without requiring physical redundancy.

### 2.1.4.4 Thermal Control Subsystem Manufacturing and Procurement Plans

To guarantee that the thermal control subsystem meets the mission's performance, cost, and timeline constraints, passive and active thermal components were selected from a combination of aerospace-proven suppliers and commercial off-the-shelf manufacturers. When possible, vendors were chosen based on previous success in space applications with companies or organizations such as NASA. The TCS's components were decided based on material and thermal reliability, compatibility with the cubic meter payload, and lead times. The table below displays primary and backup suppliers, along with lead times based on past mission documentation and available manufacturing timelines.

Manufacturing and Procurement Plans for Thermal Control Components

| Component | Supplier | Backup Supplier | Estimated Lead Times (Weeks) |
|---|---|---|---|
| Z93 - White Coating | AZ Technology | Advanced Coatings | 4 to 8 |
| Multi-Layer Insulation | Northrop Grumman | Aerospace Fabrication and Materials | 10 to 16 |
| Loop Heat Pipe (LHP) | Arquimea | Mott | 16 to 24 |
| Radiator | Arquimea | Aerospace Fabrication and Materials | 16 to 28 |
| All-Polyimide Thermofoil Heater - HAP6948-1 | Minco | Dwyer Omega | 3 |

**Component**: Z93 - White Coating
AZ Technology is the original developer of Z93, a white thermal control coating with extensive space flight success, making it the most reliable and proven source (AZ Technology, 2025). The estimated lead times of 4-8 weeks include both preparation and curing time for the coating. A possible backup supplier is Advanced Coatings. Although not as proven in spaceflight, the company can produce high-quality Parylene thermal coatings (Advanced Coating, 2025).

**Component**: Multi-Layer Insulation
Northrop Grumman has over 45 years of experience in developing, fabricating, and integrating thermal subsystems. The company promises to produce high-performance MLI blankets tailored to each customer's needs (Northrop Grumman, 2025). The estimated lead time for this component is 10-16 weeks to account for custom blanket fabrication. As a backup, Aerospace Fabrication and Materials offers customized MLI blankets and has even supplied hardware for previous NASA missions (Aerospace Fabrication and Materials, 2025).

**Component**: Loop Heat Pipe (LHP)
Arquimea specializes in high-performance thermal control systems, including LHPs, and offers reliable design and testing services for space applications (Arquimea, 2025). About 16-24 weeks are required as each unit is built based on power lever, routing, and radiator integration. A potential backup is Mott, a U.S.-based company with custom-designed LHPs and previous experience in thermal management of space vehicles (Mott Corporation, 2025).

**Component**: Radiator
Arquimea also manufactures static and deployable radiators that are integral to the functionality of the LHP. The ability to tailor the radiator to the LHP design will optimize heat rejection for the cubic meter system (Arquimea, 2025). 16-28 weeks will be needed to complete both thermal analysis and LHP interface design. As a backup, Aerospace Fabrication and Materials offers custom, adaptable radiator solutions based on the system's LHP configuration (Aerospace Fabrication and Materials, 2025).

**Component:** All-Polyimide Thermofoil Heater - HAP6948-1
Minco is a trusted provider of aerospace-grade heating components. The selected flexible heater model meets the thermal subsystem's 162.7 W power constraint while maintaining high thermal uniformity. The all-polyimide thermofoil heaters are adaptable to extreme environmental conditions and have been validated for use in space applications. Lead time for this product is approximately 3 weeks (Minco, 2025). Dwyer Omega also offers flexible heaters with similar electrical and thermal performance values (Dwyer Omega, 2025). If an issue were to arise with Minco's availability, Dwyer Omega would serve as a reliable alternative with minimal impact on the payload's performance or delivery schedule.

The selected suppliers provide a strong blend of product reliability, mission credibility, and delivery capability that aligns with the mission's integration and testing schedule. Due to customization needs, lead times range from 3 to 28 weeks. Still, with appropriate backup suppliers identified for each component, the thermal subsystem is on track to meet the milestones outlined in the project schedule.

### 2.1.4.5 Thermal Control Subsystem Verification Plans

Specific verification methods have been defined as described in the following table for each requirement as part of the thermal control subsystem.

| Req # | Requirement Summary | Verification Method | Rationale for Method | Preliminary Verification Plan |
|---|---|---|---|---|
| TH-1 | The total mass of all thermal control components shall not exceed 18 kg. | Inspection | The inspection ensures weight constraints are taken into account for. | Inspect materials used and check measurements are appropriate. |
| TH-2 | The TCS shall maintain all thermal control components within the system's dimensional constraints. | Inspection | The inspection ensures the system will not overheat or freeze. | Inspect properties of the TCS, perform regular maintenance checks. |
| TH-3 | The costs of the TCS shall not exceed $200 million. | Analysis | The analysis ensures proper allocation of funds and is not wasted. | Analyze properties and check that the amount spent is valid/reasonable. |
| TH-4 | The TCS shall withstand environmental temperatures ranging from 225.8 K to 344.7 K (at altitudes of 70 to 50 km) during the mission duration. | Test & Analysis | The test will confirm the TCS works and is able to counteract the harsh environment. Analysis will confirm by checking trends in numbers | Test TCS under similar environmental conditions. Look at trends to analyze and confirm the system has no problems. |

### 2.1.5 Payload Subsystem Overview

To accurately measure qualities affected by Venus' lack of magnetic field, superrotation phenomena, and potential for life, the spacecraft will consist of three scientific instruments. All of these instruments are capable of measuring atmospheric characteristics pertinent to the Aerobot's atmospheric location of 50-70 km above Venus' surface.

One instrument included in the payload subsystem is the Venus Monitoring Camera (VMC), which will assist in determining the influence of Venus' superrotation on its surface and atmosphere. To accomplish this, the VMC will capture images of Venus's clouds every 10 seconds in multiple wavelength ranges. The VMC consists of four different lenses, or optics, that transmit light to a single charge-coupled device (CCD). These four optics are capable of capturing spectral data in different wavelength ranges, spanning the ultraviolet to near-infrared wavelength range. The CCD will subsequently capture image data from whichever optic is currently being used, and transmit that data to a readout electronic (CRE) for processing. A digital processing unit (DPU) is also included within the instrument, which will store processed data from the CRE into its 1 gigabit mass memory (Markieciwz et al., n.d.). Following this operation, all captured images will be relayed back to Earth using the interface provided by the CDH subsystem. This collection of images will allow scientists to track cloud movements throughout the Aerobot's 48-hour descent through Venus' atmosphere. By analyzing these movements, scientists will be able to calculate how wind speeds vary with atmospheric height. If there are periodic patterns in how winds change in speed, it is likely that kelvin waves are present at specific atmospheric levels and directly influence Venus' superrotation. Furthermore, the VMC will map Venus' topography in relation to varying wind speeds by "sounding the surface through [its] 1 μm transparency window" (Markiewicz et al., n.d). The VMC's transparency window refers to a specific range of near-infrared wavelengths that can penetrate Venus' thick cloud cover, allowing the camera to detect light emitted or reflected from the surface. This makes it possible to assess the optical depth of the planet's atmospheric cloud layer. By measuring these specific atmospheric qualities, the intensity of Venus' superrotation and cloud cover can be assessed as it changes with atmospheric height. The TRL for the VMC is evaluated to be 7 since it successfully captured images of Venus and relayed them routinely back to Earth throughout the Venus Express' 8 year mission (European Space Agency, n.d.).

The two other instruments included on the Aerobot are a part of "a suite of three spectrometers in the UV and IR range… dedicated to the study of the atmosphere of Venus from ground level to the outermost hydrogen corona at more than 40,000 km" (Bertaux et al., 2007). This suite of spectrometers is known as SPICAV, whose Solar

Occultation Infrared Spectrometer (SOIR) and ultraviolet spectrometer will gather atmospheric spectroscopy data in different wavelength ranges to complete the remaining scientific objectives of the Venera Nova mission.

The ultraviolet (UV) spectrometer will determine how Venus' lack of magnetic field affects the planet's atmospheric properties. Venus' lacking magnetic field results in an increase in cosmic ray flux, which subsequently increases the amount of solar radiation interfacing with Venus' atmosphere. The UV spectrometer is a dual-mirrored instrument, consisting of a high-resolution grating and mechanical slit system capable of measuring specific quantities such as chemical composition, ionization rates, and escape rates of particles in Venus' upper atmosphere. The instrument first collects light flux "by an off-axis parabolic mirror, which reflects the light toward the entrance of the spectrometer" (Bertaux et al., 2007). The grating of the spectrometer disperses light perpendicular to the slit. From there, an image intensifier detects light and transmits it to a CCD. Image data is then processed by integrated reading electronics, and then is collected and sent back to Earth using the CDH subsystem. Additionally, "the CCD detector may be cooled by a Peltier cooler to decrease its temperature and associated dark current" (Bertaux et al., 2007). An image of SPICAV's UV spectrometer is displayed below:

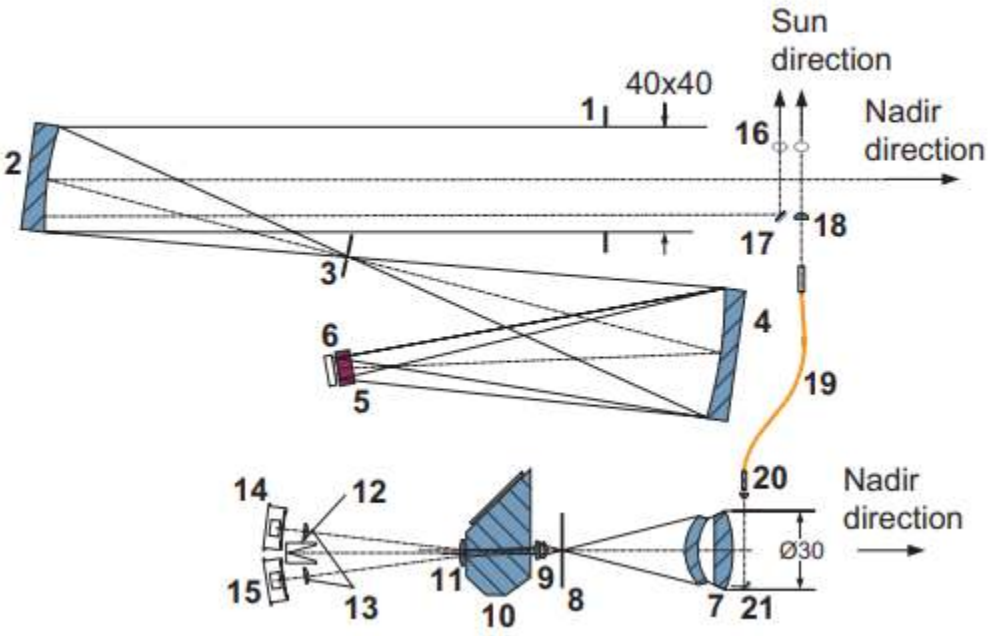



*Figure 28: Optical scheme of the UV and IR channels of SPICAV (Bertaux et al., 2007). Light enters the spectrometer from a target positioned directly below the parabolic mirror (known as the "Nadir direction"), is subsequently reflected by grating, and received by a CCD.*

This collection of spectra will assist in measuring the magnitude of cosmic ray flux present on Venus, which affects physical mechanisms such as acceleration and propagation (Paolo, 2018).

The third scientific instrument included on the Aerobot is the SOIR, used to help determine the probability of life existing within Venus' upper atmosphere. Due to the harsh weather patterns of Venus' surface, it is unlikely that life could survive on the ground; however, it is possible that Venus' upper cloud deck has conditions appropriate for fostering life. In 2020, scientists announced that they had found a potential biosignature known as phosphine in Venus' upper atmosphere, which is a strong indicator of biological processes occurring within the area (Howell, 2023). Phosphine has since been detected numerous times since its initial discovery, even in Venus' lower atmosphere (Lea, 2023). To confirm the presence and specific locations of phosphine, the SOIR will use solar occultations to search for phosphine wavelength peaks throughout Venus' main and lower atmospheric layers. The SOIR has one channel operating in the UV wavelength range and two channels measuring in the infrared range, with a diffraction grating optimized for high incidence angles (Ristic, 2009). An image of the SOIR's mechanical composition is displayed below:

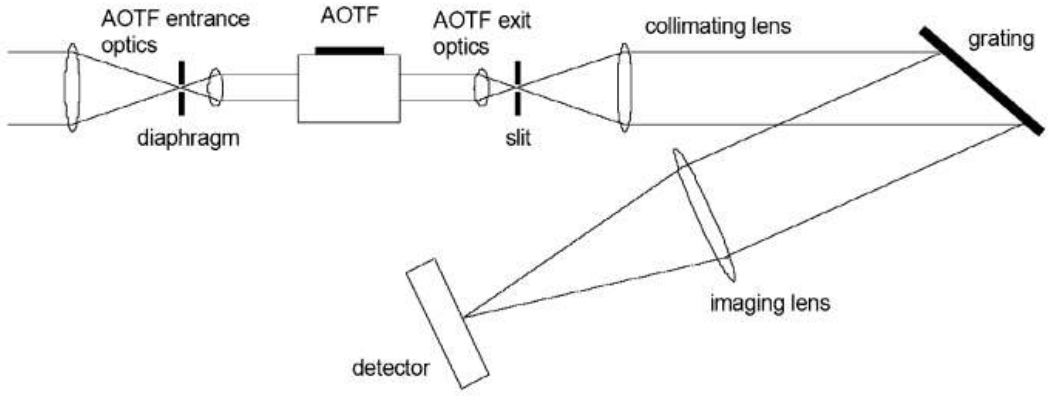



*Figure 29: Simplified optical scheme of SOIR (Bertaux et al., 2007), showing the SOIR's optical operations.*

Light first enters through an acousto-optical turntable filter (AOTF), which "is an electro-optical device that functions as an electronically tunable excitation filter to simultaneously modulate the intensity and wavelength of multiple laser lines from one or more sources" (Fellers et al., n.d.). The SOIR's entrance optics reduce the size of the incoming light to be compatible with the AOTF. Then, the AOTF processes a select part of the incoming light spectrum and sends it to the spectrometer's grating. The grating diffracts the incoming light and sends it to the AOTF's detector to be converted into

data. Overall, due to its ability to analyze spectra at particularly large angles, the SOIR will enable the Venera Nova mission to search for signs of life throughout Venus' upper—and for a short time, lower—atmospheric layer. The TRL for the entire SPICAV instrumentation suite is evaluated to be 7 since it was integrated into the Venus Express mission launched in 2006 and successfully relayed spectral data until the end of the mission in 2014. Throughout the Venus Express mission, the SOIR and UV spectrometer consistently delivered high resolution spectra of Venus' mesosphere (Royal Belgian Institute for Aeronomy, n.d.).

For these instruments to successfully operate and complete their designated scientific objectives, the Aerobot must be able to withstand the harsh temperatures and winds present throughout Venus' atmosphere. To mitigate high temperatures, the thermal control subsystem on the Aerobot will ensure that all scientific components and interfaces maintain a steady temperature state of 300 Kelvin at all times—this will be accomplished through the implementation of relevant thermal control components such as radiators and a loop heat pipe. Additionally, the mechanical subsystem will ensure that all scientific equipment remains secured within the Aerobot amongst wind vibrations. Specifically, harnesses shall be implemented within the storage space as well as compartments specifically designed to house science instrumentation securely.

A table displaying the TRL, mass, dimensions, and maximum power draw of the selected instruments for the Venera Nova mission is displayed below:

| Instrument | Maximum Power Draw (W) | Mass (kg) | Dimensions (m) | TRL |
|---|---|---|---|---|
| Ultraviolet Spectrometer | 69 | 13.915 | 0.665 x 0.542 x 0.420 | 7 |
| Solar Occultation Infrared Spectrometer (SOIR) | | | | |
| Venus Monitoring Camera (VMC) | 10.4 | 1.50 | 0.457 x 0.152 x 0.152 | 7 |

*Table including the specifications of all scientific instruments on the Aerobot. The values for the SOIR and ultraviolet spectrometer are shared, since they are within the same instrumentation suite. The maximum power draw and dimensions for the VMC were notably estimated due to instrumentation information being undisclosed for public use (section [A.1] in the appendix describes how estimations were calculated).*

Based on these values, 15.42 kg of the mass for this mission will be allocated to the payload subsystem. The overall TRL of this subsystem is 7 since all instruments successfully collected and relayed data aboard the Venus Express spacecraft, a previous mission to Venus.

### 2.1.5.1 Science Instrumentation Requirements:

Relevant science instrumentation requirements were derived based on system constraints, customer constraints, and key physical parameters. High-level operational requirements act as parent requirements, while more detailed ones that support and fulfill them are considered child requirements.

| Req # | Requirement | Rationale | Parent Req | Child Req | Verification method | Req met? |
|---|---|---|---|---|---|---|
| PAY-1 | Science instruments shall weigh no more than 30 kg. | Payload cannot be allocated any more mass or the other subsystems may not function appropriately. | SYS-001 | | Inspection | Met |
| PAY-2 | The VMC will track cloud motion every meter as the Aerobot begins descending 70 km above Venus' surface. | Vital towards determining the influence of Venus' superrotation on its surface and atmosphere. | SCI-003 | PAY-2.1 | Test & Demonstration | Met |
| PAY-3 | The VMC will measure wind speeds at every 1 meter. | Vital towards determining the influence of Venus' superrotation on its surface and atmosphere. | SCI-003 | PAY-3.1 | Test & Analysis | Met |
| PAY-4 | The VMC will record fluctuations in cloud brightness. | Vital towards determining the influence of Venus' superrotation on its surface and atmosphere. | SCI-003 | | Test & Analysis | Met |

| | | | | | | |
|---|---|---|---|---|---|---|
| PAY-5 | The UV spectrometer shall collect atmospheric spectra in the 100-400 nm range over a 1 km range every hour. | Collecting atmospheric spectra is vital towards measuring Venus' atmospheric structure as it is affected by cosmic ray flux. | 0.1 | PAY-5.1<br>PAY-5.2 | Test & Analysis | Met |
| PAY-6 | The SOIR shall detect phosphine peaks between 190 and 220 nm. | Detecting phosphine is vital towards evaluating the potential for life to exist within Venus' atmosphere. | 0.3 | PAY-6.1,<br>PAY-6.2 | Test & Analysis | Met |
| PAY-7 | The payload shall fall within dimensional constraints before deployment (1m x 1m x 1m) | Dimension Constraints | SYS-002 | | Inspection | Met |
| PAY-8 | The payload shall not exceed a cost of $70,000,000 | Cost Constraints | SYS-004 | | Analysis | Met |
| PAY-9 | All science operations shall be completed within 48 hours after deployment of EDI system. | After 48 hours the SS will uncontrollably fall and crash into Venus' surface, discontinuing science operations. | SYS-007 | | Demonstration | Met |
| PAY-2.1 | The VMC shall operate at a sensitivity of 0.3 to 1.0 μm at a spatial | Necessary performance requirement. | PAY-2 | | Test & Analysis | Met |

| | | | | | | |
|---|---|---|---|---|---|---|
| | resolution of 1.5 km. | | | | | |
| PAY-3.1 | The VMC shall operate within a wavelength range of 365-1000 nm. | Necessary performance requirement. | PAY-3 | | Test & Analysis | Met |
| PAY-5.1 | The UV spectrometer shall operate within a wavelength range of 118-320 nm. | Necessary performance requirement. | PAY-5 | | Test & Analysis | Met |
| PAY-5.2 | The UV spectrometer shall operate at a sensitivity of 200-300 nm at a spatial resolution of 100 km. | Necessary performance requirement. | PAY-5 | | Test & Analysis | Met |
| PAY-6.1 | SOIR shall operate within a wavelength range of 2.2 to 4.3 μm. | Necessary performance requirement. | PAY-6 | | Test & Analysis | Met |
| PAY-6.2 | SOIR shall operate at a sensitivity of 2257–4430 cm−1. | Necessary performance requirement. | PAY-6 | | Test & Analysis | Met |

### 2.1.5.2 Payload Subsystem Recovery and Redundancy Plans

A matrix describing potential redundancy in instrumentation is displayed below:

| Science Measurement Requirements | | Baseline Performance Requirements | | Instrument | | |
|---|---|---|---|---|---|---|
| Physical Parameters | Observables | | | VMC | UV Spectrometer | SOIR |
| Track cloud motion, brightness variations, and Kelvin waves | Capture images of Venus's clouds every 10 seconds in UV, visible, and near-infrared wavelengths. Collect wind speed data by tracking cloud motion every 10 seconds. Record periodic fluctuations in cloud brightness over the course of 48 hours. | Wavelength range: | 365 - 1000 nm | ● | ○ | ◔ |
| | | Integration time: | 0.5 ms - 30 s | | | |
| | | Sensitivity: | 0.3 to 1.0 μm | | | |
| | | Spatial resolution: | 0.2 - 45 km | | | |
| Measure the composition, ionization, and the escape rates of particles and solar photons in the upper atmosphere. | Collect ionized species such as O+, H+, CO2+ in the UV range (118-320 nm) over a 1 km range every hour. | Wavelength range: | 118 - 320 nm | ○ | ● | ○ |
| | | Integration time | 80 - 450 ms at 1 km | | | |
| | | Sensitivity: | 200 - 300 nm | | | |
| | | Spatial resolution: | 100 km | | | |
| Confirm presence of phosphine gas in the upper atmosphere. | Detect phosphine peaks between 190 and 220 nm. | Wavelength range: | 2.2 to 4.3 μm | ◔ | ○ | ● |
| | | Integration time | 5 s | | | |
| | | Sensitivity: | 2257–4430 cm−1 | | | |
| | | Spatial resolution: | 0.025-cm−1 | | | |

○ = *Does not contribute to objectives*
◔ = *Does not meet baseline*
◑ = *Partial contribution to objectives*
● = *Instrument can fully meet objectives*

Instrumentation was evaluated to minimally overlap with one another, ensuring that all instruments on the payload are essential to the Venera Nova mission.

In regards to recovery efforts, the payload subsystem relies heavily upon other subsystems for protection against the intense temperatures, wind speeds, and heat within Venus' atmosphere. While analytical and lightweight systems such as DPUs will have a backup system, there are only three instruments to be included in the payload due to mass constraints—no duplicates. For this reason, some components will need to be documented as SFPs. For instance, if the spectrometers stop functioning, the system will be unable to gather atmospheric data. To recover from potential system errors, instruments shall troubleshoot autonomously to diagnose the issue. If an instrument starts malfunctioning, the integrated software will be notified and start analyzing the system for errors immediately.

### 2.1.5.3 Payload Subsystem Manufacturing and Procurement Plans

| Component | Supplier | Back-Up | Estimated Lead Times |
|---|---|---|---|
| SOIR (SPICAV) | Belgian Institute for Space Aeronomy | OIP Space Instruments | 6 months |
| UV Spectrometer (SPICAV) | Service d'Aeronomie | Belgian Institute for Space Aeronomy | 6 months |
| VMC | Max Planck Institute for Solar System Research | Institute of Planetary Research at the German Aerospace Center | 6 months |

**Component**: SOIR (SPICAV)
6 months are required for this component to complete both research and development. The supplier chosen for SOIR is Belgian Institute for Space Aeronomy, led the Venus Express missions and oversaw the design and development of SOIR specifically. OIP Space Instruments served as a subcontractor and was responsible for developing the optical system of the spectrometer.

**Components**: UV Spectrometer (SPICAV)
6 months are required for this component to complete both research and development.

The supplier chosen for the UV Spectrometer is Service d'Aeronomie as it played a large role in the development of SPICAM. The Belgian Institute for Space Aeronomy also contributed to UV spectrometer’s development.

**Components**: VMC
6 months are required for this component to complete both research and development. The supplier chosen for VMC is Max Planck Institute for Solar System Research as it led the development of the instrument initially. Institute of Planetary Research at the German Aerospace Center also had contributed to the design and development of the instrument.

### 2.1.5.4 Payload Subsystem Verification Plans

To ensure that the payload subsystem is being developed appropriately, specific verification methods have been defined in the following table:

| Req # | Requirement Summary | Verification Method | Rationale for Method | Preliminary Verification Plan |
|---|---|---|---|---|
| PAY-1 | Science instruments shall weigh no more than 30 kg. | Inspection | Weight can be determined through the use of drawings and/or vendor specifications. | Prior to manufacturing, scientific instruments will be developed and visually displayed in numerous computer-aided drawings (CAD). Numerous characteristics such as the payload weight and dimensions can be discerned through such drawings. If the payload weight exceeds 30 kg, developers must reevaluate materials used or alter the dimensions of the instruments. |
| PAY-2 | The VMC will track cloud motion every meter as the Aerobot begins descending 70 km above Venus' surface. | Test & Demonstration | The VMC will be tested extensively within an environment that will simulate Venus' atmospheric conditions. It is necessary for the VMC to be manufactured prior to this process so its functions can be physically tested. | Instrumentation shall be tested within pressure vessels to simulate Venus' atmospheric pressure and extreme temperatures. |
| PAY-3 | The VMC will measure wind speeds at every 1 meter. | Test & Analysis | The VMC will be tested extensively within an environment that will simulate Venus' atmospheric conditions. | The VMC will be tested in a windy environment—either within a testing facility or outdoors—to test if it will be able to measure wind speeds at different altitudes. The data |

| | | | | |
|---|---|---|---|---|
| | | | It is necessary for the VMC to be manufactured prior to this process so its functions can be physically tested. | collected by VMC from this test will subsequently be evaluated for effectiveness. |
| PAY-4 | The VMC will record fluctuations in cloud brightness. | Test & Analysis | The VMC will be tested extensively within an environment that will simulate Venus' atmospheric conditions. It is necessary for the VMC to be manufactured prior to this process so its functions can be physically tested. | The VMC will be tested within a cloudy terrain to ensure it will be able to record fluctuations in cloud brightness. The data collected by VMC from this test will subsequently be evaluated for effectiveness. |
| PAY-5 | The UV spectrometer shall collect atmospheric spectra in the 100-400 nm range over a 1 km range every hour. | Test & Analysis | The UV spectrometer will be tested extensively within an environment that will simulate Venus' atmospheric conditions. It is necessary for the UV spectrometer to be manufactured prior to this process so its functions can be physically tested. | The UV spectrometer will be tested at different atmospheric altitudes outdoors to ensure that it can collect atmospheric spectra in the 100-400 nm range over a 1 km range every hour. The data collected by the UV spectrometer from this test will subsequently be evaluated for effectiveness. |
| PAY-6 | The SOIR shall detect phosphine peaks between 190 and 220 nm. | Test & Analysis | The SOIR will be tested extensively within an environment that will simulate Venus' atmospheric conditions. It is necessary for the SOIR to be manufactured prior to this process so its functions can be | Lab assistants will either develop or obtain a small sample of phosphine gas and inject it into an isolated testing chamber. The SOIR will then be placed within the chamber and be tested for phosphine peak detection. The data returned by the SOIR will subsequently be evaluated for effectiveness. |

| | | | | |
|---|---|---|---|---|
| | | | physically tested. | |
| PAY-7 | The payload shall fall within dimensional constraints before deployment (1m x 1m x 1m) | Inspection | Payload dimensions can be determined through the use of drawings and/or vendor specifications. | Prior to manufacturing, scientific instruments will be developed and visually displayed in numerous computer-aided drawings (CAD). Numerous characteristics such as the payload weight and dimensions can be discerned through such drawings. If the payload dimensions exceed 1m x 1m x 1m, developers must alter the dimensions of the instruments. |
| PAY-8 | The payload shall not exceed a cost of $70,000,000 | Analysis | Payload costs can be determined through the utilization of budgeting templates, vendor quotes, and/or appropriate estimation techniques such as the MCCET. | Prior to manufacturing, vendors will be asked for a pricing quote and materials will be selected by the science team. Payload costs will be monitored and managed by the programmatics team. Margin and inflation shall also be accounted for when finalizing payload costs. |
| PAY-9 | All science operations shall be completed within 48 hours after deployment of EDI system. | Demonstration | All scientific instruments will be tested extensively within an environment that will simulate Venus' atmospheric conditions; however, this requirement cannot be fully tested because the system shall not be integrated with other contracted subsystems prior to launch date. | Test all scientific instruments within simulated environments for at least 48 hours or more. Ensure the data collected is accurate, and appropriately prepare the subsystem for integration with other subsystems. |
| PAY-2. | The VMC shall | Test & | The VMC will be tested | Test the VMC within a pressure chamber and |

| | | | | |
|---|---|---|---|---|
| 1 | operate at a sensitivity of 0.3 to 1.0 μm at a spatial resolution of 1.5 km. | Analysis | extensively within an environment that will simulate Venus' atmospheric conditions. It is necessary for the VMC to be manufactured prior to this process so its functions can be physically tested. | outdoors to ensure that it can image targets in the nadir direction at the specified sensitivity values. Perform this test simultaneously with the PAY-3 and PAY-4 requirements tests. |
| PAY-3.1 | The VMC shall operate within a wavelength range of 365-1000 nm. | Test & Analysis | The VMC will be tested extensively within an environment that will simulate Venus' atmospheric conditions. It is necessary for the VMC to be manufactured prior to this process so its functions can be physically tested. | Test the VMC within a pressure chamber and outdoors to ensure that all four of its lenses operate within their designated wavelength ranges. Perform this test simultaneously with the PAY-3 and PAY-4 requirements tests. |
| PAY-5.1 | The UV spectrometer shall operate within a wavelength range of 118-320 nm. | Test & Analysis | The UV spectrometer will be tested extensively within an environment that will simulate Venus' atmospheric conditions. It is necessary for the UV spectrometer to be manufactured prior to this process so its functions can be physically tested. | Test the UV spectrometer outdoors to ensure that it can collect spectra within its designated wavelength range. Perform this test simultaneously with the PAY-5 requirement test. |
| PAY-5.2 | The UV spectrometer shall operate at a sensitivity of | Test & Analysis | The UV spectrometer will be tested extensively within an environment that will simulate Venus' atmospheric conditions. | Test the UV spectrometer outdoors to ensure that it can collect spectra at its specified sensitivity values. Perform this test simultaneously with the PAY-5 requirement |

| | | | | |
|---|---|---|---|---|
| | 200-300 nm at a spatial resolution of 100 km. | | It is necessary for the UV spectrometer to be manufactured prior to this process so its functions can be physically tested. | test. |
| PAY-6.1 | SOIR shall operate within a wavelength range of 2.2 to 4.3 μm. | Test & Analysis | The SOIR will be tested extensively within an environment that will simulate Venus' atmospheric conditions. It is necessary for the SOIR to be manufactured prior to this process so its functions can be physically tested. | Test the SOIR within an isolated testing chamber to ensure that it can collect phosphine peaks within its designated wavelength range. Perform this test simultaneously with the PAY-6 requirement test. |
| PAY-6.2 | SOIR shall operate at a sensitivity of 2257–4430 cm−1. | Test & Analysis | The SOIR will be tested extensively within an environment that will simulate Venus' atmospheric conditions. It is necessary for the SOIR to be manufactured prior to this process so its functions can be physically tested. | Test the SOIR within an isolated testing chamber to ensure that it can collect phosphine peaks within its designated sensitivity range. Perform this test simultaneously with the PAY-6 requirement test. |

This operational framework ensures that the payload subsystem can accomplish the scientific objectives of the Venera Nova Mission.

## 2.2 Interface Control

### N^2 Chart

The following is an N^2 diagram indicating the data flows, integrations, and data flows between the key components and functions of the Aerobot system. Each key subsystem is positioned along the diagonal of the chart, where the matrix itself highlights the dependencies and high-level interconnections needed for full functioning of the Aerobot system.

| | | | | | |
|---|---|---|---|---|---|
| Inputs from External | Mechanical, Mounting to Launch + Supporting Descent | | Data Commands from Earth (Ground Control) | | |
| | ↓ | | ↓ | | |
| Spacecraft Interfaces | | | | | |
| Internal Interfaces | Spacecraft Bus Structure + Framework | Mechanical Structure and Interface | Mechanical Protection and Stability | Mechanical Interface | Mechanical Stability |
| | | Payload System | Data from Instruments into OBC and Memory (for processing) | | |
| | | Commands to Instruments for Data Collection | CDH System | Commands for Active Cooling System | Commands for Power Distribution |

| | Thermal Protection | Thermal Control (Cooling Systems) | Thermal and Temperature Control | Thermal System | Thermal Cooling System |
|---|---|---|---|---|---|
| | Power (145 Watts) | Power (181.4 Watts) | Power (Watts) | Power (150 Watts) | Power System |
| ↓ | | | ↓ | | |
| Outputs to External | | | Processed Instrument data sent back to ground station | | |

Figure 30: N^2 diagram illustrating the functional interfaces and data flows between key components and subsystems of the Aerobot Subsystem.

## Block Diagram

The following is a block diagram illustrating the major signal pathways and connections between a lower-level breakdown of each subsystem within the Aerobot system. Each block represents a specific subsystem, while arrows indicate the flow of information, power, and control signals, highlighting functional relationships and integration points.

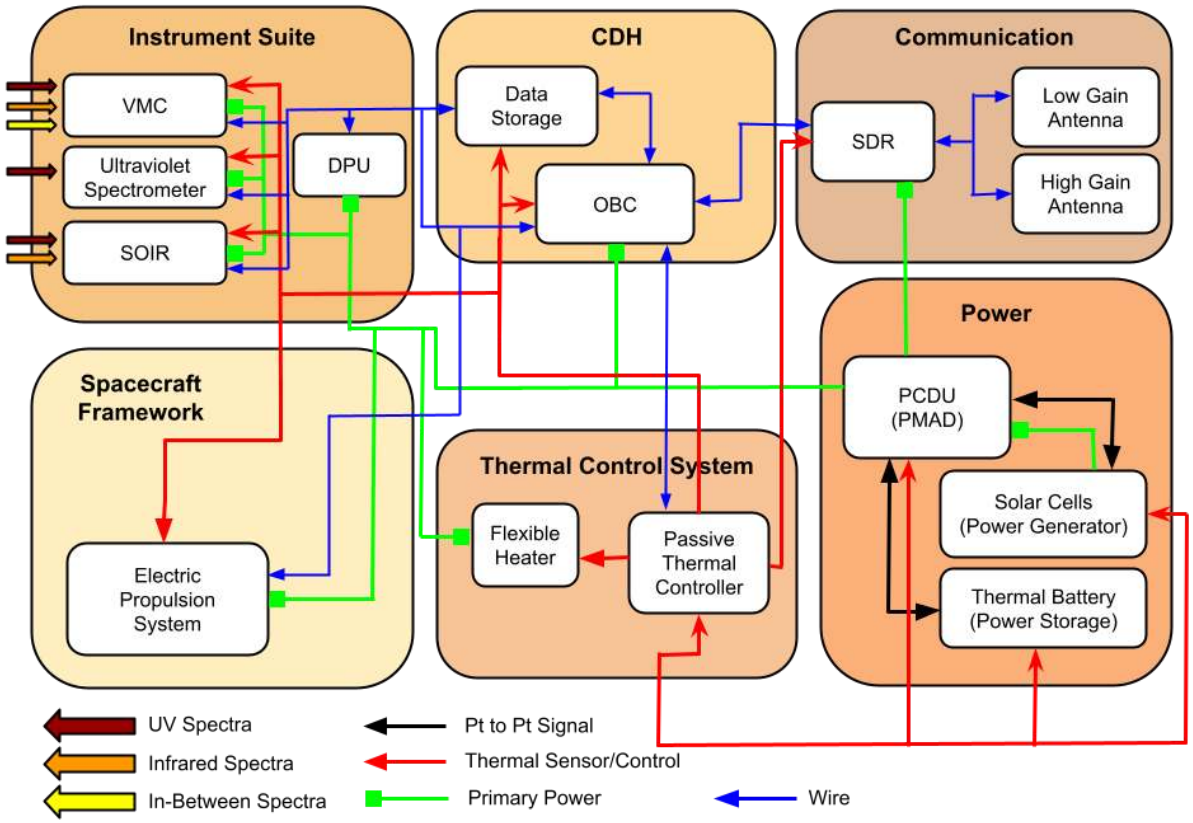



Figure 31: Block diagram depicting the major components and signal pathways within the Aerobot system.

**Interfacing Subsystem Relationships**

Each subsystem provides crucial value to supporting the scientific objectives of the Venera Nova mission. The mechanical subsystem provides the structural framework that houses and protects each subsystem, including the mechanical interfaces to attach components, ensure stability and protection, and maintain structural integrity for both the instruments and the power supplying and thermal maintaining components. Each subsystem relies on the mechanical subsystem to remain intact while launching and descending through the harsh environment and atmosphere of Venus, ensuring that each component of the Aerobot system can withstand the extreme thermal, pressure, and atmospheric conditions and still function as expected when science material is collected.

The payload system lies at the heart of the mission, allowing the science objectives to be explored and analyzed directly within Venus' environment. The payload system allows data to be collected from the three main instruments and then transmitted to the CDH system for processing and then to be stored. Further, commands from Earth (ground system) will be executed by the on-board computed, initiating data collection and specific analysis by the instruments.

The CDH subsystem, serving as the brain of the Aerobot system, handles the commands and decisions regarding all additional subsystems. This includes managing the commands for the thermal control system (initiating or stopping active cooling) in order to maintain stable, operable temperatures across the system as well as proper, efficient power distribution as required for each subsystem. As mentioned the CDH executes and inputs the commands required for the instruments to begin data collection.

The thermal system is crucial for providing the components that maintain thermal protection and control across all the subsystems. Each subsystem requires an environment with stable, operable temperatures to be able to function to their full capacity, and thus the thermal system interfaces with each other system through its active cooling system and passive systems to dissipate heat and ensure each instrument and component functions properly without overheating or deteriorating.

Finally, the power system powers each subsystem. Through the power distribution unit, commanded by the CDH, power is generated and correctly distributed to all subsystems as needed, ensuring each component receives the required voltage and current levels to operate correctly.

## 3. Science Mission Plan

### 3.1 Science Objectives

The Venera Nova mission will explore 3 scientific objectives: the influence of Venus' superrotation on its atmosphere and surface, how a lack of a magnetic field affects atmospheric composition, and the probability of life existing within Venus' super atmosphere. For the first objective,Venus' atmosphere is known to exhibit superrotation and plays a major role in Venus' climate system. The Venus Monitoring Camera will be capturing images of Venus's clouds every 10 seconds in 3 different wavelengths: Ultraviolet, Visible, and Near-Infrared. By analyzing cloud motion, brightness variations, and Kelvin waves, the mission will enable scientists to calculate horizontal wind speeds to help clarify how energy is transferred in the atmosphere.

The second objective of the Venera Nova mission is to determine how a lack of magnetic field affects atmospheric composition. Venus notably lacks a strong magnetic field, resulting in extreme cosmic radiation interfacing with the planet's atmosphere. This extreme amount of solar energy is intriguing to scientists since it differs from the atmospheres of other planets significantly. By studying the intensity of solar radiation on Venus, scientists can determine how a weak magnetic field affects atmospheric properties which can subsequently affect the planet's surface. The instrument selected to meet the objective is the SPICAV Ultraviolet Spectrometer, which will be measuring concentrations of ionized species such as O+, H+, CO2+. The measurements will also enable scientists to quantify escape rates and examine how solar photons strip away atmospheric particles, furthering knowledge and assessment of long-term habitability of planets lacking magnetic protection.

The third objective to be explored is determining the probability of life existing within Venus' upper atmosphere. Phosphine (PH3), a gas known as a potential biosignature, in Venus' atmosphere suggests the possibility of biological processes that has yet to be explored (Greaves et al., 2020). The Solar Occultation InfraRed (SOIR) will be operating in the 190-220 nm range and measuring absorption spectra to detect trace gases like phosphine. The presence of biosignatures like phosphine could provide insight into the planet's atmospheric chemistry and potentially indicate microbial activity.

### 3.2 Experimental Logic, Approach, and Method of Investigation

The Venus Monitoring Camera (VMC) will capture images of the planet's clouds and atmosphere in order to study the atmospheric superrotation. The camera has a wide field-of-view of 17.5º and a spatial resolution of 0.2–45 km per pixel (ESA, 1). VMC will capture images using four optical systems, each with a focal length of 13 mm, with

four narrowband filters in UV, near-infrared, and visible wavelengths (ESA, 1). It will begin to track cloud motion at the cloud tops and the main cloud layer at altitudes of 70km and 50 km every 10 seconds until the mission is complete (ESA, 1). This data will be used to observe atmospheric dynamical phenomena, such as wind speeds. Additionally, it will provide data on the surface terrain through the 1 μm transparency window (ESA, 2). The four optical channels have several filters, with the most critical to the mission being the NIR2 filter, which will be used to observe atmospheric motions in the main cloud deck (ESA, 3). The channels share one 1032 x 1024 pixel CCD, with each system taking one quadrant (Williams 2022). They are also connected to the same power converter, electronics, and digital processing unit (Williams 2022). The VMC will use the limb operation mode to record periodic fluctuations in cloud brightness over the entire mission duration. This mode studies the vertical structure of the atmosphere (ESA, 5). As stated in the concept of operations, after the initial calibration, the VMC will begin to collect data and send it to the orbiter every 30 seconds for the entire mission duration.

The Spectroscopy for Investigation of Characteristics of the Atmosphere of Venus (SPICAV) instrument is a spectrometer with three channels: two that image infrared radiation and one ultraviolet channel (Williams 2022). The ultraviolet channel, known as SPICAV-UV, will observe nadir, limb, and perform vertical profiling by solar and stellar occultation (Bertaux et al. 2007, 1). To collect data on SO2 and its effect on atmospheric dynamics, the spectrometer will analyze the albedo spectrum in nadir orientation (Bertaux et al. 2007, 1). SO2, along with CO2 and SO, will also be observed by measuring their density and temperature profiles in stellar occultation mode (Bertaux et al. 2007, 1). Additionally, there will be UV observations of the ionosphere through emissions of CO2+ and other species (Bertaux et al. 2007, 1). SPICAV-UV has a resolution of 0.8 nm and covers 0.118 to 0.320 microns, with a field of view of 55 x 8.7 radians (Williams 2022). Radiation is focused onto a 384 x 288 pixel CCD through the optics configuration, which is composed of an entrance slit, mirrors, and holographic diffraction grating (Williams 2022). After the first 20 minutes of the mission, the spectrometer will begin to collect data and transmit this data back to the orbiter once every hour.

One of the infrared channels from SPICAV will be utilized during the mission. Known as the Solar Occultation in the Infrared instrument (SOIR), this high-resolution spectrometer will detect phosphine gas in Venus' mesosphere. It operates in the range of 2.2-4.3 μm with a resolving power of about 15,000 (Williams 2022). SOIR features an Acousto-Optical Tunable Filter (AOTF) and an echelle grating to sort diffraction orders (Bertaux et al. 2007, 2). The optical apparatus features a Newtonian telescope with two corresponding parabolic mirrors, a tellurium oxide polarizing prism, and a zinc selenide lens (Williams 2022). The detector has a 320 x 256 array of 30 μm square pixels and was constructed out of photo-voltaic mercury cadmium telluride (Williams 2022).
By characterizing the mesosphere, SOIR aims to gain more data on the presence of phosphine on Venus, which could indicate the presence of life in the upper atmosphere.

During mission operations, SOIR will be calibrated and begin taking measurements of atmospheric spectra until mission completion. It will transmit this data back to the orbiter every 30 seconds.

### 3.3 Payload Success Criteria

The VMC (Venus Monitoring Camera) science instrument success criteria for the Venera Nova mission will determine the influence of Venus' superrotation on its surface and atmosphere by capturing images of Venus's clouds every 10 seconds in UV, visible, and near-infrared wavelengths. By measuring pressure, wind and cloud dynamics to better understand the rotation that is 60 times as its planetary rotation Venus' atmosphere expands in the sun, creating a low-pressure area that drives strong winds around the planet (Machemer, 2020)., which the VMC will be able to observe during the descending period of the aerobot. Success also involved fulfilling data handling requirements for the spacecraft and payload, including real-time processing of telemetry and commanding. Pressure, wind speed, and cloud dynamics will be measured through high-quality atmospheric data which will be obtained through the Aerobot's descent.

The UV Spectrometer involves the collection of ionized species such as O+, H+, CO2+ in the UV range (118-320 nm) over a 1 km range every hour. Measurements will determine how Venus' lack of a magnetic field affects atmospheric composition, since there is no internal geo dynamo present. Spatial resolution will be within 100 km and aims to measure wavelengths between 118-320 nm. The UV spectrometer will observe the surface in 0.8–1.0mm ‘‘windows’’ (nadir, nightside) for 10 seconds during the descending period. Measurements during the commissioning phase of the UV spectrometer will record a number of complete spectra, and then the parameters of the optimized window sets for different observation modes within the given wavelength (Bertaux et al, 2006).

The SPICAV Solar Occultation InfraRed Spectrometer will scan and detect the presence of phosphine gas in the upper atmosphere of Venus. phosphine is a molecule produced by microorganisms on Earth and can be a clue to the potential of life on Venus. The SOIR instrument will be beneficial in collecting and processing high-quality infrared data to study Venus's atmosphere. This includes thermal structure, chemical composition, and dynamics (Svedhem, 2009). SOIR ranges from 2.2–4.3mm, with a spectral resolution l/Dl 415,000, the highest on board VEX, which is a combination of an echelle grating and an AOTF crystal.

| Category | Minimum Success | Optimum Success | Stretch Goal |
|---|---|---|---|
| Investigate the influence of Venus' superrotation on its surface and atmosphere | Capture images of Venus's clouds every 10 seconds in UV wavelength | Capture images every 10 seconds in UV, visible, and near-infrared wavelengths and collect wind speed data by tracking cloud motion every 10 seconds | Record periodic fluctuations in cloud brightness over the course of 48 hours. |
| Determine how a lack of magnetic field affects atmospheric composition | Measure O+ and H+ composition and ionization in the upper atmosphere | Collect ionized species such as O+, H+, CO2+ in the UV range (118-320 nm) over a 1 km range every hour | To identify the escape rates of particles and solar photons in the upper atmosphere |
| Confirm presence of phosphine gas in the upper atmosphere | Detect phosphine peaks at any level of wavelength during descending period | Detects phosphine peaks between 190 and 220 nm. | Determine the probability of life existing within Venus' upper atmosphere |
| Study how planetary surfaces and interiors influence and interact with their host atmospheres | Get images of cloud fluctuations on Venus, detect particles in the upper atmosphere, and investigate possible phosphine within the upper surface. | Get UV and infrared wavelength images of cloud fluctuations of Venus, collect measurements of particles and solar photons in the upper atmosphere, and detect and evaluate possible phosphine within the upper surface. | Determine how Venus' planetary surface and interior influence and interact with its atmosphere. |

*Table displaying payload success criteria, ranging from minimal to optimal success.*

### 3.4 Testing and Calibration Measurements

As the VMC approaches the target, it will utilize the photon-transfer technique to specify total noise in darkness and system gain (Markiewicz et al, n.d.). This technique will calibrate solid-state imagers and camera systems, particularly the CCD, with and without light using different exposure times. The measured linearity error of the CCD is less than 1%. To separate the noise performance of the detector from that of the readout electronics, the CRE electronics were thermal-vacuum tested without the CCD. Accurate measurements of pressure, wind speed, and cloud dynamics will be obtained through the Aerobot's descent and ensured by inter-comparison of the other two science instruments located on the aerobot. Satellite and ground data will continuously be compared to validate wind speed data. Exposure time of 10 seconds is the controlled variable during testing and calibration in UV, visible, and near-infrared wavelengths. Periodic fluctuations in cloud brightness over the course of 48 hours will be evaluated to ensure accurate data.

The SPICAV UV Spectrometer will consist of two calibration aspects during the descending period of the aerobot: (i) a wavelength to pixel assignment, and (ii) a portion of emitting atmospheric light contaminating the analyzed transmission spectra. Wavelength-to-pixel assignment calibrations also known as spectral calibration which is the process of accurately mapping the spectral wavelengths to specific pixels on a detector array. This is crucial for accurate spectral analysis and utilizes spectral lines, which is needed for high-precision measurements (Evdokimova et al, 2020).
Emitting atmospheric light is a calibration method that can be measured by signal after the dark current correction.
All types of the UV emission as the nitric oxide airglow, the Lyman-a scattered light, and the sun light in the UV contained in spectra obtained by the SPICAV UV spectrometer are considered by the team as a stray light regarding to the spectral inversion problem (Gaigalas, 2009).

SPICAV SOIR (Solar Occultation in the Infrared) will perform various calibration and testing methods including dark current and thermal background detection, nonlinearity and pixel-to-pixel variability, instrument sensitivity, AOTF properties, and spectral calibration. The SOIR utilizes a combination of an echelle grating and an AOTF crystal to achieve high spectral resolution. Its primary objective is to study phosphine gas in the Venusian upper atmosphere to discover a potential for life. SOIR will have a 5s integration time and detect phosphine peaks between 190 and 220 nm (Bertaux, 2007). Signals detected by the SOIR in the upper atmosphere will depend crucially on the transparency of the cloud deck to radiation at 3mm, which seems to be the

necessary variable of this instrument. As SOIR utilizes its calibration methods to accurately detect phosphine, a beneficial way to study the lower atmosphere of Venus will be used in future studies.

### 3.5 Precision and Accuracy of Instrumentation

All payload instruments are expected to perform at their designated sensitivity values to observe and measure their respective data accurately. Sensitivity is integral to the functions of these instruments, since they must detect nearly indistinguishable changes throughout Venus' atmosphere. The VMC must operate within a sensitivity range of 0.3 to 1.0 μm to detect fluctuations in cloud brightness as well as changes in UV radiation. Detecting these fluctuations is integral to evaluate contributing factors towards Venus' superrotation—most fluctuations in brightness are challenging to detect, so a sensitivity range within micrometers is essential to the success of the Venera Nova mission.

The UV spectrometer and SOIR are also expected to operate within a high sensitivity range. The UV spectrometer must operate within a sensitivity range of 200-300 nm to appropriately detect changes in the ultraviolet spectrum and particle behavior. From these measurements, the amount of solar activity present throughout Venus' atmosphere can be determined.

The SOIR must operate at a sensitivity of 2257-4430 cm^(-1) to detect the presence of Phosphine in Venus' atmosphere. Currently, only "a handful of missions have plunged through the inhospitable, carbon dioxide-dominated atmosphere to take samples among the global layer of sulfuric acid clouds," indicating that the amount of phosphine present on Venus is undetermined (Byrne, 2020). Knowing that phosphine may be limited in amount and difficult to detect, the SOIR must be able to detect even the slightest amounts of phosphine on Venus; thus, a high sensitivity value is necessary to accurately determine the location and concentration of phosphine throughout the atmosphere.

### 3.6 Expected Data & Analysis

The Venus Monitoring Camera (VMC) will be used to collect data on the Venera Nova mission. It is similar to the Venus Express mission in a sense that it will collect cloud motion images (*see Figure 32)*. Differences include brightness variations and Kelvin waves as the aerobot descends into the upper atmosphere. One of the objectives on the Venus Express mission was to measure the spatial variations of the opacity and particle sizes on the nightside of Venus. The data together with information on the

composition will help to constrain models of the cloud formation and evolution on the planet (Titov et al, 2006). By capturing images of Venus's clouds every 10 seconds in UV, visible, and near-infrared wavelengths and collecting wind speed data by tracking cloud motion every 10 seconds, the team will identify the influence of Venus' superrotation on its surface and atmosphere. The team will utilize the VMC to capture images of Venus's clouds in Ultraviolet (365 nm), Visible (513 nm), and Near-Infrared (965 nm) wavelengths. The Venera Nova mission is slightly different from the Venus Express mission, as the team aims to investigate Venus' superrotation as the Venus Express aims to investigate the evolution of Venus. Data from the Venus Express mission can assist the team in tracking where cloud formation activity is heavy and where Venus' superrotation is the strongest.

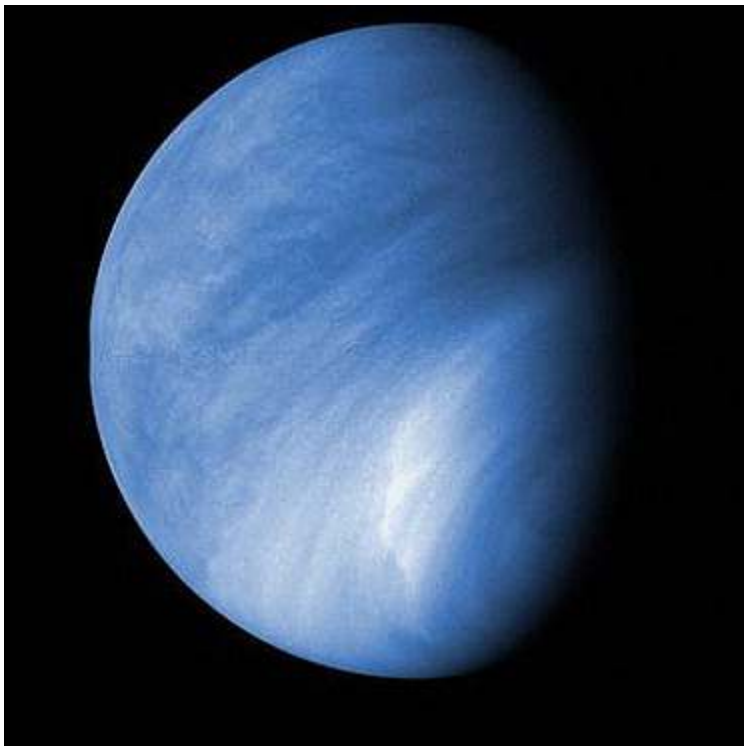


*Figure 32: An image of cloud formation made by the Venus Monitoring Camera (VMC) instrument in ultraviolet light at 365 nm from a distance of about 35,000 km (Svedhem et al, 2009).*

The SPICAV UV spectrometer can conduct observations in any direction of the sky and obtain the UV spectrum of stars or extended sources (UV light scattered by stars. The density/temperature profiles obtained with SPICAV will constrain and aid in the development of dynamical atmospheric models, from cloud top from the upper

atmosphere (Svedhem, 2009). The Venus Express mission is similar to the Venera Nova mission because the SOIR will study the ionosphere of CO, CO+, and CO2+ particles. This is essential for future missions that would rely on aerocapture spacecraft orbit injection technique that uses atmospheric drag to slow down a spacecraft, reducing the need for propulsive maneuvers and propellant and aerobraking. UV observations of the upper atmosphere will allow studies of the ionosphere through the emissions of O+, H+, CO2+ (Bertaux, 2006).

The SPICAV SOIR (Solar Occultation Infrared was used in the Venus Express to measure HDO and $H_2O$ in solar occultation to give more insight about the evolution of water on Venus. The SOIR will be utilized on the Venera Nova mission to confirm the presence of phosphine gas in the upper atmosphere. It is different in the sense that Venus Express studied isotopes of $CO_2$ and minor species, which provided a sensitive search for new species in the upper atmosphere of Venus (Bertaux, 2007).

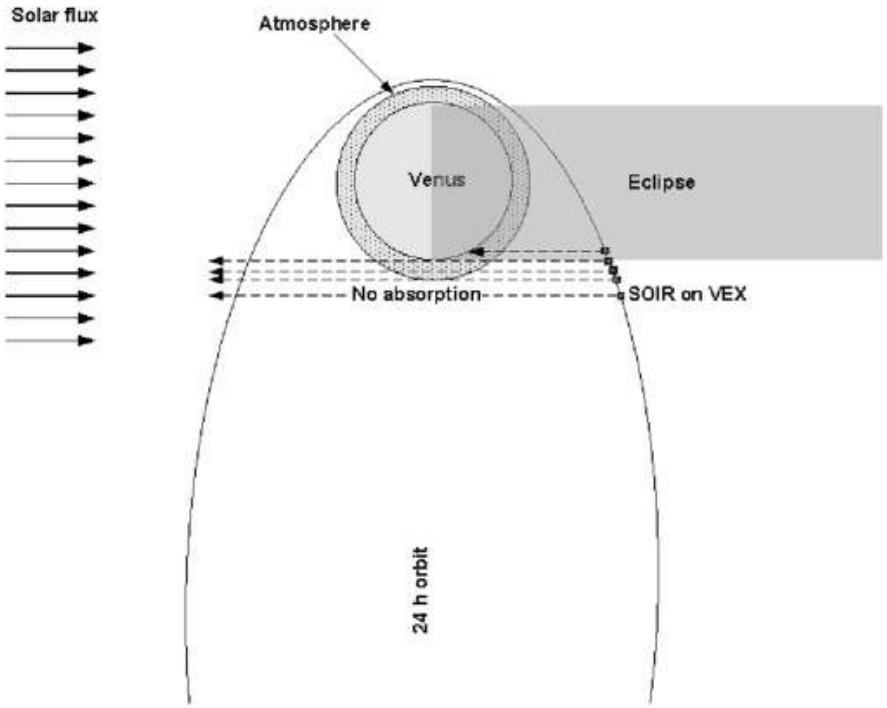



Figure 33: Diagram of Venus' atmosphere and interaction with solar flux

## 4. Mission Risk Management

### 4.1 Safety and Hazard Overview

The team has been reviewing requirements to ensure no major risks arise. Identifying

through reviewing not just the requirements but also looking back at missions before Venera Nova. A few methods used to identify and analyze risks are requirements review, scenario analysis, historical analysis, and risk matrix. When it comes to requirement review, the team takes priority in looking back at what the customer needs are to ensure all is met while maintaining a feasible margin with reasonable parameters. Scenario analysis is being used to create detailed scenarios of potential future events to assess their potential impact on the mission at hand. Regarding scenario analysis, teams will have additional meetings to bring up concerns that may arise later on. Historical analysis is being used to learn from past mistakes to improve on our mission. Things like mass constraints can be learned through other missions and seeing how they handled not being able to go over a certain weight limit. Lastly, risk matrixes have been used to list up all potential risks that may arise during or after the mission. With all this being said, the team will maintain a good risk posture by following planetary and non-planetary guidelines set in place and proactively identifying and analyzing risks to most importantly protect the organization's assets, reputation, and the safety of countless lives.

### 4.1.1. Risk Analysis

Risks are events that have negative impacts on cost, schedule, and performance. In every project, it is necessary to identify these risks to manage and mitigate them in the future to ensure the success of the project. There are many types of risks, and depending on the type, they can cause the mission success to be jeopardized; therefore, risks are important to watch and review regularly. Risks scale from high to low, likelihood of occurring, and the consequences they may bring. With this being said, the methods that will be used to identify, manage, and mitigate risks will be listed below. More will be added to this list as the team proceeds.

- **Documentation Review:** Reviewing project plans, requirements, and past reports to identify potential risks.
- **Risk Identification:** Tracking risks on a weekly or bi-weekly basis as a team.
- **Assumption Analysis:** Examine the project planning phase, draw trigger questions about potential failures or issues early to mitigate and eliminate early.
- **Root Cause Analysis:** RCA or Root Cause Analysis will be used to identify potential risks, specifically the root cause of these risks.

**Narrative:**

The MCA mission is dependent on many variables. These variables can be organized and addressed as risks. For this mission to succeed the team shall, together, plan and

predict any possible obstacles in this mission. This mission can be organized into these phases, Pre-Launch (this phase includes designing, testing, and building the sub-systems for the Aerobot), Mission Descope Phase, Post Launch Phase(this phase includes every part of the mission in between being transported to Venus and right before the Aerobot lands on the Venusian surface), and Post Landing (this phase includes every part of the mission after the Aerobot arrives at the surface of Venus including the Decommission Plan).

**Pre-Launch Phase:** The Pre-Launch Phase is very important to the mission, a lot of the important risks in this phase have to do with project planning. The MCA team shall go about this by taking into account all requirements provided by the contractor and creating a risk for each constraint in order to keep track of each respective risk. The team's mitigation approach is to keep track of these risks throughout the mission, the team can then understand how many resources are available at each given deliverable and provide resources to subteams accordingly. The risks pertaining to contractor constraints are, PROGRAMMATICS-2, PROGRAMMATICS-3, and PROGRAMMATICS-4. These risks were controlled using Gantt Charts, cost charts, and resource charts. Another risk that cannot be ignored at this time is Planetary Protection, PROGRAMMATICS-1 shall be implemented at this phase. The mitigation strategy will be to begin the sterilization of the environment around and inside the Aerobot before takeoff. This will be done frequently to avoid any chance of microbial life being transported to Venus. This is done so samples and the Venusian environment aren't contaminated due to the team's carelessness. Nicholson (2009). The success of the mission greatly depends on the condition of the team's scientific instruments. To ensure that the scientific instruments in the Aerobot's payload are protected from extreme speeds and forces, the payload systems will be connected to the rest of the subsystems. Therefore, all internal subsystems shall receive proper installation. All internal subsystems will be protected from solar radiation, extreme pressures, and lack of power. The physical environment isn't the only part of the Aerobot that can be tampered with, the digital environment can be affected as well. Hackers can breach important subsystems of the Aerobot before the launch date. NASA, having significant ties to the U.S. government, is a very targeted body when it comes to information leaks and hacks. A lot of classified information missions and information make NASA a place that is likely to hack. Meshkat (2022). To deal with this risk, the MCA Team decided on having a vulnerability analysis, this is where hardware undergoes testing to see how well information is secured. This is a very effective strategy to minimize the risk of being hacked.

**Mission Descope Phase:** Before the launch date of the Aerobot System, extensive research came back to the MCA Team highlighting the high likelihood that the corrosive properties of Venus' atmosphere could in fact damage the Aerobot balloon system. This

in turn would end the mission a lot earlier than the 48 hours that the mission was expected to last. The DSS (Descent Subsystem) Contractor offered the team two mitigation strategies. In the first mitigation strategy the DSS Contractor would increase the Aerobot balloon's wall thickness, this would add 5kg to the system and only increase the mission timeline to 36 hours. The second mitigation strategy would grant the team the entire 48 hours, but would add 5 kg of mass, 6 months dedicated to material study, and 20 million dollars will be paid to achieve all of this. After careful consideration the latter was chosen due to the amount of scientific research the mission would be able to accomplish within the full 48 hours. This Mission Descope Phase brought along 4 new risks and an update to a risk. 3 out of the 4 new risks are constraint risks pertaining to the program management of the MCA mission. MISSION DESCOPE, 3, 4, and 5 had to do with budget, mass, and time constraints. These risks were accepted due to how often deliverables were completed and these constraints were updated. Also, these risks are very similar to PROGRAMMATICS 2, 3, and 4. Similar methods shall also be utilized for the MISSION DESCOPE versions. MISSION DESCOPE-1 brought about the idea of the Aerobot Balloon popping due to corrosion. This risk was originally called CUSTOMER-19, but was named to MISSION DESCOPE-1 and closed after the mitigation strategy offered by the DSS Consultant. The last risk was MISSION DESCOPE-2, this risk was based on the notion that the MCA Team would not be able to adequately test the DSS subsystem. No mitigation strategy was implemented due to the DSS system being handled by an expert. The DSS Consultant understands what conditions the Aerobot Ballon may encounter and is tasked with finding the perfect material. The MCA Team has no material scientists and is inexperienced in this field. Therefore the risk shall be accepted.

**Post Launch Phase:** The post launch phase is to be used as an analysis of the mission to predict any risks the team can come up with. This phase shall take place between the spacecraft's launch and the Aerobot landing on the surface of Venus. Each subsystem happens to be made of components, and many common points of failure can often be found in the components that make up these systems. POWER-4, THERMAL-14, INSTRUMENTATION-17, and THERMAL-11 are all related to internal damages that can be caused by malfunctioning components. Electrical components are very complex and can very easily have unnoticeable points of failure. Zhang (2007). Since these 4 previously mentioned subsystems all include electrical components they can all use the same mitigation strategy. This strategy is to go through intense testing, if there is any possible point of failure in a component, it is imperative that it is found as soon as possible. Frequent testing increases the chance that a broken component is exposed. Systems testing will not abuse the systems, although systems must be able to be tested in environments similar to that of the MCA mission. This shall be the mitigation technique for these 4 systems. Although the entire MCA mission is based around Venus, this is not the only place that can damage the Aerobot System. Even the

environment of Space can affect the Aerobot System. The Sun is a great example of this. The Sun is capable of producing solar flares and with these solar flares can come intense amounts of radiation. The Sun can also produce this radiation without producing a solar flare. Earth is protected from this radiation with the atmosphere and the planet's surrounding magnetic field. The same magnetic field and atmosphere that the Aerobot System is flying away from. For THERMAL-13 and INSTRUMENTATION-18 these risks deal with solar radiation, solar radiation is able to damage data and systems. The best approach to mitigating radiation is by a radiation resistant material. The material used for the Aerobot System is Ti-6Al-4V. This is a titanium alloy that has very promising results. Louis (2008). The titanium alloy should be enough to shield most subsystems and ensure that most if not all data remains intact. For the next list of risks these 3 risks will pertain to Venus' unique atmospheric properties, these properties include its high pressures and highly corrosive sulfuric acid rain. The risks included are PAYLOAD-8, MECH-4, and MECH-2. MECH-2 has to do with the extreme pressures found on Venus' surface. Pressure gradients, in theory, will always try to level out, therefore any small pressure leak can greatly compromise the MCA mission. To mitigate this problem can go through testing by turning the inside of the Aerobot System into a vacuum, then using a pressure gauge to check if air is finding any way to enter the system. There are many other sensors and devices that can help identify any pressure leaks, but the simplest by far can be a simple pressure test like this. MECH-4 is a pretty common risk in the Venusian atmosphere. This risk is based on the material degradation of the Aerobot, whether it be from extreme heat or Venus' sulfuric acid rain. A lot of material analysis went into choosing the right shell for the Aerobot System. The mitigation strategy is to keep using Ti-6Al-4V. It not only is very resistant in shielding radiation but is also very heat resistant. There is a lot more information regarding this material's properties in the mechanical section. PAYLOAD-8 regards to how the payload might be affected by Venus' environment. There is not much that can be done besides strict monitoring on the enclosure that the payload is located. The final risks in this section all correlate with power management. For this part there is MECH-3, POWER-6, and POWER-5. MECH-3 has to do with particulate and matter build up. This can happen anywhere on the Aerobot System, but the part that it will affect the most is the power subsystem. MECH-3 isn't the only part limiting power, POWER-5 was implemented due to how Venus' atmosphere is so effective at breaking down light as you go deeper in the atmosphere. Landis (2021). The two of these risks can cause lots of failure to the mission, so does POWER-6. POWER-6 is a very broad risk about the power subsystem not being able to produce enough power. To mitigate all these risks, the power subteam added another solar panel and extra batteries. There are many backups in case the power system goes down. Without these backups the Aerobot System wouldn't be able to collect data or send it back.

**Post-Landing Phase:** This phase has only one major risk, CDH-8. CDH-8 is signal

interference. Once the Aerobot System is landed on the surface of Venus it is to send a signal back to us. This one action is the final risk the team must overcome to complete the MCA mission. Due to how common any emi(electro-magnetic interference) wave is, to ensure that the system has the proper circuit analysis code to remain protected against anything that can damage it. This is the plan for mitigation.

**Decommission Plan:** Once the data is collected, the Aerobot System shall enable the thrusters to slow descent speed. As the Aerobot System is slowing down data shall be uploaded to the primary spacecraft that the Aerobot System descended from. The primary spacecraft shall remain in orbit and will be ready to send the information collected back to Earth. As for the Aerobot System, the system shall use the last of its energy to slow its speed. Therefore the damage caused by the Aerobot's system shall be reduced. Once the Aerobot crashes the internal environment may incur serious damage. This damage will shut down all the subsystems and effectively kill the Aerobot. In case the internal environment is not damaged on descent the Aerobot shall have a command to shut down after coming into contact with the surface. This can be done with the Aerobot's telemetry system. Eventually, Venus' environment will begin to degrade the materials and the internal environment will be compromised. The data collected shall be sent to Earth by the primary spacecraft, which will thus conclude the MCA mission.

**Decision Making:** Many factors are involved that pose a threat to the overall mission at hand. A few to list are Venus's extreme temperature, differences in atmospheric pressures at certain levels, high wind speeds, and more. Given these factors, the team has found potential risks that could jeopardize the mission. With this being said, the team has plans in place to approach these risks in order to ensure the success of the mission. The first method the team will use is to mitigate the risk, more specifically, minimizing the likelihood and consequences of the potential harm at hand by planning or having a plan already set to lessen negative impacts. The second method the team will use is to research, researching in order to ensure safety and well-being of the people involved. The third method is to watch, monitoring risks specifically low priority risks that have a very low likelihood of occurring.  Lastly, the absolute last method team will use if reasonable, is to accept certain risks that will not jeopardize the mission if it occurs. Additionally, taking advantage of the "*NASA Lessons Learned"* webpage many possible mitigation strategies have been identified to be put to use. One of these methods used was the RIDM method. Team members were able to compare and determine alternative mitigation solutions to ultimately use the best fitting method based on the type of risk. Possible solutions were then added to a list and sorted based on complexity, price, and effectiveness. Decisions were then made and the risk log was updated accordingly. CRM was utilized after any change to the risk log. Risks were re-evaluated after a new risk was added or a mitigation strategy was updated. The risk

log would then be updated to accurately reflect the changes made to the Aerobot System

**Advanced Risk Log:**

Risks need to be accounted for in every project, using a risk log this helps track risks easier. Risk logs not only help in risk management but also project management in order to identify, analyze, and mitigate risks with ease throughout the project lifecycle.

| Rank | Risk ID # | WBS Element | Risk Owner | Category | Timeframe | Risk Title | Risk Statement | L | C | Rating | Approach | Trend | Mitigation Plans | Status of Mitigation Plans | Trigger Date | Trigger Event | Closure Date | Closure Event | Status Updates |
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| 1 | MECH-2 | Spacecraft | Isabella Benavidez Mechanical Engineer | Cost/Scope | Short Term | Aerobot System Depressurization | Given that the Aerobot System shall contain a pressurized environment that is different from the pressurized environment of Venus, there is a possibility that the Aerobot System can implode due to the pressure difference and any potential pressure leaks within the Aerobot System. Thus leading to a total mission failure with inoperable systems due to a compromised internal Aerobot environment. | 1 | 1 | Moderate | Mitigate (M) | ↓ Decrease | 1. The internal environment within the Aerobot shall be tested for any pressure leaks prior to takeoff.<br>2. An ultrasonic leak detector is proved to be quite useful in detecting pressure leaking. | Risk is identified<br>Risk is mitigated | 3/10/2025<br>3/31/2025 | SRR<br>MDR | 4/21/2025 | Project PDR | 3/06/2025: Likelihood has been updated from a 4 to 2 after completing mitigation plan 1.<br>3/31/2025: Likelihood has been updated from 2 to 1 after completing mitigation plan 2. |
| 2 | POWER-6 | Spacecraft | Andrew Hoang Electrical Engineer | Cost/Scope | Medium Term | Lack of Sufficient Power | Given that the Aerobot System shall contain a necessary amount of stored voltage at every moment of the mission also enough power input to every mandatory system. There remains a chance that power production/storage/distribution may be compromised. In turn leading to a complete internal system failure. | 1 | 1 | Moderate | Mitigate (M) | ↓ Decrease | 1. Add an extra solar panel for greater power production.<br>2. Add three extra thermal batteries for backup power. | Risk is identified<br>Risk is mitigated<br>Risk is mitigated | 3/06/2025<br>3/08/2025<br>3/29/2025 | SRR<br>MDR | 4/21/2025 | Project PDR | 3/08/2025: Likelihood has been decreased from a 4 to a 3 after completing mitigation plan 1.<br>3/29/2025: Likelihood has been updated from a 3 to a 1 after completing mitigation plan 2. |
| 3 | THERMAL-11 | Spacecraft | Elias Pedroza Thermal Engineer | Cost/Schedule | Long Term | The Overheating of System Components | Given that electronics are highly prone to overheating, there is a possibility that power distribution and system communication is greatly reduced. This can possibly lead to the failure of essential systems such as navigation and communication. This can lead to the possibility that the spacecraft misses set course entirely. | 2 | 2 | Moderate | Mitigate (M) | → Neutral | 1. Implement a thermal cooling system enclosed in a controlled environment. | Risk is identified<br>Risk is mitigated | 3/10/2025<br>3/29/2025 | SRR<br>MDR | 4/21/2025 | Project PDR | 3/29/2025: Likelihood has been updated from 4 to 2 after completeing mitigation plan 1.<br>3/31/2025: Risk ID # has been updated from Thermal-11 to CDH-7<br>4/18/2025: Risk ID # has been reverted from CDH-7 to Thermal-11 |
| 4 | INSTRUMENTATION-17 | Spacecraft | Valerie Miranda Chief Scientist | Cost/Scope | Long Term | VMC (Venus Monitoring Camera) malfunction | Failure of the system in any form would greatly reduce the scientific value of the mission. Clouds in Venus' atmosphere would be unable to be imaged and evaluated for atmospheric accelerations. | 3 | 2 | Moderate | Mitigate (M) | ↓ Decrease | 1. Test equipment in similar man-made environments to increase likelihood of survival. | Risk is identified<br>Risk is mitigated | 3/10/2025<br>3/31/2025 | SRR<br>MDR | 4/21/2025 | Project PDR | 3/12/2025: Consequences have been updated from 1 to 2 due to more research.<br>3/31/2025: Likelihood has been updated from 5 to 3 after completing mitigation plan 1.<br>3/31/2025: INSTRUMENTATION-18 and INSTRUMENTATION-17 have been merged into INSTRUMENTATION-18 to simplify risk. |
| 5 | MECH-4 | Spacecraft | Elias Pedroza Thermal Engineer | Cost | Short Term | Material Corrosion Exposure | Given the highly corrosive Venusian environment, there are extremely high possibilities of material corrosion on the outside of the Aerobot System. There is a possibility that the outer shell of the Aerobot System may deteriorate and this can lead to the internal destruction of the entire Aerobot System. | 5 | 5 | Moderate | Accept (A) | → Neutral | 1. Use material that has a high corrosion resistance. (Only needs to last 48 hours on Venus). | Risk is identified<br>Risk is mitigated<br>Risk is accepted | 3/10/2025<br>3/15/2025<br>3/31/2025 | SRR<br>MDR | 4/21/2025 | Project PDR | 3/15/2025: Consequences have been updated from a 3 to a 5 after completing mitigation plan 1.<br>3/31/2025: Risk is accepted.<br>4/21/2025: Risk is updated from THERMAL-12 to MECH-4 due to the material properties inherently being apart of mechanical. |
| 6 | THERMAL-13 | Spacecraft | Gregory Kimpo-Jabido Systems Engineer | Cost/Schedule | Short Term | Solar Radiation Exposure | Given the extreme solar radiation found in the upper atmosphere of Venus, there remains a possibility that subsystems like CDH, Power, Thermal, etc. can be damaged by solar ray emissions. Data may become corrupted and systems can lose functionality. Thus, the mission can miss out on extremely valuable data, and potentially total system failure. | 5 | 4 | Moderate | Accept (A) | → Neutral | 1. Use material that is able to shield internal systems from harmful solar emissions. | Risk is identified<br>Risk is mitigated<br>Risk is accepted | 3/08/2025<br>3/29/2025<br>3/31/2025 | SRR<br>MDR | 4/21/2025 | Project PDR | 3/29/2025: Consequences have been lowered from a 2 to 4 after mitigation plan 1.<br>3/31/2025: Risk is accepted. |
| 7 | PROGRAMMATICS-1 | Project Management | Isaac Munoz Mission Assurance Specialist | Cost/Schedule/ Scope | Short Term | Planetary Contamination | Given the arrival of the Aerobot System in the Venusian Atmosphere. There remains possible chances of contamination. The atmosphere of Earth contains small traces of microbial life, there is a possibility that this life can end up in Venus using the interior of the Aerobot System as a shelter. This can possibly affect the environment of Venus. | 5 | 4 | Low | Accept (A) | → Neutral | 1. Sterilize the environment within the Aerobot System before mission launch. | Risk is identified<br>Risk is accepted | 3/25/2025<br>3/31/2025 | SRR<br>MDR | 4/21/2025 | Project PDR | 3/31/2025: Risk is accepted. |
| 8 | POWER-5 | Spacecraft | Andrew Hoang Electrical Engineer | Schedule/Scop e | Long Term | Lack of Sunlight | Given the increasing dissipation of sunlight relative to the distance traveled from Venus' upper atmosphere. Solar panels shall lose efficiency due to lack of sunlight. Low power output can result in lower system performance. Overall leading to less time that the Aerobot System can function. | 5 | 3 | High | Mitigate (M) | ↓ Decrease | 1. Add a thermal battery to Aerobot System.<br>2. Add another solar panel to try and maintain solar power.<br>3. Add three more batteries to Aerobot System. | Risk is identified<br>Risk is mitigated<br>Risk is mitigated | 3/08/2025<br>3/08/2025<br>3/29/2025 | SRR<br>MDR | 4/21/2025 | Project PDR | 3/08/2025: Consequences are updated from 1 to 2 after mitigation plan 1.<br>3/29/2025: Consequences are updated from 2 to 3 after mitigation plan 2. |
| 9 | PROGRAMMATICS-2 | Project Management | Victor Chen Program Analyst | Cost | Long Term | Exceeding Mission Budget | Given the set amount of money granted to the MCA team, there is a chance that all granted money runs out and possibly more is spent then owned. Thus leading to upset stakeholders and quite possibly an early end to the mission. | 2 | 1 | Moderate | Mitigate (M) | → Neutral | 1. Track money spent in each deliverable. | Risk is identified<br>Risk is mitigated<br>Mission Descope | 3/10/2025<br>3/10/2025<br>3/11/2025 | SRR<br>MDR<br>Descope | 4/21/2025 | Project PDR | 3/10/2025: Likelihood is updated from a 3 to 2 after taking mitigation 1 into account. |

| Rank | Risk ID # | WBS Element | Risk Owner | Category | Timeframe | Risk Title | Risk Statement | L | C | Rating | Approach | Trend | Mitigation Plans | Status of Mitigation Plans | Trigger Date | Trigger Event | Closure Date | Closure Event | Status Updates |
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| 10 | PROGRAMMATICS-3 | Project Management | Tyree Nguyen Deputy Project Manager of Resources | Scope | Long Term | Exceeding Mass Constraint | Given the mass constraint set by stakeholder there is a chance that the spacecraft exceeds mass constraints. There is an extremely high chance that this will end the mission if it is not solved before the PDR. | 2 | 1 | Moderate | Mitigate (M) | → Neutral | 1. Track mass in each deliverable. | Risk is identified<br>Risk is mitigated<br>Mision Descope | 3/10/2025<br>3/10/2025<br>3/11/2025 | SRR<br>MDR<br>Descope | 4/21/2025 | Project PDR | 3/10/2025: Likelihood is updated from 3 to 2 after mitigation plan 1. |
| 11 | MECH-3 | Spacecraft | Manahil Syeda Lead Systems Engineer | Scope | Short Term | Dust/Particulate Build Up | Given the high winds and dusty atmosphere of the planet Venus, there is a chance that the Aerobot System shall become covered in dust. This can affect the mass, ventilation, and solar power production of the Aerobot System. This shall potentially cut down the Aerobot System's lifetime. | 3 | 5 | Low | Accept (A) | ↓ Decrease | 1. No outside ventilation. | Risk is identified<br>Risk is mitigated | 3/10/2025<br>3/15/2025 | SRR<br>MDR | 4/21/2025 | Project PDR | 3/12/2025: Likelihood is updated from a 4 to a 3 after more research of Venus.<br>3/15/2025: Consequence is updated from a 4 to 5 after mitigation plan 1. |
| 12 | PAYLOAD-1 | Spacecraft | Rachel Brown Project Lead | Scope | Medium Term | Loose Equipment | Any unsecured scientific instruments could potentially be damaged by extreme wind speeds. Intense movements can damage the functionality of these instruments and any scientific data collected. | 2 | 4 | Low | Mitigate (M) | ↓ Decrease | 1. Secure equipment to Aerobot System. | Risk is identified<br>Risk is mitigated | 3/10/2025<br>3/10/2025 | SRR<br>MDR | 4/21/2025 | Project PDR | 3/10/2025: Likelihood is updated from a 5 to 2 after mitigation plan 1.<br>3/15/2025: Risk is updated from MECH-1 to PAYLOAD-1 due to this risk being primarily a PAYLOAD problem. |
| 13 | THERMAL-14 | Spacecraft | Elias Pedroza Thermal Engineer | Cost/Schedule | Medium Term | ATC Heater/Cooler Malfunction | Given the extreme environments of Venus, the Aerobot System shall greatly rely on the ATC cooling function. It is possible that the cooling feature may experience a malfunction. This can be exceptionally devastating and could ruin the entire mission. | 3 | 1 | High | Mitigate (M) | ↓ Decrease | 1. Purchase two ATC coolers.<br>2. Test ATC cooler in a similar scenario in a testing facility. | Risk is identified<br>Risk is mitigated | 3/10/2025<br>3/31/2025 | SRR<br>MDR | 4/21/2025 | Project PDR | 3/31/2025: Likelihood is updated from a 4 to a 3 after mitigation plan 2. |
| 14 | PROGRAMMATICS-4 | Project Management | Rachel Brown Project Lead | Schedule/Scope | Long Term | Missing Launch Date | Given the strict deadline of March 1st, 2029, the Aerobot System will not be able to be launched any time before or after this date. There is a possibility that the date is missed and the entire mission is called off. | 2 | 1 | Moderate | Mitigate (M) | → Neutral | 1. Track schedule in each deliverable. | Risk is identified<br>Risk is mitigated<br>Mission Descope | 3/10/2025<br>3/10/2025<br>3/11/2025 | SRR<br>MDR<br>Descope | 4/21/2025 | Project PDR | 3/10/2025: Likelihood is updated from 3 to 2 after mitigation plan 1.<br>3/11/2025: 6 months must be allocated towards material research and balloon design. There will now be less time in between launch date and PDR. Trend has increased. |
| 15 | INSTRUMENTATION-18 | Spacecraft | Valerie Miranda Chief Scientist | Cost | Long Term | Solor Flare Exposure | Instrument detectors could be overwhelmed by increased solar radiation due to Venus's weak magnetic field. Solar flares or coronal mass ejectives could also interfere with signals, resulting in tarnished data or total data loss. | 2 | 3 | Moderate | Mitigate (M) | ↓ Decrease | 1. Test in similar man-made environment. | Risk is identified<br>Risk is mitigated | 3/31/2025 | MDR | 4/21/2025 | Project PDR | 3/31/2025: INSTRUMENTATION-18 has been updated to the Risk Title, "Solar Flare Exposure". Risk details have also been updated.<br>3/31/2025: Likelihood has been updated from a 3 to a 2 due to mitigation plan 1. |
| 16 | PAYLOAD-8 | Spacecraft | Alaina Claudio Scientist 1 | Scope | Long Term | Environmental Effects on Payload | Environmental hazards such as sulfuric acid clouds, rain, high temperatures, and extreme atmospheric pressures could result in instrumentation failure if the payload subsystem is not appropriately protected. Any subsequent data collected may be insignificant towards accomplishing the mission's scientific objectives. | 3 | 1 | High | Mitigate (M) | → Neutral | 1. Test Payload in similar conditions. | Risk is identified<br>Risk is mitigated | 3/30/2025<br>3/31/2025 | MDR | 4/21/2025 | Project PDR | 3/31/2025: Likelihood has been updated from 4 to 3 due to mitigation plan 1. |
| 17 | POWER-4 | Spacecraft | Andrew Hoang Electrical Engineer | Cost/Scope | Long Term | Electrical Components Damaged | Given the many possibilities of internal system failures, damaged electrical components can have a very vital impact on the success of data gathering and/or Aerobot functionality. Thus having the chance to ruin the mission to varying success. | 1 | 2 | Low | Accept (A) | ↓ Decrease | 1. Test electrical components in enclosed environment.<br>2. Fuses can be implemented to stop excessive current flow | Risk is identified<br>Risk is mitigated | 3/10/2025 | SRR | 4/21/2025 | Project PDR | 3/12/2025: Likelihood has been updated from 3 to 1 due to mitigation plan 1. |
| 18 | CDH-8 | Spacecraft | Gregory Kimpo-Jabido Systems Engineer | Cost/Schedule | Short Term | Signal Interference | Given the high amount of signal interference in space, communication with the Aerobot Sytsem and the subsytems may be hard to be in contact with. This can lead to system failure and lost data. Thus leading to a mission failure. | 2 | 2 | Moderate | Mitigate (M) | ↓ Decrease | 1. Use circuit analysis code to protect fragile systems | Risk is identified<br>Riskis mitigated | 3/10/2025<br>4/20/2025 | SRR | 4/21/2025 | Project PDR | 4/20/2025: Likelihood has been updated from a 4 to 2 due to mitigation plan 1. |
| 19 | CDH-9 | Spacecraft | Victor Chen Program Analyst | Cost/Schedule | Short Term | Cyber-Security Threat | Given the possibility that data can be breached and stolen by hackers, there remains a possibility that data breach/tampering can make the data collected obsolete. Functions may also not perform as they are supposed to. Data and program security can be very important to ensuring mission success. | 1 | 5 | Moderate | Mitigate (M) | ↓ Decrease | 1. Have a qualified third party conduct a computer vulnerability analysis on project material. | Risk is identified<br>Risk is mitigated | 3/10/2025<br>4/20/2025 | SRR | 4/21/2025 | Project PDR | 4/20/2025: Consequences have been updated from a 2 to 5 after more research.<br>4/20/2025: Likelihood has been updated from a 3 to a 1 after mitigation plan 1. |
| 20 | CDH-10 | Spacecraft | Gregory Kimpo-Jabido Systems Engineer | Schedule/Scope | Long Term | Software Error Bugs | Given the chance of possible bugs in CDH software, the programs that shall be implemented during the mission may run into possible failures that keep systems from operating correctly. This can disable subsystems and destroy the Aerobot from within. | 1 | 2 | Low | Mitigate (M) | → Neutral | 1. Check code with multiple experts.<br>2. Use successful code from other missions that implemented similar sub-systems. | Risk is identified<br>Risk is mitigated | 3/10/2025<br>4/20/2025 | SRR | 4/21/2025 | Project PDR | 4/20/2025: Likelihood has been updated from 2 to 1 after mitigation plan 1. |

| Rank | Risk ID # | WBS Element | Risk Owner | Category | Timeframe | Risk Title | Risk Statement | L | C | Rating | Approach | Trend | Mitigation Plans | Status of Mitigation Plans | Trigger Date | Trigger Event | Closure Date | Closure Event | Status Updates |
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| 21 | THERMAL-12 | Spacecraft | Elias Pedroza Thermal Engineer | Cost/Schedule | Short Term | Corrosion | Given Venus's toxic atmosphere, there is a possibility of corrosion of material, resulting in the breakdown of the aerobot, thus failing to safely house equipment needed for mission tasks. | 2 | 1 | Moderate | Mitigate (M) | ↓ Decrease | 1. Check code with multiple experts.<br>2. Test material used in similar scenario in a testing facility. | Risk is identified<br>Risk is mitigated | 3/10/2025<br>4/20/2025 | SRR<br>MDR | 4/21/2025 | Project PDR | 4/20/2025: Likelihood has been updated from 2 to 1 after mitigation plan 1. |
| 21 | MISSION DESCOPE-1 | Project Management | Manahil Syeda Lead Systems Engineer | Cost/Schedule/ Scope | Long Term | Balloon Material Degradation | Given the highly corrosive atmosphere of Venus, material degradation is very common. This can be a point of failure that can possible end the mission early. This in turn will limit the amount of data collected. | 1 | 2 | Low | Accept (A) | ↓ Decrease | 1. Use corrosive resistant materials on the Aerobot Balloon.<br>2. Perform additional material research, development, and testing of the balloon material to fully meet the performance requirement. | Risk is identified<br>Mission Descope | 3/10/2025<br>3/11/2025 | SRR<br>Descope | 3/11/2025 | Mission Descope | 3/11/2025: CUSTOMER-19 Risk ID # has been changed to MISSION DESCOPE-1.<br>3/11/2025: Consequences have been updated from a 5 to 2, due to Mission Descope.<br>3/11/2025: Likelihood has been updated from a 2 to 1 due to Mission Descope.<br>3/11/2025: Risk is accepted |
| 22 | MISSION DESCOPE-2 | Spacecraft | Rachel Brown Project Lead | Cost/Schedule/ Scope | Long Term | Equipment Failure Due to Limited Testing with Integration of DSS | Given the DSS system belongs to the DSS contractor, there shall be only a limited amount of time to test and integrate DSS system. Thus likelihood and consequences are unknown until DSS is tested. | 3 | 2 | Moderate | Accept (A) | → Neutral | 1. Test DSS system as soon as possible. | Risk is identified<br>Mission Descope | 3/11/2025<br>3/11/2025 | Descope<br>MDR | 4/21/2025 | Project PDR | 3/11/2025: Mission descope includes a 6 month waiting period until DSS system is to be recieved, making room for possible testing.<br>3/11/2025: Risk is accepted |
| 23 | MISSION DESCOPE-3 | Project Management | Victor Chen Program Analyst | Cost/Schedule | Medium Term | Risk of Exceeding Mission Budget | Given the 20 million dollars that must be allocated to material study and implementation, the MCA Team will run out of resources faster and due to this the team may not be able to afford mitigation strategies or parts for subsystems. Thus, making it more difficult to lower risk. | 2 | 1 | Low | Accept (A) | Closed | 1. Keep a budget list that is updated bi-weekly. | Risk is identified | 3/11/2025 | Descope | 3/11/2025 | Mission Descope | 3/11/2025: Risk is accepted |
| 24 | MISSION DESCOPE-4 | Project Management | Isaac Munoz Mission Assurance Specialist | Schedule/Scop e | Medium Term | Risk of Exceeding Aerobot System Mass | Given the extra 5 kg mass that must be added to the Aerobot System, the MCA team shall have less mass to allocate towards sub-systems this can take away from any backup components which increases the risk. | 2 | 1 | Low | Accept (A) | Closed | 1. Keep a list of mass that is updated bi-weekly. | Risk is identified | 3/11/2025 | Descope | 3/11/2025 | Mission Descope | 3/11/2025: Risk is accepted |
| 25 | MISSION DESCOPE-5 | Project Management | Isaac Munoz Mission Assurance Specialist | Schedule | Long Term | Risk of Missing Aerobot Launch Date | Given the 6 months that must be added due to the Mission Descope, the MCA Team runs the risk of going over schedule due to 6 months of time going directly to material study and implementation. Less time to work on the Aerobot System can make planning more difficult and put stress on team members. This can increase the overall risk of the mission. | 2 | 1 | Low | Accept (A) | Closed | 1.Update calendar according to the loss of the 6 months due to mission descope. | Risk is identified | 3/11/2025 | Descope | 3/11/2025 | Mission Descope | 3/11/2025: Risk is accepted |

**Risk Matrix:**

The risk matrix is a tool used to assess risks in order to evaluate and prioritize certain risks over others based on their likelihood and consequence rating. In a sense, this is a visual representation of risks that will help in aiding decision making, resource allocation, and risk mitigation.

| Risk Matrix | | | | | | |
|---|---|---|---|---|---|---|
| Likelihood | 5 | 5→ | 6→, 7→ | 8↓ | | |
| | 4 | | | | | |
| | 3 | 11↓ | | | 4↓, 22→ | 13↓, 16→ |
| | 2 | | 12↓ | 15↓ | 3→, 18↓ | 9→, 10→, 14→ |
| | 1 | 19↓ | | | 17↓, 20→, 21↓ | 1↓, 2↓ |
| | # | 5 | 4 | 3 | 2 | 1 |
| | Consequences | | | | | |

| Approach | Criticality | LxC Trend | L = Likelihood (1-5) |
|---|---|---|---|
| A - Accept | HIGH | ↓ - Decreasing (improving) | 1 = Not Likely |
| M - Mitigate | MED | ↑ - Increasing (worsening) | 5 = Extremely Likely |
| W - Watch | LOW | → - Unchanging (staying the same) | C = Consequence (1-5) |
| R - Research | | NEW - Added recently | 1 = High Consequence |
| | | | 5 = Low Consequence |

### 4.1.2. Failure Mode and Effect Analysis (FMEA)

FMEA, short for Failure Mode and Effect Analysis, is a systematic approach to identify and prioritize potential risks in a project. FMEA helps break down potential consequences of these risks/failures to integrate mitigation techniques. Using FMEA, the team has broken down and identified the top risks that can potentially cause our mission to cease operations. The risks associated with this mission may have numerous consequences on the functionality of relevant subsystems. The most crucial risks of the Venera Nova mission and their mitigation strategies are listed below:

| Function | Failure Mode (Risk Identification) | Effects | Sev | Cause | Occ | Prevention | Det | RPN | Actions |
|---|---|---|---|---|---|---|---|---|---|
| Power Subsystem | Power-5: Lack of Sunlight | With Solar cells being the Aerobot's primary source of energy, a lack of sunlight could result in not only the Power subsystem but also all other systems from receiving inadequate to no power. | 9 | Venus' dense atmosphere can cause sunlight to be blocked from reaching the spacecraft. | 8 | Ensure to research and predict the environment of Venus' atmosphere at the time of the launch date and test backup power generation systems of the overall system to ensure it can survive a lack of sunlight. | 2 | 144 | The aerobot's power subsystem does include redundant systems in order to provide stored or redundant energy sources, including backup batteries and generators. This will help provide at least a few more hours of research and data to be collected before the system fully powers down |
| | Power-6: Not | Inability to generate | 10 | Insufficient power | 2 | Ensure that the power subsystem | 1 | 20 | The power subsystem consists |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | enough power | enough power will render scientific instruments inoperable. | | generated. | | is able to generate sufficient power to support Aerobot via testing prior to launch date. | | | of 2 separate solar arrays and the recovery system will prioritize sustaining battery power by reducing output to noncritical loads. |
| Mechanical Subsystem | Mech-2: Aerobot deforms under pressure | If the structural materials of the Aerobot system are unable to handle the harsh pressures of Venus' atmosphere, the Aerbot may become deformed as it descends, compromising the ability of the internal structures of the system to function. | 9 | Venus' atmosphere shows drastic changes in atmospheric pressure throughout. | 4 | Back-Up systems are maintained in the case that the primary structural system of the Aerobot fails to function. Duplicate components are included for structural integrity, cooling, and pressure management. | 2 | 72 | Testing will be conducted on the materials of the system's structure as well as the back-up systems under high pressure environments to ensure that the system is able to fulfil its scientific objectives even when structurally compromised. |
| | Mech-3: Dust/particulate matter build up | Solar panels may malfunction due to dust, airflow may be blocked between systems, and the Aerobot may | 8 | As the Aerobot descends, it will come into contact with numerous dust particles spread | 5 | Manufacture protective equipment around crucial components to ensure matter does not build up within the | 4 | 160 | Take appropriate preventative measures to ensure dust is not able to enter the Aerobot. Perform testing that will simulate Venus' |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | be weighed down. | | throughout Venus' lower atmosphere. | | Aerobot. | | | environmental conditions. |
| Thermal Subsystem | TH-12: Corrosion | Due to Venus' harsh environment the systems' materials are susceptible to corrosion. This could result in the endangerment of the instruments and electronics housed by the system. | 9 | Could occur if PTCs are damaged, exposing the titanium to Venus' acidic atmosphere.It is also important to note that high temperatures accelerate corrosion. | 2 | Using corrosion-resistant materials such as Titanium, applying thermal coatings, and ensuring proper surface treatment and sealing of the cubic meter system. | 3 | 54 | No physical repair can be done mid-mission, but software can be implemented to shut down or reroute power away from the affected components to decelerate corrosion. |
| | TH-13: Radiation Exposure | Venus's lack of a magnetic field presents the possibility for overexposure to radiation. Such occurrences can result in irreparable damage to electrical and mechanical components inside the cubic system. | 10 | If the thermal subsytem's PTCs are not adequately designed to shield against solar radiation, excess heat may accumulate inside the system. | 4 | Can implement redundant critical systems or error correction software, or add thermal or mechanical shielding. | 2 | 80 | The system can activate the LHP and radiator to increase heat dissipation. Can also reduce power to non-essential scientific instruments and electrical components to decrease internal heat generation. |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| CDH Subsystem | CDH-8: Signal interference | The thick, dense atmosphere of Venus may result in signal interference, preventing the system from efficiently or accurately communicating with other systems, compromising data collection abilities, command execution, and system checks. | 7 | Venus has a harsh, dense atmosphere full of various particles and components capable of signal interference | 8 | Multiple methods of data communication are available in case a certain method is facing interference by Venus' environment. Autonomous programs and algorithms are further available in the case that the system cannot maintain communication with Earth. Short term and flash memory is also included to store data until communication can be restored. | 2 | 112 | Communication systems will be thoroughly tested to ensure that autonomous systems and algorithms function currently without external intervention. Further, storage systems will be tested to ensure that in case communication systems are compromised, valuable data can still be stored and extracted once systems are back online |
| | CDH-10: Software Error/Bugs | Software bugs may cause programs and algorithms to malfunction, instrument or system component performance to crash, or communications | 7 | Where certain aspects of descent in space may be unpredictable, unforeseen situations may cause bugs or | 7 | The On-Board Computer within the CDH provides a recovery software to fix any issues caused by faulty code or software, ensuring to | 1 | 49 | Thorough testing will be conducted on the system's software to fully address any potential bugs or errors in various circumstances to the extent possible. Further, as the various |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | to be compromised, sacrificing mission time and valuable data. | | errors in software to become apparent, especially in situations that had not been already tested for. | | efficiently bring systems back to functioning properly. Further, if primary communication systems fail, duplicate components are available for main hardware systems. | | | components of the system are integrated and installed with communication channels, these components alongside backup systems and algorithms will be tested. |
| Payload Subsystem | Instrumentation 17: Thermal Damage | Instrumentation components or sensors may suffer heat damage, resulting in loss of instrument functionality, loss of valuable environmental data, or degradation of communication systems. | 10 | The harsh environment and temperatures of the Venusian environment may result in overheating of system components. | 7 | While only one instance of each instrument will be present, thermal subsystems will be able to provide the necessary thermal protection to ensure each instrument can function to its full capacity without facing damage, | 2 | 140 | The integration of the thermal subsystems with the payload subsystem will be thoroughly tested to ensure the instruments can structurally withstand harsh environments while continuing to collect valuable data. This testing will ensure instruments can function in case a certain thermal system is compromised. |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | Instrumentation 18: Signal Interference | Instruments may face signal interference due to Venus' environment or other instruments, leading to further loss of data, compromised quality of data or inaccurate data, or degraded communication with other systems. Further, instrument detectors and fragile components may be overwhelmed by increased solar radiation | 7 | Venus' harsh atmosphere presents the possibility of solar flares or coronal mass ejections. Further increased solar radiation due to Venus's lack of a magnetic field presents great challenges for instrument functionality | 6 | The CDH subsystem will provide functionality to limit affects to the quality of instrument data, including systems that can directly store the data, validate data sent to other systems, and autonomously process data as needed. In case an instrument fails to function, the CDH subsystem will have the necessary algorithm to reboot and debug issues with the instrument. | 2 | 84 | The CDH and payload subsystem integration will be thoroughly tested to ensure that each component can efficiently and accurately communicate with the other. Further, recovery algorithms and autonomous systems will be tested to ensure that instruments can be rebooted and debugged in case of data collection failures or interference. |

### 4.1.3. Personnel Hazards and Mitigations

Numerous potential hazards exist throughout the manufacturing process of developing the Aerobot system. Manufacturers must work with an array of tools to produce the mechanical, electrical, thermal, CDH, and payload subsystems, which may threaten the safety of personnel. Personnel hazards will be effectively monitored and prevented through many approaches, inspections, assessments regarding safety, tools such as sensors and cameras, fire suits, chemical suits, and more will be utilized. The team will prioritize safety management throughout the project lifecycle ensuring everyone on the team is educated on safety first, reinforcing good practices, laws, protocols and regulations that are enforced by the U.S. Department of Labor. Overarching safety concerns and their mitigation strategies are included in the table below:

| **Subsystem** | **All** | | | | |
|---|---|---|---|---|---|
| **Hazard** | Breathing in metal dust and/or welding fumes. | Hearing damage from prolonged exposure to noise. | Injuries from coming into contact with sharp edges. | Injuries from manufacturing at different heights. | Improper storage can result in objects catching fire. |
| **Mitigation Strategy** | Provide protective equipment such as masks. | Provide protective equipment such as earplugs. | Implement edge protectors and padding around harsh surfaces. Provide protective gloves to manufacturers. | Ensure workers are properly supervised at all times. Establish physical barriers between the worker and the fall hazard. Provide protective equipment such as full-body harnesses and anchors. | Store all objects outside of the manufacturing space. Only objects pertinent towards manufacturing will be allowed in the area. |

The **mechanical subsystem** contains a metal framework held together by a chassis, propulsion system, thermal protection for the payload, and a power subassembly to ensure instruments operate effectively. The appropriate risks and mitigation strategies for developing these components are listed in the table below.

| **Subsystem** | **Mechanical** | | | | |
|---|---|---|---|---|---|
| **Hazard** | Electric shock from welding. | Injuries from moving objects. | Eye injuries from flying sparks. | UV radiation exposure from welding. | Fuel leaks regarding the propulsion system. |
| **Mitigation Strategy** | Only use welding machines that include insulation to protect users from shock. Insulating mats will also be implemented within the welding space. | Provide protective equipment such as face masks, sturdy shoes, and hard hats. | Provide protective equipment such as face shields. | Provide protective equipment such as radiation-blocking face shields, long sleeves, and long pants. | Inspect fuel components for signs of damage daily. Employ fuel additives to reduce harmful emissions and to keep fuel tanks clean. |

The **power subsystem** is composed of two solar arrays, a voltage regulator, a thermal battery, a battery charger/discharger, and a voltage regulator. The appropriate risks and mitigation strategies for manufacturing the power components are listed in the table below:

| **Subsystem** | **Power** | | | | |
|---|---|---|---|---|---|
| **Hazard** | Electrical shocks caused by short circuits. | Fire caused by electrical overloads. | Buildup of flammable gases during battery charging. | Long-term health issues caused by exposure to lead. | Chemicals escaping the battery's casing due to leakage. |
| **Mitigation Strategy** | Only use tools with insulated handles. Provide protective gear such as rubber gloves and shoes. | Implement overload relays to protect circuits from excess heat. Distribute electrical loads across different circuits. Use circuit breakers to stop flow of electricity during an overload. | Properly ventilate the manufacturing area. Batteries will be manufactured with built-in safety features such as vents and separators. A large-scale cooling system will be implemented to reduce the likelihood of batteries overheating. | Comply with the Occupational Safety and Health Administration's (OSHA) guidelines for battery manufacturing. All parts of the body are to be covered with protective gear to minimize lead exposure. Do not manufacture in excess of a necessary amount of lead. | Properly ground all wires and arrange the input and output of the battery supply to prevent trampling and wearing out. |

The **CDH subsystem** consists of an onboard computer, a low gain antenna, a high gain antenna, and a software defined radio. The appropriate risks and mitigation strategies for manufacturing the CDH components are listed in the table below:

| **Subsystem** | **CDH** | | | | |
|---|---|---|---|---|---|
| **Hazard** | Injuries caused by slips and falls due to cables. | Skin irritation caused by thermal paste exposure. | Long-term health issues caused by exposure to lead and mercury. | Health issues caused by the release of volatile organic compounds (VOCs) from adhesives and coating. | Electrical shocks caused by short circuits. |
| **Mitigation Strategy** | Instruct workers to regularly keep cables and other electrical components out of walkways. Place supportive mats around all manufacturing spaces. | Provide protective equipment such as face shields, long-sleeve shirts, gloves, and long pants. Instruct workers to clean potential paste spills up using alcohol and water immediately. | Comply with OSHA's guidelines for electronic manufacturing. All parts of the body are to be covered with protective gear to minimize metal exposure. Do not manufacture in excess of a necessary amount of hazardous metals. | Provide protective equipment such as face shields and masks. Minimize the use of VOC solutions by replacing them with water-based adhesives when applicable. | Only use tools with insulated handles. Provide protective gear such as rubber gloves and shoes. Regularly inspect cables for signs of damage and ensure they are properly secured. |

The **thermal subsystem** consists of a radiator, active cooler, white coating, and multi-layer insulation. The appropriate risks and mitigation strategies for manufacturing the thermal components are listed in the table below:

| **Subsystem** | **Thermal** | | | | |
|---|---|---|---|---|---|
| **Hazard** | Long-term health issues caused by exposure to lead and nickel. | Burns caused by the high surface temperature of a radiator. | Injuries caused by the repetitive straining of muscles. | Health issues caused by inhaling the radiator's white coating. | Burns caused by molten metal splashes. |
| **Mitigation Strategy** | Comply with OSHA's guidelines for radiator manufacturing. All parts of the body are to be covered with protective gear to minimize lead exposure. Do not manufacture in excess of a necessary amount of copper and aluminum. | Monitor the temperature of radiators using thermostats. Provide protective equipment such as flame-resistant gloves if contact with a high-temperatur e radiator is necessary. | Employ tools that do not rely heavily on human strength. Allow a few minutes of breaks in between manufacturing to encourage muscle recovery. | Provide protective equipment such as masks. | Ensure the environment for handling hot metals is clear of possible contaminants. Regularly monitor workers throughout manufacturing. Place barriers and shields around the soldering workstations. |

The **payload subsystem** consists of the mission's three science instruments. The appropriate risks and mitigation strategies for manufacturing the scientific instruments are listed in the table below:

| **Subsystem** | **Payload** | | | | |
|---|---|---|---|---|---|
| **Hazard** | Long-term health issues caused by exposure to lead and mercury in CCD manufacturing. | Eye and skin damage due to radiation exposure. | Electrical shocks caused by short circuits from power integration. | Fire caused by high-power lasers. | Injuries caused by assembling small spectrometer components repetitively. |
| **Mitigation Strategy** | Provide protective equipment such as masks, long-sleeve shirts, and long pants. Do not manufacture in excess of a necessary amount of harsh metals. | All parts of the body are to be covered with protective gear to minimize exposure to radiation. Perform radiation testing in an isolated area, away from as many personnel as possible. | Only use machinery with insulated handles. Provide protective gear such as rubber gloves and shoes. Ensure that all electronic components are properly grounded prior to testing. | Instruct all personnel on how to properly handle lasers. Place beam stops at the end of a laser's path to limit the potential of ignition. | Allow a few minutes of breaks in between manufacturing to encourage muscle recovery. Alternate workers between different tasks throughout the day. |

## 5. Activity Plan

### 5.1. Project Management Approach

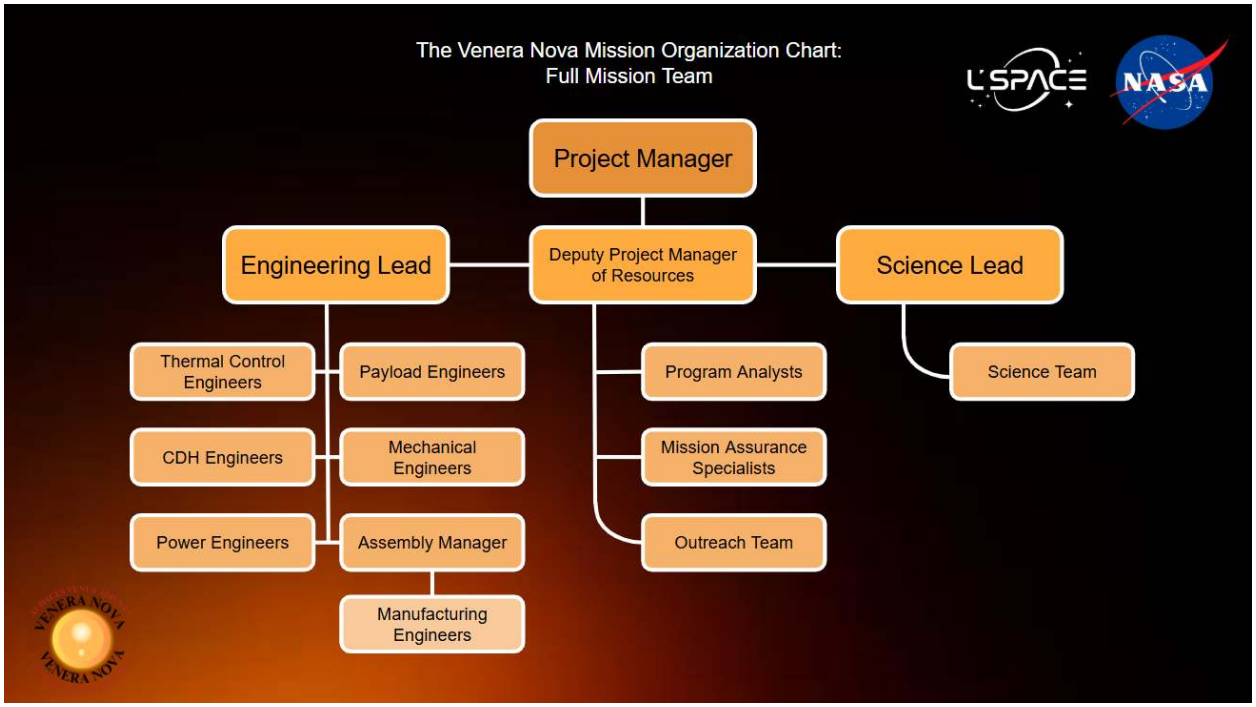



*Figure 34: Full team organizational chart. Leadership personnel are listed at the top of the chart, while all subteams are listed below.*

The Venera Nova team will consist of 3 subteams, whose personnel will vary throughout mission phases. These subteams are known as the programmatics, engineering, and science teams, who are responsible for managing different aspects of the Venera Nova mission. These teams will differ slightly in management approach, but they all will consist of one primary lead who will report to the singular project manager of the mission.

The project manager (PM) of the mission is responsible for delegating mission tasks to the programmatics, engineering, and science teams. The three subteam leads regularly report progress to the mission's project manager, who will provide feedback and streamline communication between the different teams. The project manager is responsible for overseeing all of the subteams' operations and ensuring that mission timelines are being met, as well as ensuring the wellbeing of staff.

The programmatics subteam is responsible for managing the budgeting and scheduling of the Venera Nova mission. Programmatics primarily ensures that the mission's costs do not exceed the $200,000,000 constraint, that the development of the Aerobot system remains on schedule to be ready for launch by March 1, 2029, and that

any risks that may affect the overall success of the mission are successfully evaluated and mitigated. If any subteam requires additional funding or deadline extensions, they must report to the programmatics subteam for approval. The programmatics team will then evaluate any potential risks with scheduling and either deny the request or relay the request to the project manager for final decisions. The entire subteam will consist of 10 employees who are skilled in project management, business, and budgeting throughout phases C-E of the mission. These employees will be organized into three teams: program analysts, mission assurance specialists, and outreach. Program analysts are responsible for managing the mission's budget, timelines, and changes in scope. Mission assurance specialists will evaluate any potential risks to the mission's success and come up with relevant mitigation strategies. Outreach personnel will assist in educating the public about the Venera Nova mission and updating people on the mission's progress. All of these teams will regularly communicate with a singular programmatics lead, known as the Deputy Project Manager of Resources (DPMR), who will report directly to the mission's PM about relevant programmatics operations. The DPMR is also responsible for staying updated on the science and engineering subteams' activities to ensure that risks are addressed in a timely manner.

The engineering subteam is responsible for designing subsystems that will accomplish the Venera Nova mission's scientific objectives. The Aerobot consists of the power, thermal control, mechanical, CDH, and payload subsystems which will allow the spacecraft to operate and function appropriately. The engineering team will be organized into five different groups, corresponding to the five different subsystems that the Aerobot is composed of. These groups will consist of 2-3 experienced engineers each, with skills and certifications pertinent to the development of their assigned subsystem. All subsystem personnel will create and submit necessary design blueprints throughout mission phase C, including CAD drawings and a bill of materials. A singular engineering lead will be responsible for overseeing the entire engineering team's operations throughout the entire mission life cycle and regularly communicating with the leadership team to update on progress. Any changes to engineering components in baselined subsystems must be communicated and approved by the engineering lead, who is then responsible for communicating such changes to the PM and relevant personnel. Additionally, the engineering subteam will be allocated $95,000,000 at the start of phase C for the development of the Aerobot subsystems, as estimated by the programmatics subteam with margin. Any requests to alter this allocation must be approved by the programmatics team and PM. Throughout the different mission phases, the entire engineering team will also expand in personnel to accelerate the research and development process, and to ensure that adequate testing can be observed and performed by multiple people. This process shall require a sixth group of Manufacturing Engineers and an Assembly Manager that are able to assemble purchased and specialized parts to construct the Aerobot System—this team of manufacturers shall be

monetarily compensated accordingly. Throughout manufacturing, a compound/facility shall be rented to house the Aerobot and provide space for necessary personnel to work on assigned systems. Manufacturers will be given a detailed layout of all subsystems through relevant blueprints and complete the assembly of the Aerobot within 6 months. For specialized equipment like SPICAV and the VMC, a certified professional shall be hired to oversee installation to ensure that the equipment is properly installed. For testing, many subsystem components must be transported to testing facilities. A singular driver shall be paid for each trip and to load the respective engineering components, with the Venera Nova team overlooking and assisting when necessary. The Assembly Manager and engineering lead are expected to communicate frequently throughout phases C-D of the mission to ensure that all subsystems are being developed appropriately.

The science subteam is responsible for streamlining the process for accomplishing the mission's scientific objectives. Science personnel are expected to set important scientific requirements for the mission and ensure that the fundamental goals of the Venera Nova mission are accomplished through the collection of relevant data and scientific instrumentation. The science subteam will be composed of 5-12 personnel throughout phases C-F of the mission who are skilled in scientific fields such as astrophysics, astrobiology, and spectroscopy. These personnel will be delegated tasks pertaining to their scientific specializations throughout the mission life cycle at the discretion of the subteam's science lead. This science lead, similarly to the other subteams, is responsible for delegating tasks to the science subteam and frequently communicating with the leadership team to update on scientific progress. Additionally, the science lead is expected to work closely with the engineering lead and payload subsystem personnel to confirm that scientific instrumentation is developed properly. Scientists specializing in space travel shall also plan the Aerobot's descent trajectory in Venus' atmosphere, confirming that specific mission goals are able to be conducted. Scientific research and development would likely require lab work in the form of equipment and possible material testing in conditions that are similar to the Venusian atmosphere. Lab space and technical help will be paid for in order to achieve this requirement.

Throughout the entirety of the mission, all personnel will be monetarily compensated individually. For launch, one crew will be hired to load the Aerobot system into the primary spacecraft, and a second crew shall be hired to be in charge of transportation from the assembly compound to the launch site. It is likely that these individuals shall be paid through their company—because of this, respective companies shall be paid directly. The MCA team shall occasionally require transportation to and

from facilities, which shall come in the form of air and road travel. Money shall be allocated to the team for purchasing travel tickets or car rides.

| | Phase C | Phase C-D | Phase D-E | Phase E-F |
|---|---|---|---|---|
| Science Personnel | 5 | 5 | 10 | 12 |
| Engineering Personnel | 12 | 14 | 16 | 18 |
| Technicians | 10 | 10 | 10 | 5 |
| Administration Personnel | 10 | 10 | 10 | 10 |
| Management Personnel | 4 | 4 | 4 | 4 |

*Table of personnel throughout phases C-F of the Venera Nova Mission's life cycle.*

## 5.2 Mission Schedule

### 5.2.1. Schedule Basis of Estimate

Margins and a more detailed explanation of the mission timeline are included in the Gantt Chart. Explanations of schedule margins and detailed explanations of mission timeline are included in the Mission Schedule section.

**Ground Rules**
Only Phases C-F are considered on this mission timeline. All major milestones must align with NASA's standard life-cycle reviews, such as the CDR, SIR, ORR, etc.

**Assumptions**
Aerobot manufacturing for instruments and engineering subsystems are able to be done concurrently, as the individual systems do not rely on each other and therefore can be completed at the same time. The beginning of Phase D of the Mission Schedule greatly benefits from this, as less time is required to construct all Aerobot components. Phase C is assumed to begin near the end of the MCA Academy.

**Drivers**

Dates for the start of Phase C, System Integration, and Launch are non-negotiable as they are provided as a customer constraint within the Mission Task document and are either set dates or a date after the academy ends.

### 5.2.2. Mission Schedule

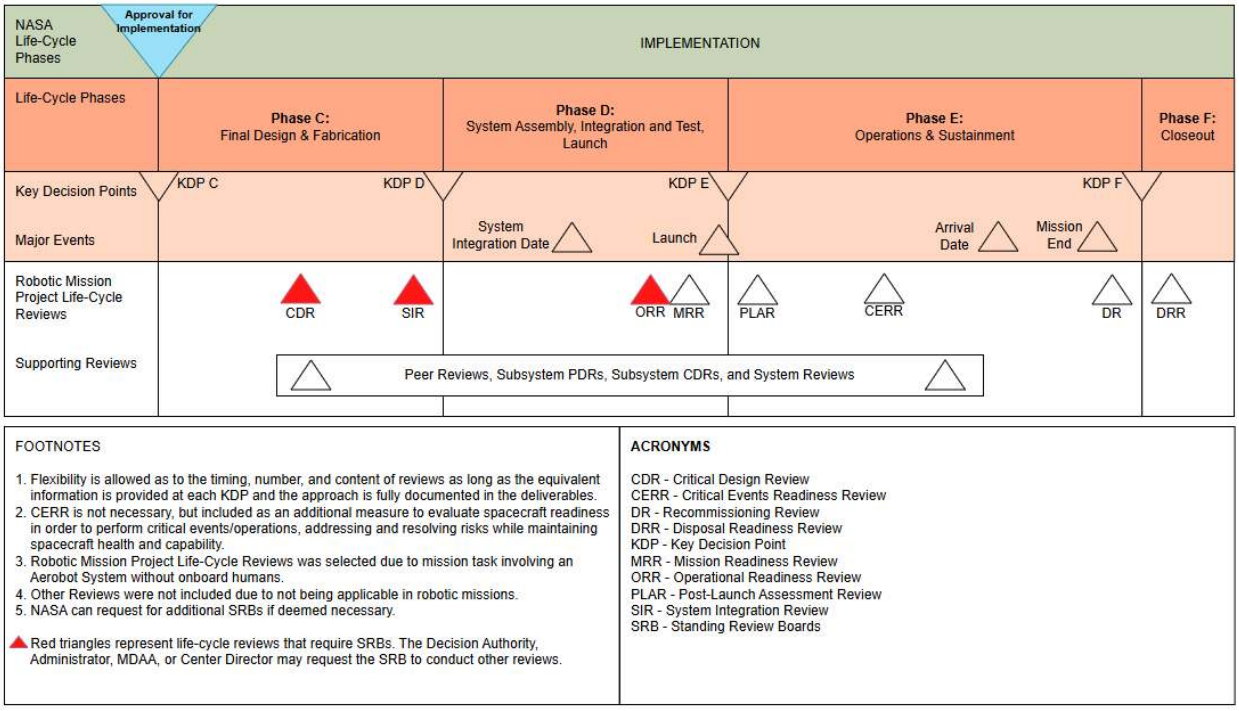
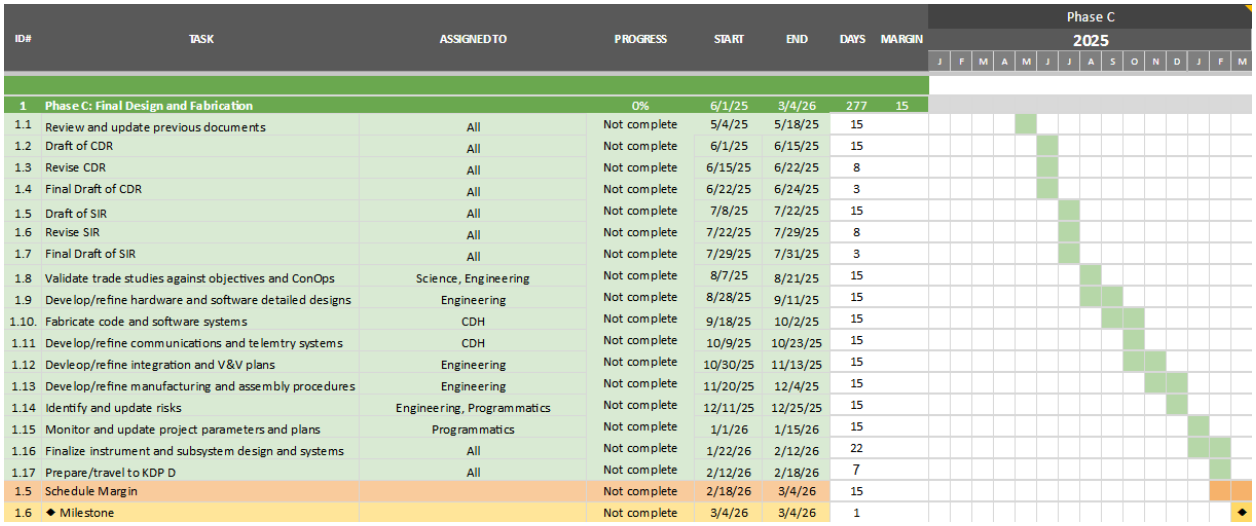



*Figure 35: Venera Nova Project Life Cycle*

The Project Life Cycle for the Venera Nova Mission provides a visual overview of the Mission's Life-Cycle Phases segmented into 3 major sections: Key Decision Points, Major Events, and the Robotic Mission Project Life-Cycle Reviews. Key Decision Points are in-person events held at NASA Headquarters in Washington D.C. that take place before the start of each phase to determine if the mission is on track to move on to the next phase. Major events such as the System Integration Date, Launch, Venus Arrival Date, and Mission End are provided by the Mission Task Document and are located within their respective phases based on the timelines. Other SRBs included in the Robotic Mission Project Life-Cycle Reviews are either requested or required by NASA based on the mission type.



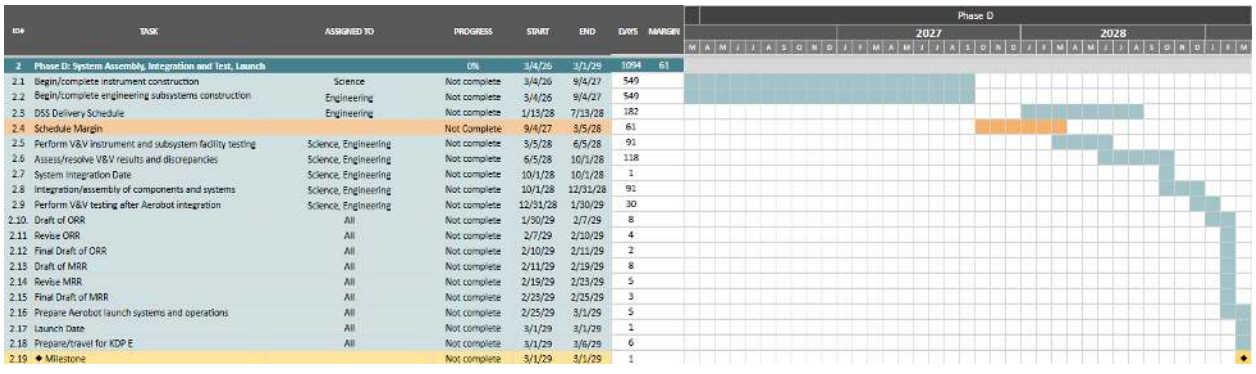
| ID# | TASK | ASSIGNED TO | PROGRESS | START | END | DAYS | MARGIN |
|---|---|---|---|---|---|---|---|
| 1 | **Phase C: Final Design and Fabrication** | | 0% | 6/1/25 | 3/4/26 | 277 | 15 |
| 1.1 | Review and update previous documents | All | Not complete | 5/4/25 | 5/18/25 | 15 | |
| 1.2 | Draft of CDR | All | Not complete | 6/1/25 | 6/15/25 | 15 | |
| 1.3 | Revise CDR | All | Not complete | 6/15/25 | 6/22/25 | 8 | |
| 1.4 | Final Draft of CDR | All | Not complete | 6/22/25 | 6/24/25 | 3 | |
| 1.5 | Draft of SIR | All | Not complete | 7/8/25 | 7/22/25 | 15 | |
| 1.6 | Revise SIR | All | Not complete | 7/22/25 | 7/29/25 | 8 | |
| 1.7 | Final Draft of SIR | All | Not complete | 7/29/25 | 7/31/25 | 3 | |
| 1.8 | Validate trade studies against objectives and ConOps | Science, Engineering | Not complete | 8/7/25 | 8/21/25 | 15 | |
| 1.9 | Develop/refine hardware and software detailed designs | Engineering | Not complete | 8/28/25 | 9/11/25 | 15 | |
| 1.10. | Fabricate code and software systems | CDH | Not complete | 9/18/25 | 10/2/25 | 15 | |
| 1.11 | Develop/refine communications and telemtry systems | CDH | Not complete | 10/9/25 | 10/23/25 | 15 | |
| 1.12 | Devleop/refine integration and V&V plans | Engineering | Not complete | 10/30/25 | 11/13/25 | 15 | |
| 1.13 | Develop/refine manufacturing and assembly procedures | Engineering | Not complete | 11/20/25 | 12/4/25 | 15 | |
| 1.14 | Identify and update risks | Engineering, Programmatics | Not complete | 12/11/25 | 12/25/25 | 15 | |
| 1.15 | Monitor and update project parameters and plans | Programmatics | Not complete | 1/1/26 | 1/15/26 | 15 | |
| 1.16 | Finalize instrument and subsystem design and systems | All | Not complete | 1/22/26 | 2/12/26 | 22 | |
| 1.17 | Prepare/travel to KDP D | All | Not complete | 2/12/26 | 2/18/26 | 7 | |
| 1.5 | Schedule Margin | | Not complete | 2/18/26 | 3/4/26 | 15 | |
| 1.6 | ◆ Milestone | | Not complete | 3/4/26 | 3/4/26 | 1 | |



*Figure 36: Gantt Chart*

**Phase C: Final Design and Fabrication**

Phase C serves as a final revision phase that focuses on completing the detailed design of the Aerobot system and all of its components, involving the fabrication and design of hardware, software systems, and the procedures to manufacture and assemble these subsystems. Deliverables, documents, and project plans are reviewed and updated accordingly to ensure the Aerobot system is ready for manufacturing and integration. The milestone for Phase C and requirements for KDP D is the completion of the CDR and SIR LCRs in addition to the review and finalization of component assembly procedures. Two week breaks are included between and after LCR deliverables while one week breaks are included between and after later tasks until the KDP D preparation in order to provide the team with ample time to complete their research and design of Aerobot subsystems. A 5% schedule margin is included for emergencies, but lowered due to forgiving task durations/intervals.



| ID# | TASK | ASSIGNED TO | PROGRESS | START | END | DAYS | MARGIN |
|---|---|---|---|---|---|---|---|
| 2 | **Phase D: System Assembly, Integration and Test, Launch** | | 0% | 3/4/26 | 3/1/29 | 1094 | 61 |
| 2.1 | Begin/complete instrument construction | Science | Not complete | 3/4/26 | 9/4/27 | 549 | |
| 2.2 | Begin/complete engineering subsystems construction | Engineering | Not complete | 3/4/26 | 9/4/27 | 549 | |
| 2.3 | DSS Delivery Schedule | Engineering | Not complete | 1/13/28 | 7/13/28 | 182 | |
| 2.4 | Schedule Margin | | Not Complete | 9/4/27 | 3/5/28 | 61 | |
| 2.5 | Perform V&V instrument and subsystem facility testing | Science, Engineering | Not complete | 3/5/28 | 6/5/28 | 91 | |
| 2.6 | Assess/resolve V&V results and discrepancies | Science, Engineering | Not complete | 6/5/28 | 10/1/28 | 118 | |
| 2.7 | System Integration Date | Science, Engineering | Not complete | 10/1/28 | 10/1/28 | 1 | |
| 2.8 | Integration/assembly of components and systems | Science, Engineering | Not complete | 10/1/28 | 12/31/28 | 91 | |
| 2.9 | Perform V&V testing after Aerobot integration | Science, Engineering | Not complete | 12/31/28 | 1/30/29 | 30 | |
| 2.10. | Draft of ORR | All | Not complete | 1/30/29 | 2/7/29 | 8 | |
| 2.11 | Revise ORR | All | Not complete | 2/7/29 | 2/10/29 | 4 | |
| 2.12 | Final Draft of ORR | All | Not complete | 2/10/29 | 2/11/29 | 2 | |
| 2.13 | Draft of MRR | All | Not complete | 2/11/29 | 2/19/29 | 8 | |
| 2.14 | Revise MRR | All | Not complete | 2/19/29 | 2/23/29 | 5 | |
| 2.15 | Final Draft of MRR | All | Not complete | 2/23/29 | 2/25/29 | 3 | |
| 2.16 | Prepare Aerobot launch systems and operations | All | Not complete | 2/25/29 | 3/1/29 | 5 | |
| 2.17 | Launch Date | All | Not complete | 3/1/29 | 3/1/29 | 1 | |
| 2.18 | Prepare/travel for KDP E | All | Not complete | 3/1/29 | 3/6/29 | 6 | |
| 2.19 | ◆ Milestone | | Not complete | 3/1/29 | 3/1/29 | 1 | |



*Figure 37: Phase D*

**Phase D: Final Design and Fabrication**

Phase D involves the construction, integration, and testing of all Aerobot instruments and subsystems, performing verification and validation and system preparation in order to ensure launch readiness within the given launch date. Instrument and subsystem construction is provided with at least 18 months, including an 11% margin of the original 18 months (549 days) in order to provide an additional 61 days to account for manufacturing or shipping delays. Facility testing requires at most 3 months (for the Thermal Vacuum) derived from MCCET facility costs and additional time is provided to assess and resolve V&V results. ORR and MRR durations were shortened due to not being a required SRB in order to meet non-negotiable launch date constraints. KDP E serves as the major milestone and will take place between Phases D and E, due to Phase E starting after launch. Major requirements to fulfill are the development, integration, and testing of the Aerobot system and completion of reviews to evaluate and ensure mission success. The 6 month DSS Delivery takes place during Phase D and has ample time before the non-negotiable System Integration Date, where the estimated time of arrival of 7/13/28 is provided by the Scope Change Overview slides. Instrument and subsystem construction is performed concurrently and do not conflict with each other.

| ID# | TASK | ASSIGNED TO | PROGRESS | START | END | DAYS | MARGIN |
|---|---|---|---|---|---|---|---|
| 3 | **Phase E: Operations and Sustainment** | | 0% | 3/1/29 | 10/29/29 | 243 | 13 |
| 3.1 | Draft of PLAR | All | Not complete | 3/1/29 | 3/15/29 | 15 | |
| 3.2 | Revise PLAR | All | Not complete | 3/15/29 | 3/22/29 | 8 | |
| 3.3 | Final Draft of PLAR | All | Not complete | 3/22/29 | 3/24/29 | 3 | |
| 3.4 | Draft of CERR | All | Not complete | 3/31/29 | 4/14/29 | 15 | |
| 3.5 | Revise CERR | All | Not complete | 4/14/29 | 4/21/29 | 8 | |
| 3.6 | Final Draft of CERR | All | Not complete | 4/21/29 | 4/23/29 | 3 | |
| 3.7 | Arrival Date to Venus | All | Not complete | 9/17/29 | 9/17/29 | 1 | |
| 3.8 | Conduct essential mission operations | Science, Engineering | Not complete | 9/17/29 | 9/19/29 | 3 | |
| 3.9 | Maintain support and analyze Aerobot health/risks | Engineering | Not complete | 9/17/29 | 9/19/29 | 3 | |
| 3.10. | Process and analyze mission data | Science, Engineering | Not complete | 9/17/29 | 9/19/29 | 3 | |
| 3.11 | Mission End | All | Not complete | 9/19/29 | 9/19/29 | 1 | |
| 3.12 | Draft of DR | All | Not complete | 9/19/29 | 10/3/29 | 15 | |
| 3.13 | Revise DR | All | Not complete | 10/3/29 | 10/10/29 | 8 | |
| 3.14 | Final Draft of DR | All | Not complete | 10/10/29 | 10/12/29 | 3 | |
| 3.15 | Prepare/travel for KDP F | All | Not complete | 10/12/29 | 10/17/29 | 6 | |
| 3.16 | Schedule Margin | | Not complete | 10/17/29 | 10/29/29 | 13 | |
| 3.17 | ◆ Milestone | | Not complete | 10/29/29 | 10/29/29 | 1 | |




*Figure 38: Phase E*

**Phase E: Operations and Sustainment**

Phase E focuses on the mission after launch, involving the completion of PLAR and CERR reviews as the spacecraft arrives in Venus' orbit after 200 days (according to the Mission Task document), where the Aerobot data collection and science objectives are being fulfilled while being monitored and supported by the team on Earth. A one week break is provided between the PLAR and CERR while a 5% margin is taken from the 231 days, using these 12 days to process mission data and work on the DR once the

Aerobot has been decommissioned. The requirements for KDP F and for the Phase E milestone is to maintain constant communication and support with the Aerobot system, fulfill team science objectives, and analyze collected mission data.

| ID# | TASK | ASSIGNED TO | PROGRESS | START | END | DAYS | MARGIN | Phase F 2029 O | N | D |
|---|---|---|---|---|---|---|---|---|---|---|
| 4 | **Phase F: Closeout** | | 0% | 10/29/29 | 12/21/29 | 54 | 3 | | | |
| 4.1 | Draft of DRR | All | Not complete | 10/29/29 | 11/12/29 | 15 | | | | |
| 4.2 | Revise DRR | All | Not complete | 11/12/29 | 11/19/29 | 8 | | | | |
| 4.3 | Final Draft of DRR | All | Not complete | 11/19/29 | 11/21/29 | 3 | | | | |
| 4.4 | Analyzed returned data | Science, Engineering | Not complete | 11/21/29 | 12/5/29 | 15 | | | | |
| 4.5 | Document lessons and final mission report | All | Not complete | 12/5/29 | 12/19/29 | 15 | | | | |
| 4.6 | Schedule Margin | | | 12/19/29 | 12/21/29 | 3 | | | | |
| 4.7 | ◆ Milestone | | Not complete | 12/21/29 | 12/21/29 | 1 | | | | ◆ |

*Figure 39: Phase F*

**Phase F: Closeout**

Phase F focuses on the end of the mission, involving the completion of the DRR review deliverable, analysis of the returned data, and final documentation regarding the Aerobot and team. The Aerobot system is not expected to survive and will not be collected after its decommission. A 5% margin of the 54 days accounts for a two day schedule margin to complete final mission summaries and data analysis before the official end of the mission.

## 5.3 Budget

### 5.3.1. Budget Basis of Estimate

Ground Rules, Assumptions, and Drivers are all based on the L'SPACE Mission Task document.

**Ground Rules**
The total budget for the mission is $200 million and will account for all phases post-PDR (Phases C-F). The Guidance, Navigation, and Control (GNC) Subsystem is not applicable in this mission.

**Assumptions**
Per Diem costs which include the Daily Lodging Rates and Meals and Incidental Expenses Rates will only account for major events such as Mission Launch and KDPs, as the total number of days spent traveling varies with margin and unexpected events. This is also true for transportation costs, as car rentals and other transportation methods can vary within each state and type of service, so a general margin and calculation are included for both sections. Daily Lodging Rates will be used as a total hotel cost. Personnel salary margin includes expenses for standard bonuses a typical employee receives. U.S. General Services Administration websites only offered calculations up to the current fiscal year (2025) and are assumed to be constant throughout the mission. System Testing is assumed to be conducted at the same facility where it was manufactured.

**Drivers**
The total mission budget is $200 million and will account for Phases C-F. The Direct Costs will be allocated into four subsystems: Mechanical, Power, Thermal, and CDH. Costs will be divided into Travel, Outreach, Direct, and Personnel. Direct costs are calculated using the MCCET, NICM, and CER tools.

**Discussion**
Ground Rules, Assumptions, and Drivers for Personnel, Direct, Travel, and Outreach Costs are further elaborated in their respective subsections. Margins are based on the Project Management skill module.

### 5.3.2. Total Mission Cost

| Additional Information | | | | | |
|---|---|---|---|---|---|
| | Phase C | Phase C-D | Phase D-E | Phase E-F | |
| # People on Team | FY 1 | FY 2 | FY 3 | FY 4 | |
| Science Personnel: | 5 | 5 | 10 | 12 | |
| Engineering Personnel: | 12 | 14 | 16 | 18 | |
| Technicians: | 10 | 10 | 10 | 5 | |
| Administration Personnel: | 10 | 10 | 10 | 10 | |
| Management Personnel: | 4 | 4 | 4 | 4 | |
| | | | | | |

# NASA L'SPACE Mission Concept Academy Budget - *Team 27*

| Mission Phase | Phase C | Phase D | Phase D | Phase E-F | |
|---|---|---|---|---|---|
| Year | Year 1 | Year 2 | Year 3 | Year 4 | Cumulative Total |
| **PERSONNEL** | | | | | |
| Science Personnel | $ 400,000 | $ 410,400 | $ 841,600 | $ 1,034,880 | $ 2,686,880 |
| Engineering Personnel | $ 960,000 | $ 1,149,120 | $ 1,346,560 | $ 1,552,320 | $ 5,008,000 |

| | | | | | |
|---|---|---|---|---|---|
| **Technicians** | **$ 600,000** | **$ 615,600** | **$ 631,200** | **$ 323,400** | **$ 2,170,200** |
| **Administration Personnel** | **$ 600,000** | **$ 615,600** | **$ 631,200** | **$ 646,800** | **$ 2,493,600** |
| **Project Management** | **$ 480,000** | **$ 492,480** | **$ 504,960** | **$ 517,440** | **$ 1,994,880** |
| | | | | | |
| **Total Salaries** | **$ 3,040,000** | **$ 3,283,200** | **$ 3,955,520** | **$ 4,074,840** | **$ 14,353,560** |
| **Total ERE** | **$ 848,464** | **$ 916,341** | **$ 1,103,986** | **$ 1,137,288** | **$ 4,006,079** |
| **Personnel Margin** | **$ 513,846** | **$ 513,846** | **$ 513,846** | **$ 513,846** | **$ 2,055,384** |
| | | | | | |
| **TOTAL PERSONNEL** | **$ 4,402,310** | **$ 4,835,935** | **$ 5,863,166** | **$ 6,172,600** | **$ 21,274,011** |
| **TRAVEL** | | | | | |
| **Total Flights Cost** | **$ 75,194** | **$ 75,194** | **$ 75,194** | **$ 75,194** | **$ 300,776** |
| **Total Hotel Cost** | **$ 22,166** | **$ 22,166** | **$ 22,166** | **$ 22,166** | **$ 88,664** |
| **Total Transportation Cost** | **$ 20,000** | **$ 20,000** | **$ 20,000** | **$ 20,000** | **$ 80,000** |
| **Total Per Diem Cost** | **$ 16,056** | **$ 16,056** | **$ 16,056** | **$ 16,056** | **$ 64,224** |
| **Travel Margin** | **$ 110,895** | **$ 110,895** | **$ 110,895** | **$ 110,895** | **$ 443,582** |
| | | | | | |
| **Total Travel Costs** | **$ 244,311** | **$ 250,663** | **$ 257,016** | **$ 263,368** | **$ 1,015,358** |

| OUTREACH | | | | | |
|---|---|---|---|---|---|
| Total Outreach Materials | $ - | $ - | $ - | $ - | $ - |
| Total Outreach Venue Costs | $ - | $ - | $ - | $ - | $ - |
| Total Outreach Travel Costs | $ 1,374 | $ 1,374 | $ 1,374 | $ 1,374 | $ 5,496 |
| Total Outreach Services Costs | $ 2,000 | $ 2,000 | $ 2,000 | $ 2,000 | $ 8,000 |
| Total Outreach Personnel Costs | $ 15,000 | $ 15,000 | $ 15,000 | $ 15,000 | $ 60,000 |
| Outreach Margin | $ 15,272 | $ 15,272 | $ 15,272 | $ 15,272 | $ 61,088 |
| | | | | | |
| Total Outreach Costs | $ 33,646 | $ 34,521 | $ 35,396 | $ 36,270 | $ 139,833 |
| **DIRECT COSTS** | | | | | |
| Mechanical Subsystem | $ 43,942 | $ 43,942 | $ 43,942 | $ 43,942 | $ 175,768 |
| Power Subsystem | $ 291,790 | $ 291,790 | $ 291,790 | $ 291,790 | $ 1,167,160 |
| Thermal Control Subsystem | $ 3,900,000 | $ 3,900,000 | $ 3,900,000 | $ 3,900,000 | $ 15,600,000 |
| Comms & Data Handling Subsystem | $ 2,182,010 | $ 2,182,010 | $ 2,182,010 | $ 2,182,010 | $ 8,728,040 |
| Guidance, Nav, & Control Subsystem | $ - | $ - | $ - | $ - | $ - |
| Science Instrumentation | $ 15,575,000 | $ 15,575,000 | $ 15,575,000 | $ 15,575,000 | $ 62,300,000 |
| Spacecraft Cost Margin | $ 2,742,055 | $ 2,742,055 | $ 2,742,055 | $ 2,742,055 | $ 10,968,220 |

| | | | | | |
|---|---|---|---|---|---|
| **Total Spacecraft Direct Costs** | **$ 24,734,797** | **$ 25,377,902** | **$ 26,021,006** | **$ 26,664,111** | **$ 102,797,816** |
| **Manufacturing Facility Cost** | **$ 4,925,000** | **$ 4,925,000** | **$ 4,925,000** | **$ 4,925,000** | **$ 19,700,000** |
| **Test Facility Cost** | **$ 3,692,500** | **$ 3,692,500** | **$ 3,692,500** | **$ 3,692,500** | **$ 14,770,000** |
| **Facility Cost Margin** | **$ 2,206,300** | **$ 2,206,300** | **$ 2,206,300** | **$ 2,206,300** | **$ 8,825,200** |
| **Total Facilities Costs** | **$ 10,823,800** | **$ 11,105,219** | **$ 11,386,638** | **$ 11,668,056** | **$ 44,983,713** |
| | | | | | |
| **Total Direct Costs** | **$ 35,558,597** | **$ 36,483,121** | **$ 37,407,644** | **$ 38,332,168** | **$ 147,781,529** |
| **Total MTDC** | **$ 24,734,797** | **$ 25,377,902** | **$ 26,021,006** | **$ 26,664,111** | **$ 102,797,816** |
| **FINAL COST CALCULATIONS** | | | | | |
| **Total F&A** | **$ 2,199,274** | **$ 2,263,585** | **$ 2,327,895** | **$ 2,392,206** | **$ 9,182,960** |
| | | | | | |
| **Total Projected Cost** | **$ 42,438,139** | **$ 43,867,825** | **$ 45,891,116** | **$ 47,196,611** | **$ 179,393,691** |
| **Total Cost Margin** | **$ 5,588,368** | **$ 5,588,368** | **$ 5,588,368** | **$ 5,588,368** | **$ 22,353,474** |
| | **13.2%** | **12.7%** | **12.2%** | **11.8%** | |
| | | | | | |
| **Total Project Cost** | **$ 42,438,139** | **$ 43,867,825** | **$ 45,891,116** | **$ 47,196,611** | **$ 179,393,691** |

Including the $20,000,000 debrief cost, the total cost is $199,393,691. The inflation rates for the following years are as follows: 0.0%, 2.6%, 5.2%, and 7.8%.

### 5.3.3. Personnel Budget

| | Phase C | Phase C-D | Phase D-E | Phase E-F |
|---|---|---|---|---|
| **# People on Team** | FY 1 | FY 2 | FY 3 | FY 4 |
| Science Personnel: | 5 | 5 | 10 | 12 |
| Engineering Personnel: | 12 | 14 | 16 | 18 |
| Technicians: | 10 | 10 | 10 | 5 |
| Administration Personnel: | 10 | 10 | 10 | 10 |
| Management Personnel: | 4 | 4 | 4 | 4 |

| PERSONNEL | | | | | |
|---|---|---|---|---|---|
| **Science Personnel** | $ 400,000 | $ 410,400 | $ 841,600 | $ 1,034,880 | $ 2,686,880 |
| **Engineering Personnel** | $ 960,000 | $ 1,149,120 | $ 1,346,560 | $ 1,552,320 | $ 5,008,000 |
| **Technicians** | $ 600,000 | $ 615,600 | $ 631,200 | $ 323,400 | $ 2,170,200 |
| **Administration Personnel** | $ 600,000 | $ 615,600 | $ 631,200 | $ 646,800 | $ 2,493,600 |
| **Project Management** | $ 480,000 | $ 492,480 | $ 504,960 | $ 517,440 | $ 1,994,880 |
| | | | | | |
| **Total Salaries** | $ 3,040,000 | $ 3,283,200 | $ 3,955,520 | $ 4,074,840 | **$ 14,353,560** |
| **Total ERE** | $ 848,464 | $ 916,341 | $ 1,103,986 | $ 1,137,288 | **$ 4,006,079** |
| **Personnel Margin** | $ 913,846 | $ 913,846 | $ 913,846 | $ 913,846 | **$ 3,655,384** |
| | | | | | |
| **TOTAL PERSONNEL** | $ 4,802,310 | $ 5,246,335 | $ 6,283,966 | $ 6,603,800 | $ 22,936,411 |

The total number of people on the team includes members of the Venera Nova mission and staff hired throughout the mission phases. The number of management and administration personnel is maintained throughout the course of the mission, as increasing or decreasing this number can greatly affect the team dynamic and teamwork, and thus should not change. Technicians are required from Phases C through the beginning of E, where instrumentation and subsystem component research, construction, and assembly is being completed, requiring technicians for specialized fields. The number of engineers increase throughout the phases as more construction and assembly requires more engineers, but also maintaining the Aerobot's communications and health alongside data collection since engineers have a greater understanding of the subsystems and mechanisms. Science personnel from Phases C-D mostly conduct research on science instrumentations and objectives, but greatly increase from Phases D-E for data analysis and collection. Management, Administration, and Technicians will work full time while Science and Engineering are working part-time.

A personnel margin of 10% is included for additional expenses and employee benefits.

### 5.3.4 Travel Budget

The following is applicable for all figures in the Travel Budget section. The costs for one-way airfares use YCA and a connecting service in order to account for the highest found costs. Airline, costs, and distance are determined from each member's nearest airport from their college available on the City Pair Program website (U.S. General Services Admission, 1980) and calculated with the assistance of Air Miles Calculator (airmilescalculator, n.d.). Distance is rounded to the nearest whole number. The costs and distance for the tables are shown as one-way airfares.

Event: Mission Launch
Date: March 1, 2029
Destination Airport: Orlando International Airport (MCO)

| Name | City | State | Origin Airport | Airline | Cost ($) | Distance (miles) |
|---|---|---|---|---|---|---|
| Tyree Nguyen | Pomona | CA | Los Angeles, CA - ONT | American | 329 | 2171 |
| Alaina Claudio | Culver City | CA | Burbank, CA - BUR | American | 333 | 2215 |
| Isabela Benavidez | Los Angeles | CA | Burbank, CA - BUR | American | 333 | 2215 |
| Gregory Jabido | Pullman | WA | Spokane, WA - GEG | Alaska | 213 | 2346 |
| Rachel Brown | Irvine | CA | Santa Ana, CA - SNA | Southwest | 293 | 2187 |
| Issac Munoz | Pomona | CA | Los Angeles, CA - ONT | American | 329 | 2171 |
| Victor Chen | Hayward | CA | Oakland, CA - OAK | Southwest | 458 | 2438 |
| Valerie Miranda | Las Vegas | NV | Los Angeles, CA - ONT | American | 329 | 2171 |
| Patrick Nguyen | Irvine | CA | Santa Ana, CA - SNA | Southwest | 293 | 2187 |
| Manahil Syeda | Berkeley | CA | Oakland, CA - OAK | Southwest | 458 | 2438 |
| Elias Pedroza | Santa Clara | CA | San Jose, CA - SJC | Southwest | 311 | 2421 |
| Andrew Hoang | Los Angeles | CA | Burbank, CA - BUR | American | 333 | 2215 |

*Figure 40: Mission Launch Air Transportation Costs Table*

All members will be arriving at the Orlando International Airport (MCO) adjacent to the Kennedy Space Center. The total cost to arrive at MCO and return to each member's origin airport is $8,024. The duration of this trip is 5 days, accounting for arriving 2 days prior and departing 2 days after the launch.

Event: Key Decision Points (KDPs)
Date: 5 days prior the next phase in the NASA Mission Life Cycle
Destination Airport: Ronald Reagan Washington National Airport (DCA)

| Name | City | State | Origin Airport | Airline | Cost ($) | Distance (miles) |
|---|---|---|---|---|---|---|
| Tyree Nguyen | Pomona | CA | Los Angeles, CA - ONT | American | 450 | 2265 |
| Alaina Claudio | Culver City | CA | Burbank, CA - BUR | Southwest | 340 | 2302 |
| Isabela Benavidez | Los Angeles | CA | Burbank, CA - BUR | Southwest | 340 | 2302 |
| Gregory Jabido | Pullman | WA | Spokane, WA - GEG | American | 329 | 2105 |
| Rachel Brown | Irvine | CA | Santa Ana, CA - SNA | American | 400 | 2288 |
| Issac Munoz | Pomona | CA | Los Angeles, CA - ONT | American | 450 | 2265 |
| Victor Chen | Hayward | CA | Oakland, CA - OAK | Southwest | 527 | 2432 |
| Valerie Miranda | Las Vegas | NV | Los Angeles, CA - ONT | American | 450 | 2265 |
| Patrick Nguyen | Irvine | CA | Santa Ana, CA - SNA | American | 400 | 2288 |
| Manahil Syeda | Berkeley | CA | Oakland, CA - OAK | Southwest | 527 | 2432 |
| Elias Pedroza | Santa Clara | CA | San Jose, CA - SJC | Southwest | 231 | 2424 |
| Andrew Hoang | Los Angeles | CA | Burbank, CA - BUR | Southwest | 340 | 2302 |

*Figure 41: KDPs Air Transportation Costs Table*

All members will be arriving at the Ronald Reagan Washington National Airport (DCA) near NASA Headquarters. The total cost of arrival and return from DCA for each member is $9,568. There are a total of 3 KDPs (KDPs C, D, and E) and the total cost of all KDPs is $28,704.

Event: Hardware Manufacturing and Testing
Date: March 4, 2026
Destination Airport: Varies on the Supplier

| Subteam | Component | Supplier | Destination Airport | Origin Airport | Airline | Cost ($) | Distance (miles) |
|---|---|---|---|---|---|---|---|
| Mechanical | Electric Propulsion System | Busek Space Propulsion and Systems | Boston, MA - BOS | Washington, DC - DCA | Southwest | 254 | 399 |
| | Closed Eye Bolt with Locking Nut ⅜-16 | McMaster-Carr (Off-the-shelf) | Los Angeles, CA - LAX | Washington, DC - DCA | American | 400 | 2310 |
| | Titanium Flat Washer (1/4") - Ultra-Corrosion-Resistant Alloy 20 Stainless Steel | McMaster-Carr (Off-the-shelf) | Los Angeles, CA - LAX | Washington, DC - DCA | American | 400 | 2310 |
| | Titanium Hex Nut (1/4"-20) | McMaster-Carr (Off-the-shelf) | Los Angeles, CA - LAX | Washington, DC - DCA | American | 400 | 2310 |
| | Grade 5 Titanium (Ti-6Al-4V) | OnlineMetals | Seattle, Washington - SEA | Washington, DC - DCA | American | 319 | 2329 |
| Power | (COTS) Triple Junction Solar Cells integrated into a custom solar panel | SpectroLab | Los Angeles, CA - LAX | Washington, DC - DCA | American | 400 | 2310 |
| | (COTS) Thermal Battery | Pumpkin Space | San Francisco, CA - SFO | Washington, DC - DCA | Southwest | 527 | 2442 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | (COTS) Power Conditioning and Distribution Unit (PCDU) | Bradford Space | Luxembourg, Luxembourg - LUX | Washington, DC - DCA | Delta | 1655 | 3983 |
| CDH | S-Band TT&C Antenna | Anywaves | Toulouse, France - TLS | Washington, DC - DCA | American | 426 | 3941 |
| | S-Band High Gain Patch Antenna | IQ Spacecom | Berlin, Germany - BER | Washington, DC - DCA | United | 457 | 4192 |
| | MicroSDR-C | Space Micro | San Diego, CA - SAN | Washington, DC - DCA | American | 305 | 2275 |
| | Crotex-M7, 384 kB RAM, 2 MB Flash Memory, CAN/LVDS Data Interface | ARM | San Jose, CA - SJC | Washington, DC - DCA | Southwest | 231 | 2423 |
| Thermal | Z93 - White Coating | AZ Technology | Huntsville, AL - HSV | Washington, DC - DCA | American | 444 | 613 |
| | Multi-Layer Insulation | Northrop Grumman | San Jose, CA - SJC | Washington, DC - DCA | Southwest | 231 | 2423 |
| | Loop Heat Pipe (LHP) | Arquimea | Los Angeles, CA - LAX | Washington, DC - DCA | American | 400 | 2310 |
| | Radiator | Arquimea | Los Angeles, CA - LAX | Washington, DC - DCA | American | 400 | 2310 |
| | All-Polyimide Thermofoil Heater - HAP6948-1 | Minco | Minneapolis/ST PAUL, MN - MSP | Washington, DC - DCA | Delta | 546 | 931 |
| Science | SOIR (SPICAV) | Belgian Institute for Space | Brussels, Belgian - BRU | Washington, DC - DCA | United | 1097 | 3879 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | Aeronomy | | | | | |
| | UV Spectrometer (SPICAV) | Service d'Aeronomie | Paris, France - PAR | Washington, DC - DCA | United | 431 | 3847 |
| | VMC | Max Planck Institute for Solar System Research | Frankfurt, Germany - FRA | Washington, DC - DCA | United | 1132 | 4068 |

*Figure 42: Hardware Manufacturing and Testing Transportation Costs Table*

Since subsystem and instrument components begin construction and testing after KDP D in Washington D.C., the origin airport for all members will be from Ronald Reagan Washington National Airport (DCA). After component construction and testing, members will be flown back to their origin airports near their colleges using the costs from the previous figure. Members on respective subteams are listed within the Team Org Chart and will be referred to in order to calculate the total flight costs. Components and suppliers are taken from the Manufacturing and Procurement Plans sections of the PDR. Construction and testing is conducted at the same facilities. Members of multiple subteams will travel for components of each subteam. The total costs from DCA Airport to the respective destination airports and back to the original near college airports is $38,466.

**Daily Lodging Rates**

Daily Lodging Rates are calculated using the U.S. General Services Administration Per Diem Website (U.S. General Services Admission, 1980) For the Mission Launch that takes place in 2029., the daily lodging rate is $169 during the month of March in Orange County for each member, excluding taxes. The combined total cost for daily lodging rates is $2,366. For the KDPs that take place in Washington D.C., there is no applicable county or state and thus the standard rate will be used, which is $110 per day. For a total of 5 days during each of 3 occurrences of 12 members, the total cost is $19,800.

**Meals and Incidental Expenses (M&IE) Rates**

Meals and Incidental Expenses are calculated using the U.S. General Services Administration Per Diem Website. For Mission Launch, the M&IE total for each member is $360 throughout the duration of the 5-day trip. The M&IE cost for each day is $80 while the first and last days are each $60. The total cost for M&IE including each member is $5,040. The M&IE cost for KDPs will use the standard rate in the state of Washington. The total for 5 days, where the first and last day of travel is $51 and M&IE total is $68 per day for each member for a total of 3 occurrences, the total cost is $11,016.

**Alternate Transportation Costs**

Costs of car rentals and other transportation methods vary heavily depending on the location, such as the city, state, and country. Therefore, a transportation of $20,000 will be allocated for all transportation expenses other than airfares. Other transportation costs will be accounted for in the margin.

| TRAVEL | | | | | |
|---|---|---|---|---|---|
| **Total Flights Cost** | $ 75,194 | $ 75,194 | $ 75,194 | $ 75,194 | **$ 300,776** |
| **Total Hotel Cost** | $ 22,166 | $ 22,166 | $ 22,166 | $ 22,166 | **$ 88,664** |
| **Total Transportation Cost** | $ 20,000 | $ 20,000 | $ 20,000 | $ 20,000 | **$ 80,000** |
| **Total Per Diem Cost** | $ 16,056 | $ 16,056 | $ 16,056 | $ 16,056 | **$ 64,224** |
| **Travel Margin** | $ 110,895 | $ 110,895 | $ 110,895 | $ 110,895 | **$ 443,582** |
| | | | | | |
| **Total Travel Costs** | $ 244,311 | $ 250,663 | $ 257,016 | $ 263,368 | **$ 1,015,358** |

*Figure 43: Travel Budget*

A 20% travel margin from the total travel costs will be applied to the original $554,477 to account for all other travel expenses and emergencies.

### 5.3.5. Outreach Budget

Outreach costs consist of social media promotion, a portable planetarium, and door to door advertising. Instagram would be the main source of mission advertisement as each post will have links to mission outlets. Instagram influencers are classified to fall into different categories according to the number of followers on their page, because Venera Nova is a future mission, each advertiser will be paid on the “micro-influencer” tier. Social media outreach consists of 15 paid promotional posts from influential figures within the aerospace industry, estimated to cost $500 per post. $2,000 will be allocated for the production of a portable planetarium, which will travel bi-weekly to different high schools for free, educational showcases. During the weekend, the planetarium show will cost $5 for entry to the public. Three tours will be done regarding the planetarium presentation with the month and year of July 2026, August 2027, December 2028. These tours are meant to influence the audience by showing what the Venera Nova mission entails and keep them motivated for the launch date. It will also influence young scholars to pursue a career in space exploration and follow mission launch date as well. Door to door advertisement will be run by volunteers, similar to a political campaign. The volunteer team will ask residents for contact information to advertise launch date and mission updates. The campaign team will utilize a QR code for all residents to scan and be transferred to the Venera Nova official website and link to the Venera Nova instagram, Facebook, and LinkedIn page. The website will explain the mission objectives and team values. Donations for mission launch advertisement will also be advertised on the Venera Nova website, but not required. Donations will go toward advertisement cost, website design, and other miscellaneous costs the team utilizes.

Influencers are categorized under Outreach Personnel. At least two influencers posting 15 paid posts of $500 each will cost $15,000. Outreach services will account for the initial planetarium cost of $2,000. The Venera Nova Team does not plan on using venues and will instead bring their personal materials for outreach causes. Volunteers will also assist with outreach work; thus,, the total cost of Outreach Materials and Venue Costs is $0.00. Outreach travel costs includes the price of gas for hauling the portable planetarium, calculated with the average price of gas in the US to be $3.16 per gallon and at least 10,000 miles with a truck of 23 MPG, will cost $1,373.91, rounded to $1,374. The industry standard margin of 20% is applied to the original cost of outreach ($76,362)  and will be $15,272.

| OUTREACH | | | | | |
|---|---|---|---|---|---|
| **Total Outreach Materials** | $ - | $ - | $ - | $ - | **$ -** |
| **Total Outreach Venue** | $ - | $ - | $ - | $ - | **$ -** |

| | | | | | |
|---|---|---|---|---|---|
| **Costs** | | | | | |
| **Total Outreach Travel Costs** | $ 1,374 | $ 1,374 | $ 1,374 | $ 1,374 | **$ 5,496** |
| **Total Outreach Services Costs** | $ 2,000 | $ 2,000 | $ 2,000 | $ 2,000 | **$ 8,000** |
| **Total Outreach Personnel Costs** | $ 15,000 | $ 15,000 | $ 15,000 | $ 15,000 | **$ 60,000** |
| **Outreach Margin** | $ 15,272 | $ 15,272 | $ 15,272 | $ 15,272 | **$ 61,088** |
| | | | | | |
| **Total Outreach Costs** | $ 33,646 | $ 34,521 | $ 35,396 | $ 36,270 | **$ 139,833** |

### 5.3.6. Direct Costs

There is a 5% Facility Cost and Spacecraft Cost Margin.

**Thermal Control Subsystem:**

| Thermal Subsystem Direct Costs | | |
|---|---|---|
| **Component** | **Cost** | **Formulas, Estimation Method?** |
| Z93 - White Coating | $900,000 | Current cost of AZ-93 white coating with no inflation and MCCET |
| Multi-Layer Insulation (MLI) | $4,000,000 | Thermal Subsystem CER and MCCET |
| Loop Heat Pipe (LHP) | $4,400,000 | Thermal Subsystem CER and MCCET |
| Radiator | $5,900,000 | Thermal Subsystem CER and MCCET |
| All-Polyimide Thermofoil Heater - HAP6948-1 | $400,000 | Current cost of flexible heater with no inflation and MCCET |
| Total | $15,600,000 | |

*Estimated Direct Costs of Thermal Control Components*

Z93 (white paint) is a passive thermal control component (0W) that will coat the side face where the radiator is located on the cubic system. To calculate the coating's final manufacturing cost ($700,000) and the final testing facility cost ($200,000), an estimated cost per square meter of Z93 and the MCCET was used. The coating's mass is negligible.

MLI is a passive thermal control component (0W) covering the top, bottom, and three sides of the cubic meter system. Based on a thermal mass estimation of 0.7 kg/$m^2$, a value of 3.5 kg was substituted into the thermal/fluids subsystem CER (Birur and Swanson, 2003). The CER's output value was then substituted into the MCCET to obtain a final manufacturing cost of $3,100,000 and a final testing facility cost of $900,000.

An LHP is a two-phase passive thermal control device (0W) that transfers heat using capillary action to circulate fluid in a closed loop. The fluid evaporates at the heat source and condenses at an attached radiator. The exact mass of Team 27's LHP is difficult to estimate as LHPs are highly customizable; therefore, an approximation of 4 kg was made based on prior space missions (Thayer et al., 2023). This value was substituted into the MCCET, resulting in a final manufacturing cost of $3,400,000 and a final testing facility cost of $1,000,000.

A radiator is a passive thermal control system (0W) placed on one side of the cubic meter system. It will be coated in Z93 and made of 6061 Al. A mass of 6.36 kg was found through measurements from CAD. From the MCCET, the final manufacturing cost ($4,500,000) and the final testing facility costs ($1,400,000) were estimated.

The All-Polyimide Thermofoil Heater - HAP6948-1 is an active thermal control component that serves as the heater for the cubic system. The flexible heater’s low mass of 0.00318 kg caused unrealistic outputs from the thermal CER and MCCET. Therefore, the product’s current price ($179.99) with no inflation was substituted into the MCCET. Resulting in a final manufacturing cost of $300,000 and a final testing facility cost of $100,000.

All MCCET calculations, except for the Z93 - White Coating and the Thermofoil Heater, utilized an inflation rate of 170.53% relative to 2004 and 2025. The total manufacturing cost is $13,000,000 and the total testing cost is $3,900,000.

**Mechanical Subsystem:**

| Mechanical Subsystem Direct Costs | | |
|---|---|---|
| **Component** | **Cost** | **Formulas, Estimation Method?** |
| Electric Propulsion System BHT-600 Hall Effect Thruster | $150,000 | Busek Propulsion Product |

| | | |
|---|---|---|
| Closed Eye Bolt with Locking Nut ⅜-16 | $55.36 | Price found on McMaster-Carr's website. |
| Titanium Flat Washer (1/4") - Ultra-Corrosion-Resistant Alloy 20 Stainless Steel | $116.40 | Price found on McMaster-Carr's website |
| Titanium Hex Nut (1/4"-20) | $163.8 | Price found on McMaster-Carr's website |
| Grade 5 Titanium (Ti-6Al-4V) | $25,441.0 | Price found on Online Metal's website |
| Total | $175,766.56 | |

*Estimated Direct Costs of Mechanical Subsystem*

For the electric propulsion system, the BHT-600 Hall Effect Thruster is the model that will be used. While the company does not specify their pricing, the cost of the thruster was estimated to be $150,000 based on thrusters with a similar design that were used in the aerospace industry. For the closed eye bolt with locking nut, titanium flat washer, and titanium hex nut, each component will be purchased directly from McMaster-Carr. The corresponding prices for each component were determined by the price listed on the manufacturers website multiplied by the units required. The bolt is priced at $13.84 for each unit, making the total cost $55.36 for four bolts. The washer is $1.94 per unit and 60 will be purchased, leading to a total cost of $116.4. The hex nut is priced at $2.73 per unit, with a total cost of $163.8 for 60 units. The Grade 5 Titanium (Ti-6Al-4V) will be purchased from OnlineMetals. The cost of one sheet with the required thickness and similar dimensions to that of the Aerobot is $1,839.63. Multiplying this price by 6 for each side and rounding up for a slight difference in cost due to dimensioning results in a total cost of $11,100.00. For the chassis, the price of one square meter of Ti-6Al-4V is $4,070.74. This cost was multiplied by the area of the chassis, which was 3.52 $m^2$, making the total cost $14,341. Adding all these costs together results in a total cost of $175,766.56.

**Power Subsystem:**

| **Power Subsystem Direct Costs** | | |
|---|---|---|
| **Component** | **Cost** | **Formulas, Estimation Method?** |
| SpectroLab XTE-SF | $214,000 | 1000W required: 856 cells (1.17W/cell) * |

| | | |
|---|---|---|
| | | $250/W = $214,000 |
| Space Panel Manufacturing from SpectroLab | $539,410 | Wiring: $100/cell * 856 cells = $85,600<br>Aluminum honeycomb structure: $1250/m^2 * 2.31 m^2 = $2887.50<br>Bypass diodes: $30/cell * 856 cells = $25,680<br>Thermal coating/insulation: $1750/m^2 * 2.31 m^2 = $4042.50<br>Cleanroom assembly: $200/cell * 856 cells = $171,200<br>Testing (Vibration, Thermal, Radiation, Deployment): $125,000/panel * 2 panels = $250,000 |
| Pumpkin BM2 (4S2P Configuration) | $69,000 | 1000W required: 6 units (168W/unit) * $11,500/unit = $69,000 |
| Bradford Space SuperNova PCDU | $345,000 | Single unit cost |
| Total | $1,167,160 | |

*Estimated Direct Costs of Electrical Power Subsystem*

The total cost for the development of the Electrical Power System is derived from the Mission Concept Cost Estimate Tool (MCCET) to be $5,574,289.75. The individual component's estimated total cost is $1,167,160, with the difference coming from personnel, facilities, and manufacturing/testing costs. The Power Generation Subsystem costs $753,410 with $214,000 coming from the cost of 856 SpectroLab's Triple Junction Solar Cells. The mission specifically requires 856 solar cells since the target power generation is 1000W and each cell generates 1.17W. The specific cost of the solar cell is not publicly available, so the cost is estimated to be $250/cell coming from a lower estimate of $100/W = 117/cell to an upper estimate of $400/W = 468/cell. (Techno-Economic Analysis of Three Different Substrate Removal and Reuse Strategies for III-V Solar Cells.) The cost for SpectroLab to manufacture the solar panel, integrating the 856 solar cells into a deployable solar panel is $539,410 coming from the cost of the wiring, aluminum honeycomb base structure, bypass diodes, insulation, assembly, and testing. The cost of the Power Storage Subsystem comes from the Pumpkin BM2 Thermal Batteries totaling $69,000. With a target of 1000W storage, there must be 6 batteries, each costing $11,500, as each holds 168W/unit. The Power Management and Distribution Subsystem costs $345,000 consisting of a single Bradford Space SuperNova PCDU.

**CDH Subsystem:**

| Power Subsystem Direct Costs | | |
|---|---|---|
| **Component** | **Cost** | **Formulas, Estimation Method?** |
| ANYWAVES S-Band TT&C Antenna | $1,766,264.31 | MCCET |
| IQ Spacecom S-Band High Gain Patch Antenna | $2,209,165.15 | MCCET |
| Space Micro MicroSDR-C | $6,330,398.37 | MCCET |
| ARM Crotex-M7, 384 kB RAM, 2 MB Flash Memory, CAN/LVDS Data Interface | $1,440,424.16 | MCCET |
| Total | $8,728,039.20 | |

*Estimated Direct Costs of CDH Subsystem*

Initial cost estimated ANYWAVES S-Band TT&C Antenna to be $7154.04, IQ Spacecom S-Band High Gain Patch Antenna to be $8000, Space Micro MicroSDR-C to be $2709 and ARM Crotex-M7, 384 kB RAM, 2 MB Flash Memory, CAN/LVDS Data Interface to be $30. When calculated via MCCET, the low and high gain antennas, ANYWAVES S-Band TT&C Antenna costs $1,766,264.31 and IQ Spacecom S-Band High Gain Patch Antenna costs $2,209,165.15. Furthermore, Space MicroSDR-C costs $6,330,398.37 and ARM Crotex-M7, 384 kB RAM, 2 MB Flash Memory, CAN/LVDS Data Interface costs $1,440,424.16. This totals out to $8,728,039.20. Calculations were performed with an inflation rate of 2.8%.

Management costs were calculated via MCCET, and totaled to $430,000.00. System engineering costs totaled to $360,000.00, and production assurance costs totaled to $430,000.00. Integration & test costs totaled to $1,520,000.00. This gave a total cost per unit of $11,500,000.00. Test facility costs were calculated via MCCET. TVAC totaled to $1,730,000.00, EMI totaled to $860,000.00, VIBE totaled to $430,000.00 and Ambient costs totaled to $430,000.00.

For phase B, manufacturing costs totaled to $460,000.00 and testing costs totaled to $140,000.00. Phase C manufacturing costs totaled to $6,555,000.00 and

testing costs totaled to $1,995,000.00. Phase D manufacturing costs totaled to $4,485,000.00 and testing costs totaled to $1,365,000.00.

MCCET calculated schedule estimate relation for planetary, remote sensing instruments (Optical and microwave) to be 118.0 months. And fields and particles calculated to be 154.9 months. Planetary in situ instruments were calculated to be 60.7 months.

**Payload Subsystem:**

| Payload Subsystem Direct Costs | | |
|---|---|---|
| **Component** | **Cost** | **Formulas, Estimation Method?** |
| VMC | $9,400,000 | Payload Subsystem CER and MCCET |
| SPICAV | $52,900,000 | Payload Subsystem CER and MCCET |
| Total | $62,300,000 | |

*Estimated Direct Costs of Payload Subsystem*

Payload costs were estimated using the MCCET and relevant CER equation. The Optical Planetary Instruments CER was used to estimate the VMC's estimated cost without wraps and adjusted for inflation. Subsequently, the wrap cost for VMC was calculated to be $7,200,000 and testing facility cost was estimated to be $2,200,000. In total, the VMC is estimated to cost $9,400,000.

SPICAV's cost was estimated similarly to VMC, with the optical Planetary Instruments CER yielding a cost of $30,396,547.20 without wraps. When considered with wraps and manufacturing costs, SPICAV was estimated to cost $40,700,000. Facility testing costs were estimated to be $12,200,000; thus, the total estimated cost for SPICAV is $52,900,000.

## 5.4 Scope Management

### 5.4.1. Change Control Management

**Change Control**
**RFAs & ADVs:**
The team will take note and discuss as a team regarding requests for action items (RFAs) and advisory action items (ADVs) to make immediate changes if required with utmost importance. The team will have meetings 1-2 meetings every week on discord to discuss any changes, new information, or problems that arise. If members are not able to make the meeting, notes will be taken and posted on a google drive and shared to all.

**Team-Requested/Internal Change:**

If the team plans on requesting a change, the team will first discuss it amongst the whole team and document the change to be made to show stakeholders/customers. Then, the project manager and other required members of the team will sit down with stakeholders to discuss such changes to work out an agreement.

**Customer-Requested Change:**
In the case of customer changes, the project manager and specific team leads will have a meeting to discuss with the customer to identify and come up with the best solution for such changes.

Regarding implementing the change, the team will follow a strict change control process below:
**Process for change control:**

- Identify and analyze the change to be made and how it will affect the system.
- Document the change to be made by creating a change request form or creating a change plan.
- Wait for approval from stakeholders or higher-ups.
- Review and implement the change.
- Validate change to ensure no problems are present.
- Finalize changes, then carry out a post-implementation review of the change.

**Tracking Decisions:**
The team will first identify the problem/task at hand, splitting into our respective subteams to discuss and create trade studies or research if necessary. The leads will then come together after taking note of the decisions/changes to be made, providing a list for the project manager and subteam members to see. Depending on the decision/change, stakeholders or higher-ups will be notified moving forward. Once approval is given for the team to discuss decisions/changes, the team will have a

meeting where they will vote on the best fit change regarding the task at hand. In the case of time-sensitive decisions, an autocratic approach will be taken if required. Changes will be tracked by documentation from team leads, followed by announcements by the project manager. Regarding what items will be prioritized when developing plans to address changes, the most important item that will be regarded is budget/cost as money is required in order to operate with no funding or being over budget the mission would be delayed which would affect the second item, scheduling. These two will be prioritized over all other items, but other items like science, risks, and more will be taken account of as they are equally important to success in projects.

### 5.4.2. Scope Control Management

**Scope Management Overview:**
Scope control management is a huge part of projects and can cause issues if not handled/planned accordingly. The project manager will monitor the project and in the case of changes that need to be made to the scope baseline, the team will have a meeting to discuss whether to add changes or reduce parts as necessary. More specifically, in the case of downscoping or an increase in scope, the team will have meetings to define and make adjustments if needed to prevent issues down the line. Some potential issues that may occur are changes in objectives/goals that may affect scheduling, budgeting, manufacturing problems, and more. During this meeting, the team will go over the two possible scenarios and will proceed as planned following the change control plan and making adjustments as the team progresses. In short, the team will analyze project objectives/goals to find a boundary (in scope or out of scope) that the team can maintain, defining what that scope baseline is to come up with possible constraints, assumptions, changes in scheduling, and changes in budgets.

**Downscoping/Increase in scope:**
In either case of reducing/limiting or increasing/adding certain aspects, the team will have a meeting to analyze the current situation to break down the issue at hand. Regarding potential changes/adjustments that need to be made, the team will continue with the same process for change control. The team will first analyze and identify the change to be made and how it will affect the whole project lifecycle. Then, document all important and necessary information to create a change form if necessary. In the case of a change form being required, the team shall continue working on parts of the project that do not require changes while waiting for approval from stakeholders to absolutely change certain parts of the project. After the change is processed and made, the team will review the change(s) before implementing it to ensure no problems arise that could jeopardize the project. In the case of issues that may pose a threat, the team will do a thorough investigation to find what can be done to prevent this issue. If there is no issue, the team will proceed with implementation and review the change one last time to

ensure no problems arise after implementation. Additionally, The team will proceed attentively, monitoring the budget to ensure it is being spent efficiently and making changes to schedules if required to ensure enough time is given for tasks. In the case of extra funds or the need for more, the team will allocate/recalculate parts of the mission that may or may not need the current budget. Depending on the current stage of the project, these plans may change accordingly to adapt to the current timeline.

**Actions for Scope Management:**

- With more funding would allow the team to broaden our scope, providing high quality outcomes and faster progress since the team is not limited by budget. More skilled personnel, advanced machinery, and facilities would be able to aid in the success of the project.
    - Regarding funding, things like constraints would be accounted for easily as funding would open up new opportunities. For example, the team had a mass constraint problem as the team is not able to use a lighter yet more durable material, as the team was being limited to using a heavy material as cost was a concern.
- With more time the team would be able to plan better and provide higher quality outcomes which would account for unforeseen delays. Not to mention, allocation of resources would be used more effectively and efficiently which would greatly contribute towards success.
    - Regarding scheduling, there would be more time to plan, schedules that fit more with the team, and stress would ultimately go down as the team would not have to be scattering to meet deadlines and handle individual responsibilities outside of work.
- With more workers, the team would be able to handle larger workloads, complete tasks faster, and adapt to problems that arise with ease.
    - Regarding more workers, extra workers would be assigned to different teams depending on their experience, with more hands to work with efficiency would go up when handling heavy amounts of workload. The amount of workers ultimately go down as the mission is completed and comes to an end.

### 5.5 Outreach Summary

Outreach of the Venera Nova mission will consist of door-to-door (D2D) advertisement, high school outreach and social media sponsorships. The D2D advertisement will educate residents on the mission and objective of Venera Nova and showcase the capabilities of the team's aerobot. D2D will be on a volunteer basis open to interns and motivated citizens who have a love for space, similar to what is seen during a campaign, the recruitment/volunteer process will happen around phase C to promote this project but will not fall short to present potential future projects as well. Not only that, this will also provide a great platform for building long lasting relationships with customers that can be valuable in the long run.

High school outreach will consist of a mobile planetarium, showcasing the mission plan. The three science goals will be accomplished while a planetarium instructor is explaining the mission in detail.This will motivate high school students to follow the mission timeline as years go by and potentially influence them to pursue a space career field.

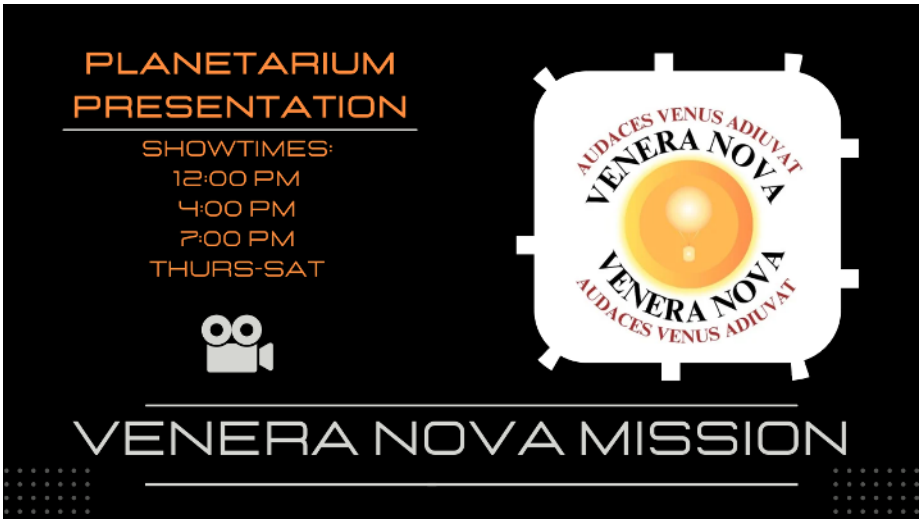



*Figure 44: Graphic representation of a planetarium promotional poster.*

Social media sponsorships will be paid partnerships with famous actors, influencers, and well known members of the space community that will advertise the mission and launch date. An average sponsorship post on Instagram is estimated to cost about $500 and the team is expected to have 15 paid advertisements from individuals mentioned above. All outreach events will happen late phase C into early phase D (stages of planning and development, early to finalization). Three tours will be done regarding the planetarium presentation with the month and year of July 2026, August 2027, December 2028 prior to finalization stage. Through these outreach events, the team will be able to

not only educate and engage with the public but also interact with current and future potential stakeholders creating a strong trust/bond that can open up more opportunities and relations that will benefit everyone involved.




*Figure 45: Graphic representation of a social-media based promotional poster.*

## 6. Conclusion

The Venera Nova team plans on exploring mysterious phenomena on Venus. This will be achieved by studying Venus' strange super-rotation, lack of a planetary magnetic field, and possible principles to support life within the upper atmosphere as the team's aerobot descends into the Venus atmosphere. A range of instruments will be utilized to study winds, material composition, magnetic fields, cloud forms, and solar winds in the mission's location of interest, Ovda Regio. Ovda Regio is located within the Aphrodite Terra region, a continent-sized highland region, south of Venus' equator. This area would be ideal for studying wind speed and will greatly contribute to accomplishing the three science objectives of this mission.The team's design shall be complete by October 1, 2028 and ready to launch by March 1, 2029. Venus' extreme conditions will also be factored in and countermeasures will be appropriated. All instruments within or outside of the aerobot will be protected long enough to perform their specific function. The aerobot will first enter Venus' atmosphere near its anti-solar point and utilize the three specific instruments to accomplish objectives.

If more time was provided for this mission, the team would like to try designing a rover that could withstand Venusian conditions. Similarly to NASA's Perseverance Rover, the team intends to collect data from another planet so close in proximity to Earth. The extreme pressures, weather, and temperature are nothing like Earth's and would make for great research. The team's plan of action for the Mission Definition Review is to determine the maturity of the Venera Nova mission by evaluating outreach protocol, mission cost, and variables that will contribute to a successful launch and mission.

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# Appendix

**[A.1]** The maximum power draw for VMC was calculated based on known power efficiency values (0.5-0.75) and total secondary power consumption (5.2 W). From these values, maximum power draw was calculated using the formula

$$P_{primary} = P_{secondary} / \text{efficiency} = 5.2 / 0.5 = 10.4 \text{ W}$$

The dimensions for the VMC was approximated using the following image:



*Image of the VMC (Markiewicz et al., n.d.).*

The team estimated the VMC dimensions relative to the red pair of pliers in the background of the image. The pliers are 7 inches in length, resulting in an estimate of VMC being roughly 2 pliers tall and one plier in both width and length (Lowes, n.d.).

**[A.2]** Full derivation of hot and cold case heat flow maps for thermal control systems.

# Heat Flow Map

- Going to want system to operate at 300 K (≈ room temp) bc electronics will most likely be operational at that T

- $A = 1\,m^2 \rightarrow$ 6 sides
  - will be experiencing $Q_{rad}$ on all surfaces

$Q_{cond} = \frac{kA(T_1 - T_2)}{L}$, $Q_{rad} = \varepsilon\sigma FA(T_1^4 - T_2^4)$, $Q_{conv} = hA(T_1 - T_2)$

- $AU_{sun\ to\ venus} = 0.723\ AU$ → https://lco.global/spacebook/solar-system/venus/#:~:text=Venus%20is%20the%20second%20closest,1.082%20x%20108%20km.

**Hot Case** → $T_{venus} = 394.7$ K
- → 50 km altitude

**Cold Case** → $T_{venus} = 225.8$ K
- → 70 km altitude

$Q_{int} = 79.4$ W

**MLI** : top + 3 sides
- $\alpha = 0.05$
- $\varepsilon = 0.05$

**White Paint (293)** : bottom + 1 side
- → $\varepsilon = 0.92, \alpha = 0.16$

White coating emissivity and absorptivity (p42)

**Hot Case** $T_{venus,hot} = 394.7$ K, 50 km altitude, $Q_{internal} = 79.4$ W

**Solar Load**
- assumption: only affects top face ∴ use $\alpha_{MLI}$
- for hot case assume F=1

$Q_{solar} = q_{solar\ flux} A \alpha_{MLI} F \rightarrow q_{solar\ flux} = \frac{1367}{(AU)^2}$ → Astronomical Units, $AU_{sun\ to\ venus} = 0.723\ AU \rightarrow q_{solar\ flux} = \frac{1367}{(0.723)^2} \Rightarrow q_{solar\ flux} = 2615.122$

$\rightarrow Q_{solar} = 2615.122(1)(0.05)(1) \Rightarrow$ **$Q_{solar} = 130.76$ W** into sys

**Albedo**
- similar to solar load, except the albedo is the sunlight that reflects off the surface.
  - Venus reflects ~75% of the sunlight that hits it ∴ albedo of 0.75 (can't be ignored bc high value)
  - Hot Case (50 km) → assume full albedo bc denser atmosphere, which reflects more sunlight onto sys → albedo = 0.75
  - Cold Case (70 km) → assume half albedo bc of atmosphere is ab half as dense ∴ albedo = 0.375 ← $\cdot\frac{1}{2}$
- assumptions for faces
  - Top Face, viewing factor, F=0 (neglect)
  - Side Faces, $F_{side} = 0.3$ → 3 side face, $\alpha_{MLI}$
    → 1 side face, $\alpha_{293}$
  - Bottom Face, $F_{bottom} = 1$ & $\alpha_{MLI}$

$Q_{albedo} = q_{solar\ flux} \cdot albedo_{50\,km} \cdot A \cdot \alpha \cdot F \rightarrow$ A, $q_{solar}$, & albedo const. for all $\rightarrow Q_{albedo} = q_{solar\ flux} \cdot A \cdot albedo_{solar} \left[3(\alpha_{MLI} \cdot F_{side}) + 1(\alpha_{293} \cdot F_{side}) + 1(\alpha_{MLI} F_{bottom})\right]$

$\Rightarrow$ **$Q_{albedo} = 280.47$ W** into sys

**Radiative HT**
- assume F=1 for hot & cold case bc system will be surrounded by Venus' radiative atmosphere
- use $T_{venus,hot}$ & $T_{sys}$ & Stefan-Boltzmann constant, $\sigma = 5.67 \cdot 10^{-8}\ \frac{W}{m^2 \cdot K}$
- 5 faces w/ MLI & 1 faces w/ 293

$Q_{rad,venus} = \varepsilon\sigma FA(T_{sys}^4 - T_{venus,hot}^4) \rightarrow \sigma, F, A$ & $(T_{sys}^4 - T_{venus,hot}^4)$ are const. $\rightarrow Q_{rad,venus} = \sigma FA(T_{sys}^4 - T_{venus,hot}^4)(5\varepsilon_{MLI} + 1\varepsilon_{293})$

$\Rightarrow$ **$Q_{rad,Venus} = 399.31$ W** into sys
- ↑ is actually −

## Convective HT

- for h, assume sys is moving at same speed as Venus' winds, ∴ its relative velocity is 0. → $Re = \frac{\rho V \delta}{\mu} = 0$

$\therefore Nu = 0.3 + \frac{0.62 Re^{1/2} Pr^{1/3}}{[1+(0.4/Pr)^{2/3}]^{1/4}} \left[1 + \left(\frac{Re}{282000}\right)^{5/8}\right]^{4/5} \Rightarrow Nu = 0.3$ ∴ $h = Nu \frac{k}{D}$, let D = length from one end of the face to the other, so D = 1m ∴ $h = Nu \cdot k$

- for convection use k for air, not surface. Venus' atmosphere is 96% $CO_2$ ∴ use thermal conductivity of $CO_2$ → $k_{CO_2}$ @ 344.7 K = 0.0305 $\frac{W}{m \cdot K}$

https://www.britannica.com/place/Venus-planet/The-atmosphere for % of $CO_2$

https://www.engineeringtoolbox.com/carbon-dioxide-thermal-conductivity-temperature-pressure-d_2019.html for $k_{CO_2}$ values

- use $T_{venus,hot}$ & $T_{sys}$

$Q_{conv} = hA(T_1 - T_2) = G \cdot Nu k_{CO_2, 344.7K} A(T_{sys} - T_{venus,hot}) \Rightarrow Q_{conv} = 1.65$ W into the sys

---

Cold Case $T_{venus,cold} = 225.8$ K, 50 km altitude, $Q_{internal} = 79.4$ W

**Solar Load** · assumption: for coldest case assume F = 0 for solar load

$Q_{solar} = q_{\substack{solar \\ flux}} A \alpha_{MLI} F \Rightarrow Q_{solar} = 0$ W

**Albedo**

- similar to solar load, except the albedo is the sunlight that reflects off the surface.
  - Venus reflects ~75% of the sunlight that hits it, ∴ albedo of 0.75 (can't be ignored be high value)
  - Hot Case (10 km) → assume full albedo be denser atmosphere, which reflects more sunlight onto sys → albedo = 0.75
  - Cold Case (70 km) → assume half albedo bc of atmosphere is ab half as dense, ∴ albedo = 0.375 ← ·½
- assumptions for faces
  - Top Face, viewing factor, F = 0 (neglect)
  - Side Faces, $F_{side} = 0.3$ → 3 side face, $\alpha_{MLI}$
    → 1 side face, $\alpha_{293}$
  - Bottom Face, $F_{bottom} = 1$ & $\alpha_{MLI}$

$Q_{albedo} = q_{\substack{solar \\ flux}} \cdot albedo_{\substack{venus}} \cdot A \cdot \alpha \cdot F \rightarrow$ A, $q_{solar\ flux}$, & albedo const. for all $\rightarrow Q_{albedo} = q_{\substack{solar \\ flux}} \cdot A \cdot albedo_{\substack{venus}} \left[3(\alpha_{MLI} \cdot F_{side}) + 1(\alpha_{293} \cdot F_{side}) + 1(\alpha_{MLI} F_{bottom})\right]$

$\Rightarrow Q_{albedo} = 140.24$ W into sys

## Radiative HT

- assume F = 1 for hot & cold case bc systems will be surrounded by Venus' radiative atmosphere
- use $T_{venus,cold}$ & $T_{sys}$ & Stefan-Boltzmann constant, $\sigma = 5.67 \cdot 10^{-8} \frac{W}{m^2 \cdot K}$
- 4 faces w/ MLI & 2 faces w/ 293

$Q_{rad,venus} = \varepsilon \sigma FA(T_{sys}^4 - T_{\substack{venus \\ cold}}^4) \rightarrow \sigma, F, A$ & $(T_{sys}^4 - T_{venus,hot}^4)$ are const. $\rightarrow Q_{\substack{rad, \\ venus}} = \sigma FA(T_{sys}^4 - T_{\substack{venus, \\ cold}}^4)(5\varepsilon_{MLI} + 1\varepsilon_{293})$

$\Rightarrow Q_{rad,Venus} = 364.90$ W out of sys

## Convective HT

- for h, assume sys is moving at same speed as Venus' winds, ∴ its relative velocity is 0. → $Re = \frac{\rho V \delta}{\mu} = 0$

$\therefore Nu = 0.3 + \frac{0.62 Re^{1/2} Pr^{1/3}}{[1+(0.4/Pr)^{2/3}]^{1/4}} \left[1 + \left(\frac{Re}{282000}\right)^{5/8}\right]^{4/5} \Rightarrow Nu = 0.3$ ∴ $h = Nu \frac{k}{D}$, let D = length from one end of the face to the other, so D = 1m ∴ $h = Nu \cdot k$

- for convection use k for air, not surface. Venus' atmosphere is 96% $CO_2$ ∴ use thermal conductivity of $CO_2$ → $k_{CO_2}$ @225.8 K = 0.0113 $\frac{W}{m \cdot K}$

https://www.britannica.com/place/Venus-planet/The-atmosphere

https://www.engineeringtoolbox.com/carbon-dioxide-thermal-conductivity-temperature-pressure-d_2019.html

- use $T_{venus,cold}$ & $T_{sys}$

$Q_{conv} = hA(T_1 - T_2) = G \cdot Nu k_{CO_2, 225.8K} A(T_{sys} - T_{venus,cold}) \Rightarrow Q_{conv} = 1.51$ W out of sys.

---

## Net Heat Loads

**Hot Case** $Q_{int} + Q_{solar} + Q_{albedo} + Q_{rad} + Q_{conv}$

$79.4 + 130.74 + 280.47 + 299.31 + 1.65 = 891.49$ W

Need ~892 W of cooling

$\sum Q_{in} = Q_{int} + Q_{solar} + Q_{albedo} + Q_{rad} + Q_{conv}$

$= 891.49$ W

$\sum Q_{out} = Q_{cooler}$

$= 892$ W

**Cold Case** $Q_{int} + Q_{solar} + Q_{albedo} + Q_{rad} + Q_{conv}$

$79.4 + 0 + 140.24 + (-364.90) + (-1.51) = -146.77$ W out of sys

need ~147 W of heating

$\sum Q_{in} = Q_{int} + Q_{solar} + Q_{albedo} + Q_{heater}$

$= 219.64 + Q_{heater}$ ← =? $= 366.64$ W

let = 147 W

$\sum Q_{out} = Q_{rad} + Q_{conv}$

$= 366.41$ W

$\sum Q_{net} = 0$ W